

Kidney Replacement Therapy: Dialysis

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12.1 Indications for Acute/Chronic Dialysis

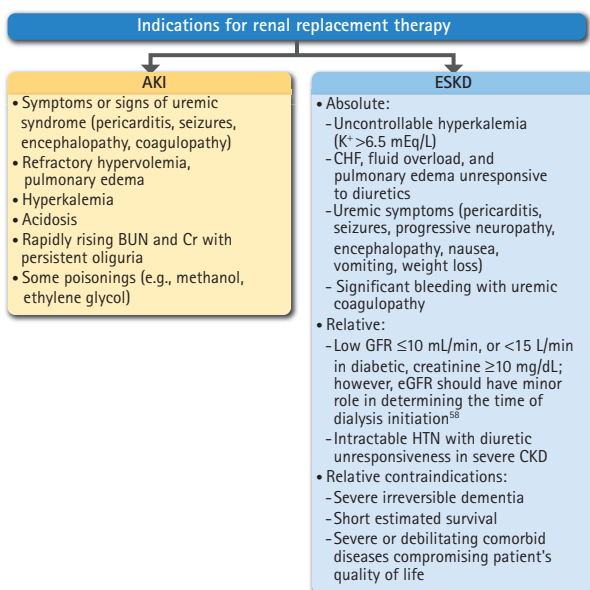
Treatment options for both end-stage kidney disease (ESKD) and acute kidney injury (AKI) include extracorporeal hemodialysis-based options, and peritoneal dialysis (PD).¹ Important elements in the decision to initiate dialysis include interpreting the signs and symptoms of uremic syndrome (Table 12.1).

Assessment should include the specific indication(s) for initiating dialysis, the appropriateness of initiating dialysis as opposed to conservative management, the best method of dialysis, and the appropriate timing of initiation of dialysis.²⁻⁴

In some cases, the initiation should be done on a trial basis, leaving open the possibility of later withdrawing the patient from dialysis support. An increasingly popular strategy in the chronic setting is active conservative chronic kidney disease (CKD) management, to forestall the need for dialysis. In addition, withholding of dialysis is increasingly a part of the nephrologist's evaluation in the AKI or ESKD setting.^{5,6} Examples include patients with profound, irreversible neurological impairment, terminal illness, inability to cooperate with the procedure, and those with markedly impaired functional status.² Such a decision requires appropriate input from the patient/healthcare proxy/family members, including the option for them to pursue a second opinion.

TABLE 12.1 ■ Symptoms and Signs of the Uremic Syndrome

Symptoms
• Poor appetite, weight loss • Diminished taste • Pruritus • Muscle cramps • Fatigue • Nausea/vomiting • Cold intolerance • Sleep disturbance • Impaired cognition
Signs
• Hypervolemia, edema, congestive heart failure • Neuropathy • Pericarditis • Encephalopathy, confusion, seizures • Coagulopathy, bleeding



AKI, Acute kidney injury; BUN, blood urea nitrogen; CHF, congestive heart failure; CKD, chronic kidney disease; Cr, creatinine; eGFR, estimated glomerular filtration rate; ESKD, end-stage kidney disease; HTN, hypertension.

Figure 12.1 Indications for renal replacement therapy.

12.1.1 INDICATIONS FOR DIALYSIS INITIATION

Several indications for initiating dialysis are common, and they are similar in acute or chronic renal failure: to alleviate symptoms or signs of uremia, correct abnormalities of electrolyte or acid-base balance, or reverse uncontrolled hypervolemia (Figure 12.1).

Dialysis may be started in severe acute renal failure when no quick recovery is expected, to avoid “impending uremia.” Typical indications for initiating renal replacement therapy (RRT) are related to hyperkalemia, acidemia, volume overload, and symptomatic azotemia.

12.2 Dialysis Modalities for Acute Kidney Injury

Guidelines state that intermittent hemodialysis (IHD) and continuous renal replacement therapy (CRRT) are complementary therapies, with no clear superiority of one over the other (PD is currently rarely used for AKI). In a recent meta-analysis of 21 studies comparing clinical outcomes among critically ill adults with AKI treated with CRRT, IHD, or sustained low efficiency dialysis (SLED), for example, no modality had a distinct advantage.⁷ For clinically stable AKI patients, IHD is the standard modality. For more compromised patients in the intensive care unit (ICU) in whom either hemodialysis (HD) or CRRT is a consideration, continuous therapies do not appear to result in superior key outcomes such as patient survival, ICU length of stay, or ventilator days.⁸

Meta-analyses have compared outcomes without an overall survival benefit to either modality. Several, but not all, analyses of observational data have shown a higher rate of nonrecovery from AKI among survivors treated with IHD.⁹ Conventional HD may be assumed to be contraindicated in severely compromised ICU patients (e.g., those with hypotensive circulatory failure, for

whom CRRT is the standard modality). In patients with acute brain injury or fulminant hepatic failure, better preservation of cerebral perfusion may be achieved with CRRT.

Recent guidelines from the International Society for Peritoneal Dialysis suggested that PD should be considered a suitable modality for treatment of AKI in all settings,¹⁰ citing studies that demonstrated equivalent survival and possibly a shorter need for RRT with PD compared to other modalities.^{11,12}

12.2.1 APPROPRIATENESS OF DIALYSIS INITIATION

Whether or not to offer dialysis must be addressed in the AKI patient, just as it is in the ESKD patient. While the vast majority of patients could undergo dialysis with current technology, feasibility must first be addressed. The potential for dialysis recovery should be evaluated. Advance directives must be accommodated. The patient must be informed about the procedure and potential outcomes, including the possible dependence on chronic dialysis.¹³ Professional guidelines should be reviewed.¹⁴

12.2.2 TIMING OF INITIATION OF DIALYSIS IN AKI

The evidence regarding timing of dialysis initiation in sick patients with AKI remains relatively weak, and guidelines remain nonspecific.¹⁵ It is generally accepted that dialysis is urgently indicated when the AKI patient has pericarditis, altered mental status, seizures, severe metabolic derangements, uncontrollable symptomatic hypervolemia, or treatable intoxications.

When the case to initiate dialysis is less compelling, considerations may include the likelihood of recovery and the risks of doing dialysis. Elective initiation of dialysis for AKI may be required when symptoms or laboratory derangements are less severe. Multiple clinical trials have compared an early initiation approach when there is no specific indication, with a delayed approach (i.e., initiating when relevant but nonsevere signs or symptoms) have developed.

In recent years, a few randomized controlled trials have attempted to resolve the issue of initiation of dialysis for AKI with mixed results (Table 12.2).¹⁶

Two randomized controlled trials have recently reported data on the timing of RRT in critically ill patients with AKI, while demonstrating the difficulties of such trials. In the Artificial Kidney Initiation in Kidney Injury (AKIKI)¹⁷ multicenter trial from France, 620 patients were initiated either “early” (KDIGO stage 3) or “late” (KDIGO stage 3 and clinical indications), to determine whether late dialysis initiation improved mortality risk. There was no difference in the primary outcome of 60-day all-cause mortality. Importantly, however, half of patients randomized to late initiation did not receive RRT, either recovering kidney function or dying. Nonetheless, a post-hoc analysis indicated that, for patients in the late-start group actually treated with RRT, mortality was higher in comparison with the early group.

In the Standard Versus Accelerated Initiation of Renal-Replacement Therapy in Acute Kidney Injury (STARRT-AKI) trial,¹⁸ the most definitive study to date, over 3000 AKI patients without a specific indication for dialysis were randomized. Dialysis was initiated in most of the early-initiation patients, but only about two-thirds of the late-initiation patients. Adverse events, continued dialysis dependence, and rehospitalization, but not mortality, were higher in the early-initiation group. In the AKIKI-2 trial, critically ill AKI patients were allowed to be more azotemic before randomization to early or late strategies, with mixed results.¹⁹

The single-site Early Versus Late Initiation of KRT in Critically Ill Patients With AKI (ELAIN) trial in Germany²⁰ compared early (KDIGO AKI stage 2) with late (KDIGO AKI stage 3). To diminish the likelihood that enrollees would actually recover kidney function and not need dialysis, clinical AKI criteria were combined with levels of the biomarker NGAL. In the late group, only 5% recovered from AKI without needing dialysis. The early group had a lower 90-day all-cause mortality; the ELAIN trial is the only recent randomized trial to demonstrate mortality benefit.

TABLE 12.2 ■ Comparison of Recent Randomized Clinical Trials of Timing of KRT in AKI

Variable ^a	ELAIN (N = 231)	AKIKI (N = 620)	IDEAL-ICU (N = 488)	STARTR-AKI (N = 3019)
Study design	Single-center RCT	Multicenter RCT	Multicenter RCT	Multicenter multinational RCT
Early group	Within 8 hr of stage 2 AKI and NGAL > 150 ng/mL; median: 6 hr	Within 6 hr of stage 3 AKI; median: 2 hr	Within 12 hr of stage 3 AKI; median: 7.6 hr	Within 12 hr of stage 2 or 3 AKI; median: 6.1 hr
Delayed/standard group (time from eligibility)	Within 12 hr of stage 3 AKI or clinical indication; median: 25.5 hr	After 72 hr in stage 3 AKI or clinical indication; median: 57 hr	After 48 hr in stage 3 AKI or clinical indication; median: 51.5 hr	After 72 hr in AKI or clinical indication; median: 31.1 hr
Key exclusions	eGFR < 30	Severe lab abnormalities	Emergent need for dialysis	eGFR < 20
KRT protocol	Prescribed protocol using CVVHDF	Discretion of site providers, guideline based	Discretion of site providers, guideline based	Recommendations provided based on guidelines
Primary outcome	90-day mortality: 39.3% vs. 54.7% ($P = 0.03$)	60-day mortality: 48.5% vs. 49.7% ($P = 0.79$)	90-day mortality: 57.7% vs. 53.8% ($P = 0.38$)	90-day mortality: 43.9% vs. 43.7%; RR, 1.00 (95% CI, 0.93–1.09)
Proportion not requiring KRT	0% vs. 9.2%	1.9% vs. 49.0%	2.8% vs. 38.4%	3.2% vs. 38.2%
Dialysis dependence (among survivors)	At 90 days: 13.4% vs. 15.1% ($P = 0.80$)	At 60 days: 1.9% vs. 5.2% ($P = 0.12$)	At 90 days: 2.0% vs. 2.7% ($P = 1.00$)	At 90 days: 10.4% vs. 6.0%; RR, 1.74 (95% CI, 1.24–2.43)
Adverse events	No significant differences	CRBSIs: 10.0% vs. 5.2%; $P = 0.03$	Hyperkalemia: 0% vs. 4.1%; $P = 0.03$	Any: 23.0% vs. 16.5% ($P < 0.001$); no difference in serious adverse events

^aAll comparisons are early versus delayed/standard groups.

AKI, Acute kidney injury; CI, confidence interval; CRBSIs, catheter-related bloodstream infections; CVVHDF, continuous veno-venous hemodiafiltration; eGFR, estimated glomerular filtration (in mL/min/1.73 m²); KRT, kidney replacement therapy; NGAL, neutrophil gelatinase-associated lipocalin; RCT, randomized controlled trial; RR, relative risk. (Table 1.¹⁰)

Few studies have examined initiation of dialysis for AKI in specific settings. For elective initiation of dialysis in cases of AKI after major abdominal surgery, there is a suggestion that early initiation of dialysis might be beneficial to patient mortality.²¹

Finally, a recent meta-analysis showed that mortality did not differ significantly according to whether dialysis was initiated early or delayed in AKI, and suggested initiation only when a specific clinical indication emerges as an acceptable approach.²²

Taken together, these studies have diminished enthusiasm for aggressive initiation of dialysis for AKI in the absence of a compelling indication.

12.2.3 WITHDRAWAL FROM DIALYSIS

The principal justification for dialysis is to improve quality of life and to prolong life. However, dialysis in vulnerable patients may be associated with deteriorating physical function, increasing dependence, and deteriorating quality of life. Withdrawal from dialysis may then be appropriately discussed with the patient and family, with emphasis placed on symptom control and addressing end-of-life goals.¹³

12.2.4 RECOVERY FROM AKI

The dialysis- or CRRT-dependent AKI patient should be monitored for recovery of kidney function, but the optimal strategy and timing for actually weaning from dialysis remain uncertain.²³

The obvious clinical clues include increase in urine output (UO) before dialysis (usually detected in the interdialytic interval), slowed postdialytic rise in creatinine (Cr), or a functional test such as responsiveness to diuretics. In the future, renal biomarkers may be shown to be independent predictors of recovery.

12.2.5 ADEQUACY OF CONVENTIONAL HEMODIALYSIS IN AKI

Assessment of adequacy of IHD is conventionally based on urea kinetic modeling. In the chronic setting, a calculated single-pool Kt/V urea of 1.2 is accepted as the minimal goal for adequate dialysis. Kt/V (urea) (the fractional urea clearance) is regarded as the most clinically valid small-solute measure of dialysis dose at the current time. It is a biochemical outcome measure. Kt/V (urea) refers to clearance of urea in a given time period ($Kt = \text{clearance} \times \text{time}$), individualized to patient size, with this calculation based on the patient's total body water volume (V) as estimated from the patient's weight, leading to the expression Kt/V. In the setting of AKI, however, the volume of distribution of urea may be highly variable. Kidney Disease: Improving Global Outcomes (KDIGO) guidelines recommend delivering a Kt/V of 3.9 per week for IHD.²⁴ The Veterans Administration/National Institutes of Health (VA/NIH) ATN study²⁵ randomized patients to receive either intensive or less intensive dialysis; for those on intermittent dialysis, the less intensive group were treated 3 times per week, and the more intensive group 6 times per week. Intensive renal support did not improve outcomes. The best evidence, such as from the Acute Renal Failure Trial Network (ATN) Study (VA/NIH), suggests that patients with dialysis-dependent AKI should receive at least 3 HD treatments per week, each with a delivered Kt/V value of 1.2. A simpler approach may be to target a urea reduction ratio ($URR = [\text{predialysis BUN} - \text{postdialysis BUN}] / \text{predialysis BUN}$) of >0.67 .²⁶

12.2.6 CONTINUOUS RENAL REPLACEMENT THERAPY

The choice of renal replacement modality is commonly determined by the patient's hemodynamic status. However, CRRT for AKI in gradually undergoing a paradigm shift beyond just

those patients too unstable to tolerate conventional HD, to becoming the treatment of choice for all with multiorgan failure.²⁷ CRRT is an ICU-based therapy intended to be applied on a continuous but short-term basis.

CRRT is the preferred dialysis modality in patients with AKI and hemodynamic instability.²⁸ A typical indication for CRRT is the need for RRT (fluid overload, uremia, uncorrectable acidosis, hyperkalemia, some intoxications) in a hemodynamically unstable patient. It may also have advantages for the AKI patient with increased intracranial pressure,²⁹ severe hypo/hypernatremia,³⁰ or drug intoxications. Typical indications for CRRT are shown in Table 12.3.

There are multiple CRRT options, as shown below in Table 12.4.

Most CRRT is done by continuous veno-venous hemofiltration (CVVH), rather than continuous arterio-venous hemofiltration (CAVH), as CVVH without dialysis or with concomitant dialysis, through a process called continuous veno-venous hemodialysis or hemodiafiltration (CVVHD). Balanced hemofiltration and dialysis with larger volumes of hemofiltration and large volumes of replacement fluid, together with dialysis, can be achieved with CVVHDF (also known as continuous hemodiafiltration) when the hemofiltration rate is set at 20% and the effluent target ranges from 20 to 35 mL/kg/hr. They should be viewed as complementary therapies for the AKI/ESKD population.

CRRT offers several advantages over intermittent HD in critically ill patients: greater hemodynamic stability, increased fluid removal, and continuous solute (uremic toxins, electrolytes) control.³¹ CRRT clearance is based on a combination of diffusion (dialysis), convection (ultrafiltration), and to a lesser degree, adsorption by the membrane. It is similar to clearance by intermittent dialysis, although less efficient per unit time.

Compared to HD, blood flow is slower, the throughput of fluid (i.e., administered and removed) is much larger than for HD, and the dialysate rate is slower. Because filter clearance for small solutes is equal to the product of the total effluent (i.e., postfilter fluid) amount and

TABLE 12.3 ■ Typical Indications for CRRT

- Oliguria with hypotension
- AKI/ESKD with fluid overload
- Cardiogenic shock with pulmonary edema
- AKI/ESKD with acute brain injury
- ARDS
- Severe sepsis
- Refractory congestive heart failure

AKI, Acute kidney injury; ARDS, acute respiratory distress syndrome; CRRT, continuous renal replacement therapy; ESKD, end-stage kidney disease.

TABLE 12.4 ■ CRRT Options

Option	Clearance	Dialysis	Replacement	UF Goals	Clearance
SCUF	Convection	No	No	Modest	Low
CVVH	Convection	No	Yes	Large	Medium
CVVHD	Convection and diffusion	Yes	No	Modest	Higher
CVVHDF	Convection and diffusion	Yes	Yes	Large	Highest

CRRT, Continuous renal replacement therapy; CVVH, continuous veno-venous hemofiltration; CVVHD, continuous veno-venous hemodialysis; CVVHDF, continuous veno-venous hemodiafiltration; SCUF, slow continuous ultrafiltration; UF, ultrafiltration.

TABLE 12.5 ■ Examples of Available Substitution Fluid and Dialysate Solutions in the United States for CRRT by Manufacturer

	GAMBRO (BAXTER)		NXSTAGE	B. BRAUN
	^a PrismaSol BGK/B22K/BK	^b PrismaSATE3 BGK/B22K/BK	^b RFP 400–456	^b Duosol 4551– 4556
Na ⁺ , mEq/L	140	140	130–140	140–136
K ⁺ , mEq/L	0–4	0–2–4	0–4	0–4
Cl [–] , mEq/L	108–113	108–120.5	108.5–120.5	109–117
Lactate, mEq/L	3	3	0	0
Bicarbonate, mEq/L	22–32	22–32	25–35	35–25
Ca ⁺⁺ , mEq/L	0–2.5–3.5	0–2.5–3.5	0–3	3–0
Mg ⁺ , mEq/L	1.0–1.2–1.5	1.0–1.2–1.5	1–1.5	1–1.5
Dextrose, g/dL	0–1	0–1.1	1	1–0

^aSubstitution fluid.
^bDialysate solutions.
CRRT, Continuous renal replacement therapy.
From Macedo E, Mehta RL. Continuous dialysis therapies: core curriculum 2016. *Am J Kidney Dis*. 2016 Oct;68(4):645–657. doi:10.1053/j.ajkd.2016.03.427.

the concentration of solute in that effluent is relative to plasma (i.e., the sieving coefficient)—which reaches 1 for small solutes—clearance adequacy can be measured as actual solute removal. Adequacy is therefore determined by the total effluent fluid rate (replacement fluid, dialysate, net fluid removal), which is recommended to be 25 mL/kg body weight per hour.

The fluids utilized for replacement and dialysate are premixed, sterile, and free of bacterial endotoxins. Depending on the product, they generally contain calcium, magnesium, sodium, potassium, bicarbonate, lactate, and dextrose. An additional fluid replacement option contains phosphate. A few of the options do not contain dextrose; loss of endogenous glucose in the ultrafiltrate may place some critically ill, catabolic patients at risk for euglycemic ketosis during CRRT.³² Typical replacement and dialysate solutions are shown in Table 12.5.²⁷

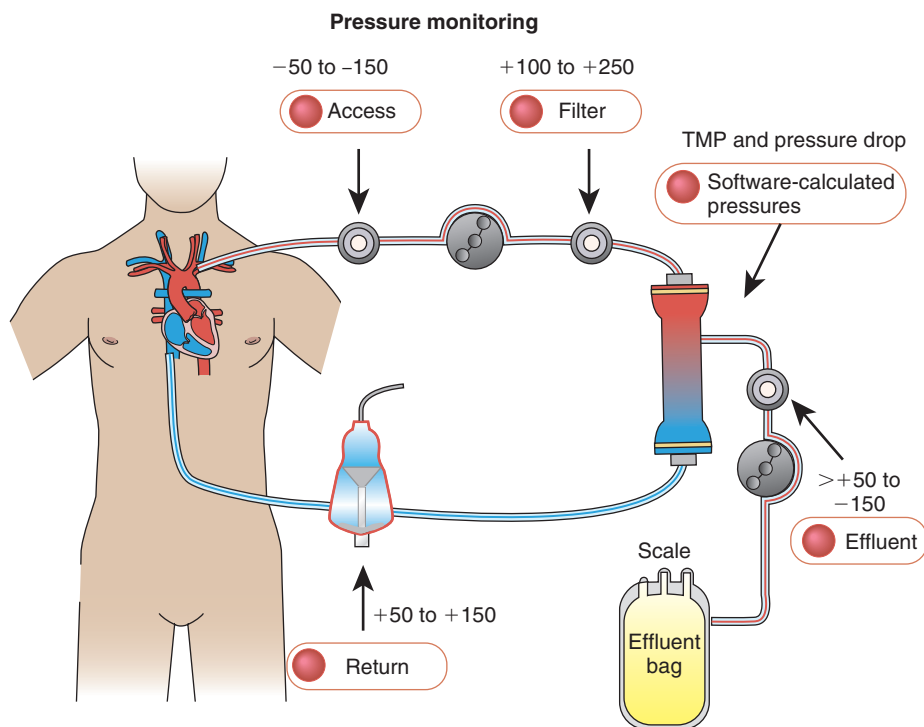
A typical order set for CVVHDF would include blood flow rate 100 to 250 mL/min; dialysate flow rate 1000 to 2000 mL/hr; replacement flow rate 1000 to 2000 mL/hr; and ultrafiltration rate 33 to 66 mL/min.²⁷

Replacement fluids may be administered pre- or postfilter. Prefilter administration is associated with decreased filter clotting and increased filter lifespan. (The blood pump will usually have its rate increased to compensate for the added fluid and to achieve sufficient clearances.) Postfilter administration is associated with improved solute clearances.

As a motor-driven system involving flow of fluid, key pressure measurements are determined at four points in the CRRT extracorporeal circuit. Abnormal pressure alarms require clinical correlation. Pressures are used to calculate the transmembrane pressure (TMP) and pressure drop from one end of the filter to the other. Pressure measurements that may require troubleshooting are shown in Figure 12.2.

12.2.7 CHARACTERISTIC PRESSURE MEASUREMENTS AND COMMON ABNORMALITIES

- Access line: Preblood pump; pressure always negative (–50 to –150); high due to catheter clotting, line kinked.



TMP, Transmembrane pressure.

Figure 12.2 The CRRT extracorporeal circuit and typical pressure readings obtained at the points in the circuit shown to determine the status of the dialysis access outflow, the CRRT filter, the effluent collection system, and the dialysis access venous return.

- Filter (prefilter): At postblood pump; always positive (+50 to +150); detects resistance to flow to patient, or clotting.
- Effluent line: Before effluent pump; usually negative but may be positive; -150 to +50; used to calculate TMP.
- Return line: Pressure of return circuit before catheter; high pressure due to catheter clotted/line kinked.

The filtration fraction is the fractional volume of plasma removed from the dialyzed blood in the process of ultrafiltration (i.e., the ratio of the ultrafiltration rate to the plasma flow rate). Filtration fraction is calculated as follows:

12.2.8 FILTRATION FRACTION

- $FF(\%) = (UFR \times 100)/Q_p$
- where $Q_p = BFR \times (1 - hct) =$ filter plasma flow rate in mL/min (FF, filtration fraction; UFR, ultrafiltration rate; Hct, hematocrit)
- Example: $BFR = 100 \text{ mL/min}$
 - $Hct = 30\%$
 - $UFR \text{ of } .21 (21\%) = FF 30\%$

Experts recommend keeping the filtration fraction below 20% to 30% in order to reduce the risk of premature filter clotting.³³ However, there are limitations of using the filtration fraction

as an index of risk of hemofilter clotting, including the rate of administration of prefilter replacement fluids. So the use of end-of-hemofilter hematocrit is an alternative, but needs to be validated.

Removal of medications by CRRT, since it is a form of continuous dialysis, may be significant, and may even exceed that of normal kidney function. Multiple citations are available as guides to drug dosing in CRRT.³⁴

Some antimicrobial agents do not require dose modification during CRRT (Table 12.6)³⁴.

CRRT allows for increased volumes of nutritional support but may result in loss of some nutrients.

The CRRT system typically requires anticoagulation, in the form of regional citrate anticoagulation with calcium replacement (Figure 12.3)²⁷, or, less commonly, systemic or regional heparin anticoagulation (Figure 12.4).

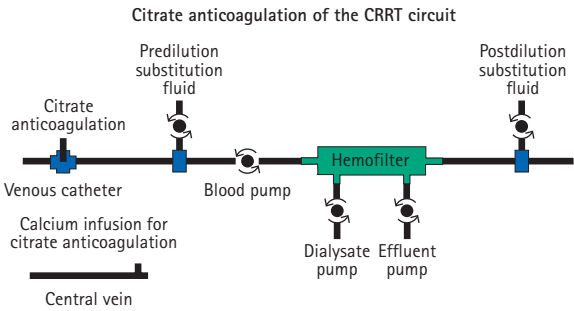
Citrate infused into the arterial port of the circuit chelates calcium, resulting in anticoagulation of the filter. Replacement parenteral calcium is administered separately. Guidelines now advocate citrate anticoagulation as first-line therapy for CRRT.²⁴ Trisodium citrate is typically administered prefilter at a fixed ratio between blood and citrate infusions. Replacement calcium

TABLE 12.6 ■ Antimicrobials With No Dosage Adjustment in CRRT

- Amphotericin
- Azithromycin
- Ceftriaxone
- Clindamycin
- Doxycycline
- Linezolid
- Metronidazole
- Micafungin
- Oxacillin
- Rifampin
- Tigecycline
- Voriconazole

CRRT, Continuous renal replacement therapy.

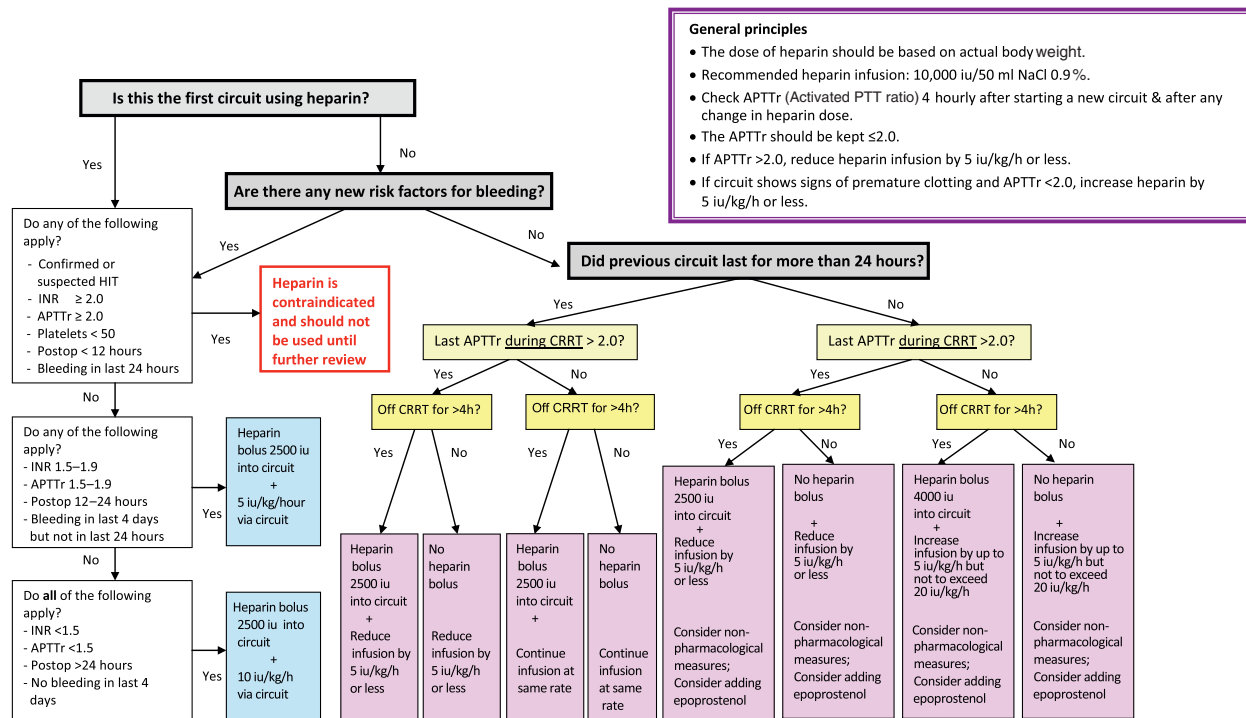
From Thompson AJ. Drug dosing during continuous renal replacement therapies. *J Pediatr Pharmacol Ther.* 2008 Apr;13(2):99–113. doi:10.5863/1551-6776-13.2.99.



CRRT, Continuous renal replacement therapy.

Figure 12.3 Citrate anticoagulation of the CRRT circuit. (Modified from Macedo E, Mehta RL. Continuous dialysis therapies: core curriculum 2016. *Am J Kidney Dis.* 2016 Oct;68[4]:645–657. doi:10.1053/j.ajkd.2016.03.427.)

Suggested protocol for heparin anticoagulation in CRRT



CRRT, Continuous renal replacement therapy.

Figure 12.4 Suggested protocol for heparin anticoagulation in CRRT. HIT, heparin-induced thrombocytopenia. (From Dickie H, Tovey L, Berry W, Ostermann M. Revised algorithm for heparin anticoagulation during continuous renal replacement therapy. *Crit Care*. 2015 Oct 27;19:376. doi:10.1186/s13054-015-1099-y.)

TABLE 12.7 ■ Common Complications of CRRT

• Hypotension • Arrhythmias • Bleeding • Citrate intoxication • Hypothermia • Underdosing of drugs • Hypocalcemia • Hypophosphatemia • Hypokalemia • Nutrient losses
<i>CRRT</i> , Continuous renal replacement therapy.

is administered systemically. Dosing protocols for using unfractionated heparin vary widely; a typical approach would be a systemic loading dose of 500 to 1000 IU followed by 500 units hourly, and adjusted according to the activated partial thromboplastin time (Figure 12.4).

The results of a recent large randomized trial comparing anticoagulation strategies for CRRT favored citrate compared to heparin anticoagulation, with greater efficacy (filter lifespan) and safety (lower risk of bleeding complications).³⁵ Several regional citrate anticoagulation protocols are available.³⁶ CRRT in COVID-19 patients may be complicated by thromboembolic events and disseminated intravascular coagulation. Frequent clotting of CRRT filters and dialysis catheters in many cases has required both systemic heparin and prefilter citrate anticoagulation as well.^{37,38}

From the above description of the CRRT procedure, several complications would be predicted. Common complications are shown in Table 12.7.

CRRT is intensive and complex therapy. Key quality metrics in CRRT commonly used include delivery of adequate dose of dialysis, adequate fluid removal, actual compared to prescribed dialysis dose, filter life, and preventable significant errors. Strategies have been described to bring process-based metrics to quality improvement of CRRT delivery.³⁹

Few studies have evaluated strategies for weaning the AKI patient off of CRRT.⁴⁰ Clinical considerations should include evidence of impending renal recovery (UO monitoring, improvement in kidney function tests), as well as goals of care. Increased UO before CRRT cessation is a common predictor.

Between CVVH and CVVHDF, there is little evidence to support one modality over the other in the overall CRRT population. Several studies have addressed both renal recovery and mortality outcomes of CRRT relative to HD in AKI, with no consensus.

While no improvement with CVVH in mortality was shown in a meta-analysis, hemofiltration did increase the clearance of medium to larger molecules.⁴¹

Data on mortality and other outcomes related to the amount of CRRT delivered are limited.⁴² The RENAL study assessed CRRT at two levels of intensity in critically ill patients with AKI (replacement): lower (effluent flow of 25 mL/kg/hr) or higher (40 mL/kg/hr).⁴³ High intensity (35–48 mL/kg/hr) was not associated with benefit in terms of mortality or renal recovery in a meta-analysis.⁴⁴

A summary of basic reminders for the dialysis prescription is shown in Table 12.8.

12.2.9 SUSTAINED LOW-EFFICIENCY DIALYSIS

SLED has been increasingly used recently, related to patient surges exceeding the availability of CRRT machines in the ICU setting. The 2012 KDIGO guidelines for AKI noted that SLED

TABLE 12.8 ■ What the Clinician Needs to Know to Write Orders for CRRT

Pumps: - Blood flow rate 120–250 (average 180) mL/min. - UF rate 1–2 L/hr total, but net UF depends on patient volume status and overall goals of treatment. Replacement fluids: - If standard commercial replacement fluid is not available, peritoneal dialysis fluid, Ringer's lactate, normal saline, or pharmacy made solutions can be used. Dialysis fluid: - The composition of dialysis fluid is selected based on the same principles as for intermittent hemodialysis; peritoneal dialysate solutions are often used, though calcium-free dialysate may be preferable for citrate anticoagulation. Anticoagulation: - Systemic heparin or regional citrate anticoagulation. - Citrate anticoagulation: - ACD-A solution of 3% trisodium citrate infused at a rate of 150% in mL/hr of the blood flow rate (Q_B) in mL/min (e.g., 225 mL/hr for a Q_B of 150 mL/min) Ca^{++} infusion rate is calculated based on the principle of giving about 1 mmol of Ca for each mmol of citrate, then adjust infusion rate based on ionized calcium (iCa^{++}) level (e.g., calcium gluconate 20 g in 500 mL D5W at 30 mL/hr with an increase for $iCa^{++} < 1.0$ mmol/L or a decrease for $iCa^{++} > 1.1$ mmol/L). Parameters to monitor: - Serum electrolytes, iCa^{++}, Mg^{++} every 12–24 hr. - For citrate anticoagulation, iCa^{++} every 4 hr initially until stable, then every 6 hr and total calcium every 12 hr to assess for citrate toxicity (by an increasing gap between total and ionized calcium levels). - Activated clotting time measured at the post-filter maintained between 180 and 200 sec with either heparin or citrate is usually adequate.
<i>CRRT</i>, Continuous renal replacement therapy; <i>UF</i>, ultrafiltration.

TABLE 12.9 ■ Suggested SLED Prescription

Blood Flow	Dialysate Flow	Duration	UF Rate	Frequency
150–300 mL/min	100–300 mL/min	6–12 hr	Up to 500 mL/hr	3–6 times per week
<i>SLED</i> , Sustained low efficiency dialysis; <i>UF</i> , ultrafiltration.				

was suitable for hemodynamically unstable patients, but noted the lack of scientific evidence necessary to support it as a modality alternative to CRRT.²⁴ At some centers, it is employed as a prolonged therapy instead of CRRT.

The same dialysis machines for intermittent HD are used for SLED; alternatively, CRRT machines may be used for this hybrid therapy. A typical SLED prescription would include blood flow 250 to 300 mL/min, dialysate flow 100 to 300 mL/min, a duration of 6 to 12 hours, fluid removal less than 500 mL/hr, and treatment frequency 3 to 6 times per week. Metabolic clearance goals are achieved in less time than with CRRT. There is less need for anticoagulation than with CRRT (i.e., prefilter heparin 1000 u bolus, then 500 u/hr would be typical). Potential advantages of SLED include faster correction of acidemia, less anticoagulation, and uninterrupted therapy (i.e., CRRT patients have the treatments paused not infrequently for radiology, etc.) so that clearance goals are more likely to be met than with CRRT.

Typical parameters for a SLED prescription are shown in [Table 12.9](#).

SLED and CRRT have demonstrated equivalence in outcomes such as hemodynamic tolerance, fluid removal, ICU stays, renal recovery, pressor requirement, and mortality rates.^{45,46}

Practical disadvantages include cost of supplies, limited flow rates, and dependence on dialysis nursing because conventional HD machines are used.

12.2.10 PERITONEAL DIALYSIS FOR AKI

PD should be considered a suitable modality for treatment of AKI. In fact, potential advantages include simplicity, lower cost, gradual solute removal, no need for anticoagulation, better hemodynamic tolerance, and possibly shorter time to renal recovery. Disadvantages may include catheter-related mechanical problems, risk of peritonitis, lack of adequacy in hypercatabolic or hypoperfused patients, undesirable effects on mechanical ventilation, glucose absorption, and protein loss. The main concern of physicians may be inadequate clearances with PD.⁴⁷

Because of insufficient good-quality clinical data, selection of PD for AKI patients needs to be individualized.

Commercially available PD solutions typically contain significant lactate, which may accumulate in critically ill patients or those with liver failure. Emergent placement of PD catheters should be used where resources and expertise exist.⁴⁸

The appropriate dose of PD in the AKI patient remains uncertain. The application of standard urea kinetic modeling has not been validated for use of PD in AKI. Recent guidelines indicate that targeting a weekly Kt/V urea of 3.5 provides outcomes comparable to those of daily HD in critically ill patients.⁴⁸

Recent meta-analyses have shown that PD is not inferior to other extracorporeal therapies in the management of patients with AKI,⁴⁷ and additional experience has been gained during the COVID-19 pandemic,^{49,50} though the need for prone positioning to improve oxygenation posed a challenge for some COVID-19 patients.

12.3 Dialysis for ESKD

Characteristics of currently available dialysis modalities are shown in Table 12.10. (adapted from Table 1⁵¹).

TABLE 12.10 ■ Characteristics of Dialysis Modalities

Characteristic	Conventional Hemodialysis	Short Daily Hemodialysis	Nocturnal Hemodialysis	Continuous Ambulatory Peritoneal Dialysis	Automated Peritoneal Dialysis
Location	In-center	In-center or home	In-center or home	At home	At home
Sessions per week	3	5–6	5–6	Daily	Daily
Time per session	4 hr	2–3 hr	6–9 hr	3–5 hr per exchange	8–9 hr at night
Contraindications	No vascular access; CV instability	No vascular access; CV instability; inadequate home support	No vascular access; CV instability; inadequate home support	Abdominal surgeries; nonfunctional membrane; inadequate home support	Abdominal surgeries; nonfunctional membrane; inadequate home support

CV, Cardiovascular.

Conventional HD is by far the most common modality. Compared to the standard 3 weekly in-center treatments, frequent (6 weekly) treatments appear to reduce long-term mortality and benefit select patients.⁵² Longer session times appear to improve hospitalization and mortality rates.⁵³ HD intensification with extended-duration nocturnal treatments modestly improves overall metabolic profiles of uremic toxins⁵⁴ but is more effective in clearing small toxins, and studies suggest improved quality of life and patient survival.⁵⁵ Results from recent studies have revealed concerns inherent in the 3-times weekly intermittent HD schedule, which obligates one longer, 3-day interdialytic interval each week. These include increased arrhythmias, hospitalizations, and mortality on the day of the dialysis treatment that follows the longer interdialytic interval.⁵⁶

12.3.1 TIMING OF DIALYSIS INITIATION IN ESKD

Older studies established a trend for early initiation of dialysis in ESKD.⁵⁷ The definition of early and late was based on the degree of renal dysfunction, measured by creatinine clearance (CrCl) or Cr-based estimated glomerular filtration rate (eGFR). More recently there has been much uncertainty about the timing of chronic dialysis initiation (early vs. late start).

Several studies have found that there is no benefit of early initiation of dialysis, and that it may lead to dialysis of patients who might not require it. Furthermore, there is a suggested benefit of a late start based on symptoms or signs of uremia. The results of the Initiating Dialysis Early and Late (IDEAL) study, the only randomized trial to have tested the impact of dialysis initiation at two different levels of kidney function on outcomes, demonstrated no significant difference in survival or other patient-centered outcomes between treatment groups.⁵⁸ Another large retrospective analysis concluded that late initiation of dialysis (classified by eGFR at the time of initiation) was associated with a reduced risk of mortality.⁵⁹

These data have challenged the established paradigm of using estimates of glomerular filtration as the primary guide for initiation of maintenance dialysis and illustrate the compelling need for research to optimize the high-risk transition period from CKD to ESKD. That is, the solute and volume control achievable by dialysis may be offset by the potential risks of dialysis. Risk equations for predicting the time frame to needing maintenance kidney replacement therapy have been developed.⁶⁰

12.3.2 APPROPRIATENESS OF DIALYSIS INITIATION

The appropriateness of starting dialysis in a particular patient applies especially to the chronic setting, and should be based upon two considerations: expected patient survival with or without dialysis, and quality of life.^{61,62}

Elderly patients (>80 years) with significant comorbidities might need to be informed that chronic HD may extend life only 2 to 3 months more than conservative medical management without improving quality of life, although that should be decided individually on a case-by-case basis.^{63,64}

12.3.3 CONSERVATIVE MANAGEMENT OF THE ESKD PATIENT

Conservative (nondialytic) care of the patient with advanced kidney failure is increasingly recognized as an alternative and appropriate approach for certain patients.^{65,66} Key features while attempting to delay late progression of kidney disease include minimizing ESKD complications and improving symptom burden and life satisfaction. Advanced care planning and shared decision-making should be incorporated.

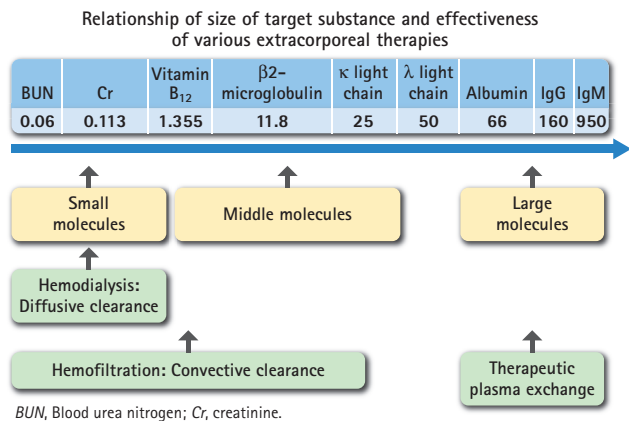


Figure 12.5 Relationship of size of target substance and effectiveness of various extracorporeal therapies.⁶⁹

12.3.4 DIALYSIS MEMBRANES

HD membranes are manufactured with cellulose, modified cellulose, or synthetic polymers.⁶⁷ A variety of membrane compositions continue to be tested in order to bring solute removal closer to that of native kidneys.

Dialyzer clearance specifications are determined by the surface area (A) and mass transfer coefficient of a given solute, which is a function of the membrane composition. Ideal features of an HD membrane include nonthrombogenic, selective permeability, biologically compatible, adequate surface area, tight pore size distribution, minimal thickness, and affordability.⁶⁸ Noncellulosic membranes are characterized as high flux (i.e., they are better able to remove fluid and moderate-sized molecules).

Solute removal in conventional HD occurs primarily by diffusion. The effectiveness of extracorporeal therapies in general is related to the size of the target substances being removed.⁶⁹ While smaller molecules are effectively removed by HD, clearance of “middle molecules” (0.5–60 kD), typically represented by beta2-microglobulin, is less effective (Figure 12.5)⁶⁹.

Middle molecules have been associated with inflammation, cardiovascular events (CVEs), and other ESKD comorbidities.⁷⁰ Compared to the previous generation of low-flux dialysis membranes, currently used high-flux membranes provide greater clearance of larger solutes. The clinical benefit of even newer “medium cutoff” dialysis membranes, which help remove larger middle molecules with minimal albumin leak, is under investigation.⁷¹ This device has recently received FDA approval in the United States.

Dialysis membrane biocompatibility refers to specific interactions between the dialysis membrane and the patient’s blood. The material of the dialysis membrane is the major determinant of the biological response to it.⁷² Improved, biocompatible membranes elicit less of an inflammatory response. Synthetic membranes have higher biocompatibility than older cellulosic membranes.

Membrane characteristics are shown in Figure 12.6.

12.3.5 VASCULAR ACCESS FOR HEMODIALYSIS

While the intent of chronic dialysis is to clear uremic toxins from the entire body, quality HD starts with the existence of a reliable access to the patient’s vascular system.⁷³ There are three main

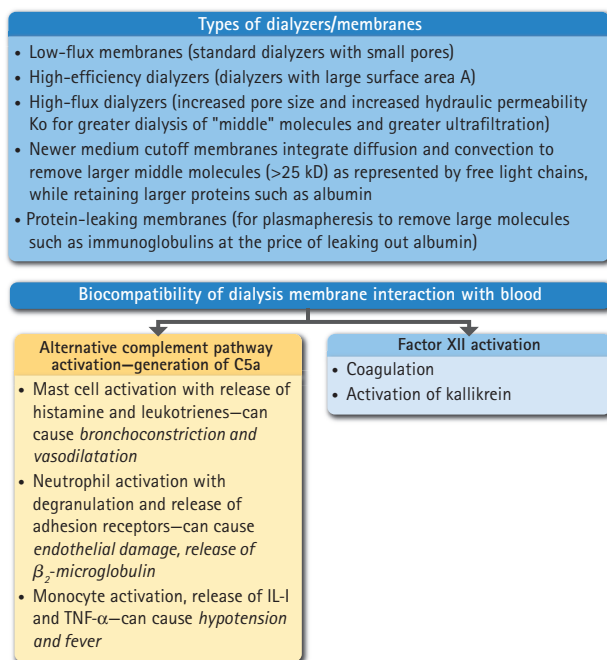


Figure 12.6 Types of dialyzers/membranes.

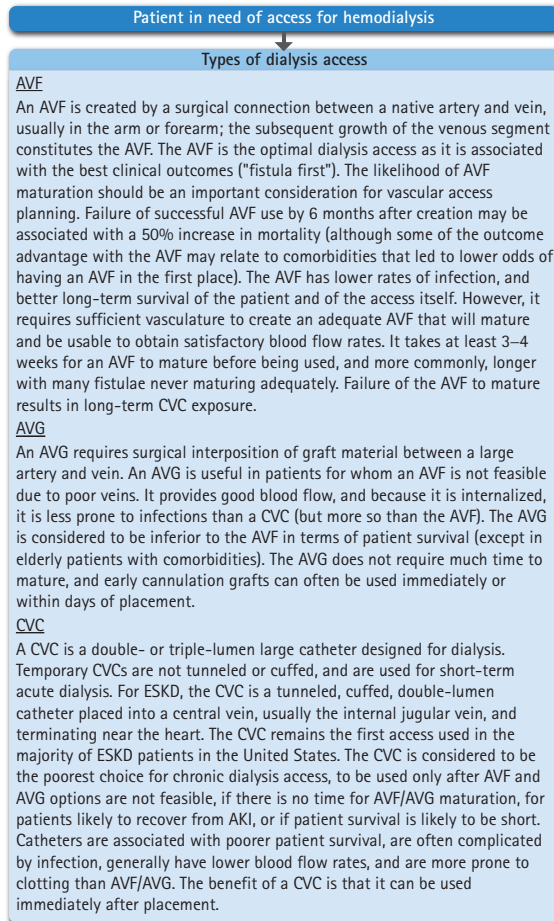
types of vascular access for HD: arteriovenous fistula (AVF), arteriovenous graft (AVG), and central venous catheter (CVC; tunneled or not). (Note that a "hybrid" access called the Hemodialysis Reliable Outflow [HeRO] involves a graft at one end, typically sutured to the brachial artery, and a tunneled central venous segment at the other end; it is designed to allow bypass of a central venous occlusion). The standard types of dialysis access are described below (Figure 12.7).

The "Fistula First" initiative emphasized the AVF as the optimal vascular access type,⁷⁴ while reevaluation of this strategy for all patients has increased awareness that this may not always be the best strategy, such as in elderly patients.⁷⁵ HD access should be able to provide a blood flow of at least 300 mL/min. Recent recommendations call for a more patient-focused approach to access planning and management.⁷⁶

12.4 Dialysis Access Planning

Access care decisions involve protocols and an interdisciplinary team but must be individualized in order to determine the best access use for each patient^{73,77} since there are advantages and disadvantages of all vascular access types.⁷⁸ In recent years, there have been shifts from AVGs to fistulas and from distal to upper arm fistula locations. Patients referred to the nephrologist prior to the need for dialysis are more likely to start dialysis with an AV fistula as their permanent access. Unfortunately, the CVC remains the first access used in as many as 75% of patients initiating dialysis in the United States.

Dialysis access planning begins with educating the patient about the basic types of kidney replacement therapy (HD, PD, transplantation). In patients not certain to choose PD or those unlikely to receive a timely kidney transplant, evaluation for an AVF should be initiated, while at the same time CKD management should continue, and the patient should understand that this does not mean dialysis is imminent.



AKI, Acute kidney injury; AVF, arteriovenous fistula; AVG, arteriovenous graft; CVC, central venous catheter; ESKD, end-stage kidney disease.

Figure 12.7 Types of dialysis access.

Since fistula maturation (i.e., maturation means the AVF can be cannulated and provide prescribed dialysis) may take at least 1 month, commonly, 3 to 12 months, and may require interventional procedures before use, each patient with progressive and/or an eGFR of 15 to 20 mL/min/1.73 m² should therefore begin access planning, including patient education and referral to the access surgery team.⁷⁹ (It is important that the patient understand that veins for potential AVF placement should not be used for venipuncture, placement of intravenous catheters, or overlying blood pressure [BP] cuff placement.) Imaging studies to determine the suitability of vessels for AV access creation, such as Doppler ultrasound, may be required.

12.4.1 ACCESS DYSFUNCTION/MONITORING/SURVEILLANCE

Access dysfunction occurs in several ways, most importantly resulting in reduction in access flow (Table 12.11).

TABLE 12.11 ■ Dysfunction of the Dialysis Fistula/Graft

- Inadequate dialysis achieved
- Excessive bleeding from needle sites
- Loss of thrill/bruit
- Pseudoaneurysms
- Localized arm swelling
- Inadequate blood flow delivered
- Recirculation (e.g., visible recirculation of saline)
- Arterial “steal” syndrome (i.e., vascular compromise below the access)

TABLE 12.12 ■ Dialysis Access Surveillance (Clinical Assessment) and Monitoring (Special Testing)

- Goal is to identify access with high likelihood of stenosis
- Patient education for signs and symptoms of access dysfunction
- Physical exam of pulse and bruit of AVF/AVG for abnormality
- Failure of AVF to collapse when the upper extremity is elevated
- Static venous pressure measurements at dialysis (at blood flow rate (Q^b) 200 mL/min, VP should be <130 mmHg with 15-g needle, <150 mmHg with 16-g needle)
- Recirculation study indicated when blood flow achieved during dialysis or dialysis adequacy (decreased Kt/V) is poor
- Direct Doppler volume blood flow measurements of AVF/AVG
- Angiography if surveillance/monitoring suggest abnormality

AVF, Arteriovenous fistula; AVG, arteriovenous graft; VP, venous pressure.

Recent guidelines emphasize the primary importance of clinical monitoring strategies such as inspection, palpation, and auscultation in early detection of access dysfunction.

Access recommendations now conclude that evidence supporting technical measures such as access blood flow, pressure monitoring, or imaging for stenosis as surveillance tools is inadequate.^{73,80} Likewise, guidelines are not to recommend preemptive angioplasty of fistulas or grafts which have stenosis that is not associated with clinical indicators.

Key features of dialysis access surveillance and monitoring are shown in Table 12.12.

12.5 Hemodialysis Catheters

For both ESKD and (more commonly) AKI requiring urgent HD, the most efficient and achievable access remains the HD catheter. Tunneled dialysis catheters are in fact the most common vascular access used to initiate maintenance dialysis in many patients with ESKD, and may be appropriate in cases of AKI with some likelihood of recovery from longer-term dialysis dependence. The ideal dialysis catheter would deliver adequate blood flow, be resistant to infection and thrombosis, and not cause vascular injury,⁸¹ but currently available catheters are imperfect.

Duration of optimal catheter use to minimize infection risk is shown in Table 12.13.

TABLE 12.13 ■ Standard Duration of Use for Hemodialysis Catheters

- Temporary (not tunneled) femoral vein catheters, inserted with sterile technique and meticulously cleaned daily in bedbound patients, can usually be left in place safely for 3–7 days, occasionally longer, but are not suitable for ambulatory patients and may be undesirable in morbidly obese patients.
- Subclavian and internal jugular temporary catheters (not tunneled) may be left in place for 2–4 weeks.
- Silastic/silicone cuffed catheters (tunneled) are suitable for long-term use.

Early complications of catheter insertion are mainly related to insertion, as shown in [Table 12.14](#). Subsequent complications of the HD catheter use are shown in [Table 12.15](#).

An additional management challenge is a poorly functioning catheter. The differential diagnosis may include not only thrombotic occlusion, but also growth of a fibrin sheath, malpositioning of the catheter tip, or kinking of the catheter. Mechanical problems are more common shortly after insertion whereas thrombotic occlusion or fibrin sheath formation is more likely 2 weeks or more after catheter insertion.⁸² The approach to malfunction is indicated in [Figure 12.8](#).

Note that evidence does not support the routine use of prophylactic systemic anticoagulation in order to prevent catheter thrombosis episodes.

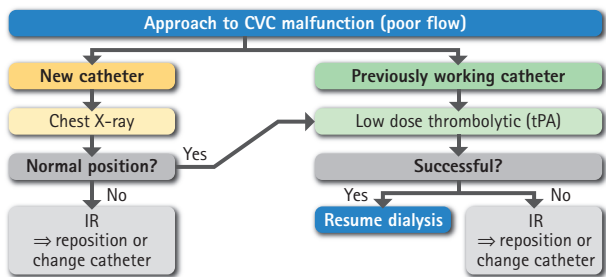
TABLE 12.14 ■ Complications of Hemodialysis Catheter Insertion

• Arterial, rather than venous, puncture • Exogenous or subcutaneous hemorrhage • Hemothorax • Pneumothorax • Air embolism • Atrial and ventricular arrhythmias • Perforation of central vein or cardiac chamber • Pericardial tamponade • Venous thrombosis
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TABLE 12.15 ■ Later Complications of Hemodialysis Catheters

• Infection (see more detailed discussion below). • Clotting of catheter or formation of a fibrin sheath. Minimize by instillation of heparin or tPA following hemodialysis. Unclear role of systemic warfarin or low molecular weight heparin. Treatment: tPA - 2 mg of rtPA into each port. Some cases may require catheter stripping to remove clot and fibrin sheaths that adhere to the catheter, or exchanging the catheter over a guidewire. • Central vein thrombosis or stricture (30%–40% in subclavian, 2%–10% IJ), which may cause loss of ipsilateral arm use for AVF. KDOQI recommends avoidance of subclavian vein unless no other options exist, or the ipsilateral extremity can no longer be used for permanent dialysis access (AVF or AVG) if occlusion occurs. • IVC thrombosis with lumbar catheter. • Arteriovenous fistula formation as rare complication.
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AVF, Arteriovenous fistula; AVG, arteriovenous graft; IJ, internal jugular vein; IVC, inferior vena cava; KDOQI, Kidney Disease Outcomes Quality Initiative; rtPA, recombinant tissue plasminogen activator; tPA, tissue plasminogen activator.



CVC, Central venous catheter; IR, interventional radiology; tPA, tissue plasminogen activator.

Figure 12.8 Approach to CVC malfunction.

12.6 Hemodialysis Catheter Infections

Prophylaxis: Catheter infection may occur via contamination of external or internal tunneled catheter surfaces. In addition to routine surveillance and monitoring, methods to prevent catheter-related bloodstream infections (CRBSIs) include extraluminal strategies and the use of intraluminal antibiotic lock solutions.⁷³ The approach to prophylaxis is shown in [Table 12.16](#).

CRBSIs contribute significantly to morbidity and mortality in HD patients.⁸³ Common risk factors include impaired immune system, skin colonization with *Staphylococcus*, prolonged catheter duration, and a history of prior catheter-related bacteremia episodes.

Blood cultures should be drawn as an initial step to confirm the presence of bloodstream infection. Determination that bacteremia is secondary to the HD catheter is likely when there is obvious infection of the external portion of the catheter by the same organism.

Confirmation that the catheter is the culprit in a bacteremic patient, according to recent guidelines, may also be determined by criteria involving cultures obtained from blood samples through the catheter or directly from the catheter tip (at removal), compared to results from a peripheral venous culture.⁷³

There are four main options for CVC management (in addition to systemic antibiotic treatment) in the setting of CRBSI, as shown in [Table 12.17](#).

Immediate empiric catheter removal may be indicated in the presence of high-grade bacteremia, severe sepsis, metastatic infection, or concurrent exit-site/tunnel infection.

Empiric therapy should include both Gram-positive (vancomycin) and Gram-negative (ceftazidime/aminoglycoside) coverage.

Subsequent antibiotic coverage should be based on identification and antibiotic sensitivity of the offending pathogen.⁸⁴

The KDOQI algorithm for CVC-related infection is shown in [Figure 12.9](#).⁷³

TABLE 12.16 ■ Hemodialysis CVC Infection Prophylaxis

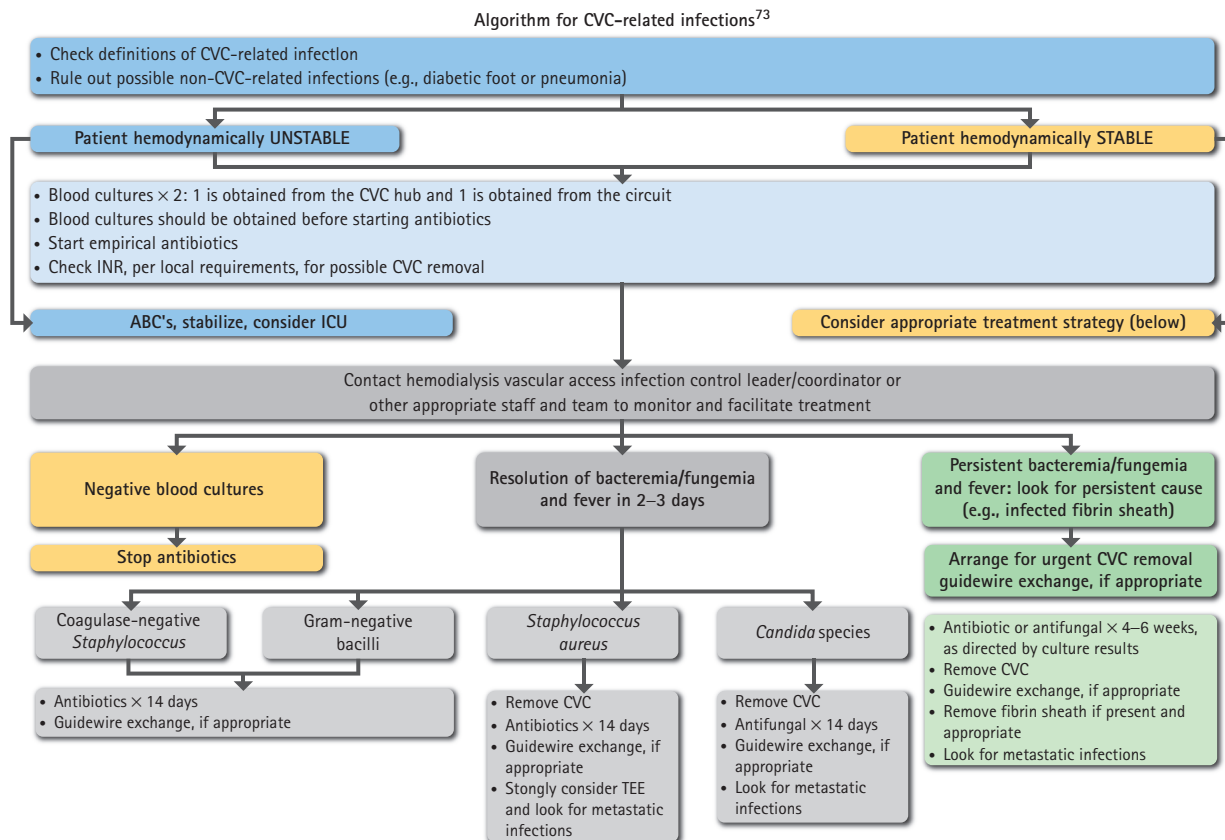
- Using tunneled rather than nontunneled dialysis catheters
- Strict aseptic technique during connection and disconnection procedures
- Exit site and hub care with a chlorhexidine-based solution or alcohol swabs
- Topical triple antibiotic ointment or mupirocin ointment to exit site with each dressing change
- Antimicrobial barrier catheter caps
- Protection of exit site from wet and dirty environments
- Filling the lumen with antibiotic-anticoagulant after dialysis (“antibiotic lock”) in high-risk patients (a large number of antibiotic CVC lock solutions are available)

CVC, Central venous catheter.

TABLE 12.17 ■ Management Options for Catheter-Related Infection

1. CVC removal with exchange over a guidewire at the same site
2. CVC removal with a new CVC placed at a new site
3. CVC removal with delayed placement of a new CVC
4. Retention of CVC, without or with use of antibiotic lock solution

CVC, Central venous catheter.



ABC's, Airway/Breathing/Circulation; CVC, Central venous catheter; ICU, intensive care unit; TEE, transesophageal echocardiography.

Figure 12.9 Algorithm for CVC-related infections.

12.7 Anticoagulation in Hemodialysis

Anticoagulation is an important part of the dialysis procedure itself.⁸⁵ Most common options for anticoagulation are unfractionated heparin and, less commonly, low molecular weight heparin. While there is no need to adjust dosing of unfractionated heparin in ESKD, dosage reduction is necessary for low molecular weight heparin. Other agents are available when heparin is not an option (e.g., in heparin-induced thrombocytopenia). The pharmacokinetic properties of anticoagulants in the setting of ESKD are shown in Table 12.18.

The need for systemic anticoagulation is also common in ESKD apart from the dialysis procedure. A common example is atrial fibrillation. The proper use of anticoagulation unrelated to the dialysis procedure in ESKD begins with assessment of bleeding risk (Table 12.19).

The site of action of oral anticoagulants in the coagulation cascade is noted in Figure 12.10.⁸⁶

The use of vitamin K antagonists has long been the standard of care in CKD patients at risk for thromboembolic complications, but the risk of bleeding is disproportionately increased.⁸⁷ For ESKD patients requiring systemic anticoagulation such as atrial fibrillation, initiation and maintenance of anticoagulation is complex. For example, doses of warfarin required may be 20% lower,

TABLE 12.18 ■ Pharmacokinetic Properties of Anticoagulants in Renal Failure

Anticoagulant	Chemical Composition	Mechanism	Follow-Up Indicator
Unfractionated heparin	Mixture of glycosaminoglycan chains 5000–30,000 Da; renal excretion 10%	Binds to antithrombin III, inhibits clotting factors IIa and Xa	PTT; anti-Xa heparin assay (therapeutic range 0.3–0.7 U/mL); 1%–5% heparin-induced thrombocytopenia
Low molecular weight heparin	Fragments of larger heparins, 4000–8000 Da; renal excretion 40%	Binds to antithrombin III, inhibits clotting factor Xa	Anti-Xa (therapeutic range 0.5–1.2 U/mL, prophylactic 0.2–0.5 U/mL); 0%–3% heparin-induced thrombocytopenia
Citrate	Citrate 2.4 mEq/L in dialysate acid concentrate or trisodium citrate solution, 3% (ACD-A solution)	Citrate chelates calcium to disrupt the coagulation cascade	Visual inspection for clots in dialysis circuit; ionized calcium (therapeutic <0.4 mmol/L in the dialyzer) or an activated clotting time of 1.5–2.0 times baseline (180–250 sec)

PTT, Partial thromboplastin time.

TABLE 12.19 ■ Assessing Bleeding Risk During Hemodialysis Anticoagulation

Serious risk	Moderate risk
Major surgery <72 hr	Already anticoagulated
Active bleeding	Recent tunneled hemodialysis catheter placed
Bleeding diathesis	Pericarditis
Recent intracranial hemorrhage	Minor surgery <72 hr
	Major surgery <7 days

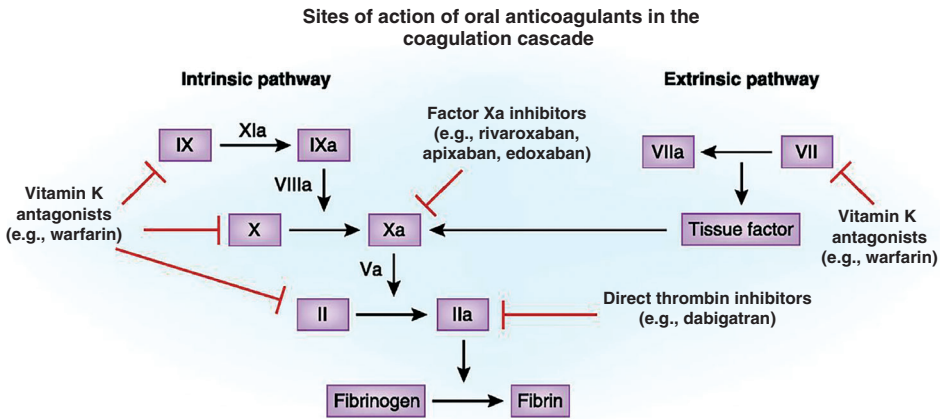


Figure 12.10 Sites of action of oral anticoagulants in the coagulation cascade. (From Jain N, Reilly RF. Clinical pharmacology of oral anticoagulants in patients with kidney disease. *Clin J Am Soc Nephrol*. 2019 Feb 7;14[2]:278–287. doi:10.2215/CJN.02170218.)

and ESKD patients spend less time in the therapeutic INR range. Note that warfarin is considered a risk factor for calcific uremic arteriolopathy (calciphylaxis). Non-vitamin K-dependent oral anticoagulants are increasingly used, primarily for atrial fibrillation, and would be expected to have an improved risk–benefit profile.⁸⁸ Apixaban 2.5–5 mg twice daily may be administered in ESKD, but the dosage should be reduced to 2.5 mg twice daily if the patient is over 80 years old or weighs less than 60 kg. Apixaban appears to be associated with reduced mortality and embolic events and is safer than warfarin treatment in dialysis patients. In a recent randomized trial, rivaroxaban reduced fatal/nonfatal CVEs and major bleeding complications compared to a vitamin K antagonist in HD patients with atrial fibrillation.⁸⁹

The pharmacokinetic properties of oral anticoagulants are shown in Table 12.20.⁸⁶

12.8 Hemodialysis Emergencies

Maintenance HD safety has improved significantly over time and is now considered a routine procedure.⁹⁰ Nonetheless, serious complications can occur and require emergency evaluation and management. Some are reviewed below (Table 12.21).

12.8.1 DIALYZER REACTIONS

Allergic or even anaphylactic reactions may occur in the dialysis patient, usually within the first 5 to 10 minutes (Type A, commonly related to ethylene oxide used as a sterilant, or less commonly to the dialyzer membrane), but in some cases up to 30 minutes after initiation. Dialyzer reactions occurring after 30 minutes and up to 60 minutes after initiation (Type B) are usually less severe and most commonly related to reactions to the dialyzer membrane that are often complement mediated. Less commonly, reactions are related not to the sterilant or membrane, but to heparin or other medications administered, such as iron, rarely erythropoiesis stimulating agents (ESAs), or blood products.

Presentation may include acute bronchial constriction with respiratory distress or chest tightness, vasodilation with hypotension, anxiety, abdominal cramping/diarrhea, or in the worst cases, cardiac arrest.⁹¹

TABLE 12.20 ■ Summary of Pharmacokinetic and Pharmacodynamic Properties of Commonly Used Oral Anticoagulants (adapted from Table 1⁸⁶)

OAC	Type	Prodrug	PHARMACOKINETICS			Pharmacodynamics: Binding to Effector
			Metabolism	Renal Dose Adjustment	Dialyzable	
Warfarin	Vitamin K–dependent factor inhibitor	No	Extensive metabolism by CYP2C9	Minor	No	Irreversible
Dabigatran	Direct thrombin inhibitor	Yes	Metabolized by esterases, 80% excreted by kidney	Yes	Yes	Reversible
Apixaban	Free and clot-bound Xa inhibitor	No	Metabolized in liver by CYP3A4, then excreted in feces and kidney (25%), no active metabolite	Yes for some	Small	Reversible
Rivaroxaban	Free and clot-bound Xa inhibitor	No	66% excreted by kidney, 36% unchanged, minimal in feces	Yes	No	Reversible
Edoxaban	Free Xa inhibitor	No	50% excreted unchanged by the kidney, 10% hydrolyzed by carboxyesterase 1	Yes	No	Reversible

CYP2C9, Cytochrome P450 type 2C9; CYP3A4, cytochrome P450 type 3A4; OAC, oral anticoagulant; Xa, factor Xa.

TABLE 12.21 ■ Major Hemodialysis Emergencies

- Dialyzer reaction
- Dialysis disequilibrium syndrome
- Venous air embolism
- Pyrogenic reactions
- Arrhythmias

Dialysis in such severe cases should be terminated without returning blood to the patient; provide aggressive fluids for hypotension, call for emergency assistance, and administer antihistamines, corticosteroids, or epinephrine.

For subsequent episodes, switch to a dialyzer with a non-ethylene oxide sterilizer, such as a gamma ray–sterilized dialyzer; use an alternative dialysis membrane; prime the dialyzer well, and pretreat with antihistamines/steroids as needed.

12.8.2 DIALYSIS DISEQUILIBRIUM SYNDROME

Neurological signs and symptoms of dialysis disequilibrium syndrome can include nausea, vomiting, confusion, or seizures during or shortly after the initial dialysis session(s).

Risk factors are very high BUN/Cr, rapid reduction of BUN/Cr by the dialysis treatment, older age, liver disease, abnormal serum sodium, and metabolic acidosis.⁹²

Prevention measures for the initial 3 HD treatments include lower blood flow, shortened dialysis sessions, and higher dialysate sodium, or consideration of CRRT.⁹⁰

12.8.3 VENOUS AIR EMBOLISM

A venous air embolism occurs when air enters the patient's bloodstream from the extracorporeal circuit due to a loose connection or defect in the components of the blood circuit.

A venous air embolism may cause hypoxia, hypotension, pulmonary hypertension (HTN), or cardiac arrest; if it coexists with a patent foramen ovale, it may enter the arterial circulation and result in ischemic injury, usually cerebral.

In cases of venous air embolisms, stop dialysis, place the patient in the left lateral recumbent position, and treat for hypoxia and hypotension.

12.8.4 PYROGENIC REACTIONS

Symptoms of pyrogenic reactions include fever, chills, hypotension, headache, and myalgias during HD. Such reactions are usually caused by Gram-negative bacteria that produce endotoxins in the dialysate. A pyrogenic reaction may lead to systemic infection and death. The dialysis machine, extracorporeal circuit including dialysis membrane, and water treatment system should be evaluated.

12.8.5 ARRHYTHMIAS

Arrhythmias and cardiac arrest account for almost one-third of deaths in ESKD patients, the most common single cause. Furthermore, the risk of sudden cardiac death is highest following the longer interdialytic period. Unlike in the general population, bradyarrhythmias appear to be more common than ventricular arrhythmias.⁹⁰ Potential triggering factors include rapid electrolyte changes (potassium, calcium).⁹³

12.9 ESKD Comorbidities/Coexisting Conditions

ESKD is a state of multiple comorbidities. Conditions variously included as comorbidities in the incident/prevalent ESKD patient population include HTN, diabetes, ischemic heart disease, congestive heart failure, hyperlipidemia, cerebrovascular disease, anemia, gout, and others.

12.9.1 ISCHEMIC HEART DISEASE

Atherosclerosis is accelerated in patients with ESKD, due to both traditional and nontraditional (uremia-related) factors, such as vascular calcification, endothelial dysfunction, inflammation, anemia, and insulin resistance.⁹⁴ Changes in ventricular function chronically or temporally related to the dialysis procedure (see Intradialytic Hypotension section below) may be a result of myocardial hypoperfusion. ESKD patients with severe coronary artery disease may require coronary bypass surgery.

12.9.2 HYPERTENSION

HTN affects a majority of ESKD patients. Multiple mechanisms are involved, although the predominant one is sodium and volume excess. HTN contributes to cardiovascular morbidity and mortality in ESKD. Control of extracellular volume is the primary therapeutic strategy. The linear relationship between systolic BP and mortality in the general population is not seen in ESKD. In fact, data suggest a U-shaped relationship, with increased mortality risk at low systolic (<~110 mmHg) and at high systolic (>165 mmHg).⁹⁵

Control of HTN in ESKD patients, especially if documented in home or ambulatory readings, is associated with improvement in cardiovascular outcomes. Dialysis patients also suffer greater adverse consequences of hypotension than the general population. Trials done in ESKD patients suggest that treatment may be important, but have not led to firmly established BP treatment targets. Guidelines suggest that predialysis BP should be 140/90 and postdialysis BP 130/80, but goals must be individualized.

Medications commonly used are listed in [Table 12.22](#).⁹⁶

TABLE 12.22 ■ Antihypertensive Medications Used in Hemodialysis⁹⁶

Medication	Usual Dose Range	Removal With Hemodialysis (%)
Alpha-blockers		
Doxazosin	1–16 mg daily	None
Prazosin	1–10 mg twice daily	None
Terazosin	1–20 mg daily	None
ACEi		
Benazepril	5–40 mg daily	20–50
Enalapril	2.5–10 mg daily	35–50
Fosinopril	10–80 mg daily	None
Lisinopril	2.5–40 mg every 24–48 hr	50
Perindopril	2–8 mg daily	50
Quinapril	10–20 mg daily	25
Ramipril	2.5–10 mg daily	20–30
Trandolapril	0.5–4 mg daily	30
ARBs		
Candesartan	4–32 mg daily	None
Irbesartan	75–300 mg daily	None
Losartan	50–100 mg daily	None
Olmesartan	10–40 mg daily	None
Telmisartan	40–80 mg daily	None
Valsartan	80–320 mg daily	None
Beta-blockers		
Atenolol	25–100 mg every 24–48 hr	50–75
Carvedilol	6.25–25 mg twice daily	None

Continued on following page

TABLE 12.22 ■ Antihypertensive Medications Used in Hemodialysis (Continued)

Medication	Usual Dose Range	Removal With Hemodialysis (%)
Labetalol	100–1200 mg twice daily	None
Metoprolol	50–200 mg twice daily	High
Nadolol	40–240 mg daily	50
Propranolol	40–160 mg twice daily	None
CCBs		
Amlodipine	2.5–10 mg daily	None
Diltiazem CD	180–360 mg daily	None
Felodipine	2.5–10 mg daily	None
Nifedipine XL	30–120 mg daily	None
Verapamil	180–360 mg daily	None
Diuretics		
Bumetanide	0.5–2 mg twice daily	None
Ethacrynic acid	50–200 mg twice daily*	None
Furosemide	40–160 mg twice daily	None
Metolazone	5–10 mg daily	None
MRAs		
Eplerenone	50–100 mg daily	None
Spironolactone	25–100 mg daily	None
Others		
Clonidine (oral)	0.1–0.4 mg three times daily	5
Hydralazine	10–100 mg three times daily	None
Minoxidil	2.5–30 mg daily	Yes

*Ethacrynic acid is not generally used due to concern for hearing loss.
ACEi, Angiotensin-converting-enzyme inhibitors; *ARBs*, angiotensin II receptor blockers; *CCBs*, calcium channel blockers; *MRAs*, mineralocorticoid receptor antagonists.

Drug selection should take into consideration patient comorbidities for which specific agents are also indicated. The approach to drug selection should include factors such as comorbid conditions, adverse effects, pharmacokinetics in ESKD, and effects of the dialysis treatment on BP.

Note that most angiotensin-converting-enzyme inhibitors (ACEi), metoprolol, atenolol, and nadolol are removed by HD.

12.9.3 CHRONIC FLUID OVERLOAD

Controlling fluid overload in ESKD patients should be viewed as another component of dialysis adequacy. Pulmonary congestion is a predictor of cardiac events and mortality in ESKD.⁹⁷ Clinical assessment of fluid overload may be difficult. While fluid overload carries risk, so does the need for excessive ultrafiltration during the dialysis procedure in an ongoing attempt to achieve euvolemia. Fluid removal rates higher than 13 mL/hr/kg body weight are associated with

poor outcomes.⁹⁸ The use of diuretics between HD treatments to control hypervolemia was initially considered controversial, but they are now being used successfully by many nephrologists.

Dialysis can potentially relieve heart failure. There is no consensus for treatment of systolic heart failure. The benefit of beta-blocker therapy was indicated in a recent observational study showing lower mortality in HD patients with heart failure.⁹⁹ Atenolol, metoprolol, nadolol, and bisoprolol are dialyzed off. Some studies have shown increased risk for diabetes, stroke, or hyperkalemia with beta blockade.

12.9.4 PULMONARY HYPERTENSION

Pulmonary HTN is a progressive pulmonary circulatory disease that accompanies either left or right ventricular failure and volume overload.¹⁰⁰ It is underrecognized in ESKD patients, though causes and mechanisms of right heart failure have now been well described.¹⁰¹ However, awareness of right-sided heart failure and pulmonary HTN in the ESKD population is increasing.¹⁰²

Treatment consists of control of HTN, use of beta-blockers, ACEi/angiotensin II receptor blockers (ARBs), mineralocorticoid receptor antagonists (MRAs), correction of anemia, reassessment of target weight, treatment of arrhythmias, and evaluation and treatment of coronary ischemia.

12.9.5 PERIPHERAL ARTERY DISEASE

The prevalence of peripheral artery disease (PAD) is 25% to 35% in incident patients. The risk in prevalent patients is higher in association with intradialytic hypotension (IDH).¹⁰³ Predictors include advanced age, male sex, diabetes, current smoking, coronary artery disease, and malnutrition.¹⁰⁴ Presentation includes gangrene, ischemic ulceration, and pain at rest; classic symptoms of claudication are uncommon. Guidelines recommend evaluation for PAD at initiation of chronic dialysis. Calcification of vessels may reduce the success of endovascular therapy. Increased postoperative mortality after surgical revascularization is a risk.

12.9.6 INTRADIALYTIC HYPOTENSION

Intradialytic hypotension (IDH) is one of the most frequent complications of HD, affecting 10% to 30% or more of treatments, depending on its definition. During an HD treatment, the majority of patients experience some overall decline in BP. Commonly used definitions of IDH include a decrease in systolic BP of 20 mmHg, a nadir systolic BP below 90 mmHg, or any hypotensive event that requires an immediate corrective measure. The ultrafiltration goals for an HD treatment in the United States are commonly up to 3 L (roughly 1 total plasma volume).¹⁰⁵

The evaluation should begin with a full assessment of the patient's condition and comorbidities that may contribute to hypotension with HD fluid removal (Figure 12.11).¹⁰⁶

The mechanism of IDH in the absence of the underlying conditions described above usually involves failed compensations of cardiac output and arteriolar tone for fluid removal. Compensation may be hindered by impaired sympathetic activation. Major risk factors are diabetes, large interdialytic weight gains, and lower body weight.¹⁰⁷

The most common symptoms of IDH include muscle cramps, nausea, vomiting, and dizziness. Severe episodes may be complicated by loss of residual renal function, hypotensive seizures, stroke events, coronary and cerebral ischemia, vascular access occlusion, and increased mortality risk.¹⁰⁸ Repeated episodes are now known to result in damage to the heart with myocardial fibrosis, central nervous system (CNS) cerebral ischemia, kidneys, and gastrointestinal (GI) tract related to end-organ hypoperfusion.^{109,110}

Multiple interventions may be utilized to prevent IDH, but because of inadequate longitudinal trials, treatment is individualized and frequently empiric. Refractory patients should be evaluated for undiagnosed cardiac disease and undergo autonomic testing. Note that the full

Differential diagnosis of intradialytic hypotension

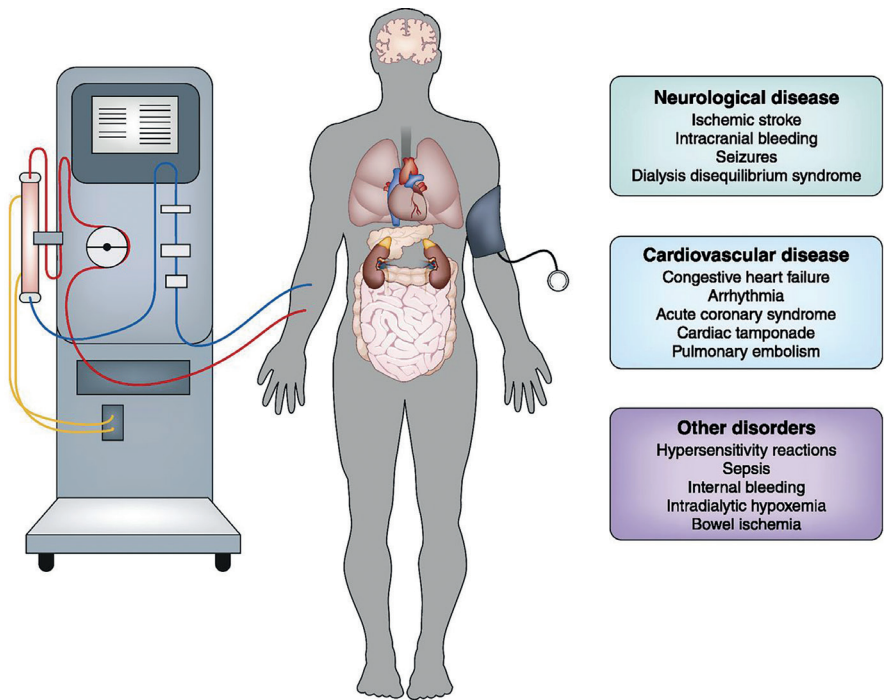


Figure 12.11 Underlying factors to be considered in the patient with HD treatments complicated by IDH.

TABLE 12.23 ■ Prevention and Treatment of Intradialytic Hypotension

Dialysis-Related Modalities	Extra-Dialysis-Related Modalities
Confirm accuracy of blood pressure reading	Adjust antihypertensive dosing and timing
Establish an accurate target weight	Limit intradialytic weight gain
Reduce ultrafiltration rate	Improve anemia
Reduce intradialytic blood flow	Predialysis midodrine
IV fluid administration	Predialysis pseudoephedrine
Ultrafiltration modeling	Predialysis caffeine consumption
Prohibit food consumption during dialysis	Fludrocortisone (extrarenal effect)
Oxygen administration	Oral carnitine (if deficient)
Increase dialysate calcium	
Cool dialysate to 36°C	

Trendelenburg position is no longer recommended. Both hyperoncotic (25%) and isooncotic (5%) intravenous albumin have been used in place of isotonic saline for hypotension, but resultant volume expansion appears to be ineffective despite its oncotic properties.¹¹¹

Prevention and treatment of IDH is outlined in [Table 12.23](#).

12.10 Anemia Management

Comorbid anemia occurs in over half of ESKD patients, and a substantial portion of ESKD patients initiate dialysis with very low hemoglobin (Hgb) levels.¹¹² Anemia at initiation of dialysis is associated with increased fatigue, weakness, and an increased risk of cardiovascular mortality. The multifactorial etiology includes decreased endogenous erythropoietin (EPO) production, functional or absolute iron deficiency, resistance to EPO in uremia related to suppression of hypoxia-inducible factors, and shortened red cell lifespan.

Initial anemia evaluation should be similar to that in the non-ESKD patient, and should consider factors such as the patient's history, complete blood count (CBC), iron studies, and reticulocyte count (Table 12.24).

Determination of iron deficiency is of relevance not only as a primary diagnosis, but as a factor in the response to ESAs.

Anemia is treated with ESAs (e.g., epoetin), intravenous iron, and, infrequently, blood transfusions. Note that management in the dialysis facility is usually facilitated by the use of approved protocols. KDOQI Guideline Recommendations for HD Patients¹¹³ are shown in Table 12.25.

Concerns regarding toxicity of ESAs began to emerge with their initial regulatory trials. Several adverse effects of ESAs were reported early, when Hgb targets were near normal or normal, including CVEs, malignancy, seizures, exacerbation of HTN, dialysis access thrombosis, and mortality.

Harmful effects were related to the use of large ESA doses and maintaining higher Hgb levels.¹¹⁴ As a result, clinically acceptable target for Hgb with ESAs are in the range of 10 to 12 g/dL. Several treatments are now available to treat patients with ESKD. Doses are titrated based on the patient's response to ESA treatment.

ESA hyporesponsiveness is found in up to 20% of HD patients, and is associated with higher ESA requirements, increased major CVEs, and mortality. Causes include iron deficiency, inflammation, hyperparathyroidism, and systemic infection. Rarely, neutralizing antibodies to ESAs will result in severe anemia.¹¹⁵

Parenteral iron is also known to raise or help sustain Hgb levels in ESKD.

Current treatment options for anemia management in the ESKD patient are shown in Table 12.26.¹¹⁵⁻¹²¹

TABLE 12.24 ■ Evaluation of Anemia in the ESKD Patient

Initial anemia evaluation (Hgb <12 g/dL)
Iron deficiency—iron indices (transferrin saturation <20%; serum ferritin <200 ng/mL)
Reticulocyte count
RBC indices
Serum folate level
Vitamin B₁₂ level
Stool for occult blood

ESKD, End-stage kidney disease; *Hgb*, hemoglobin; *RBC*, red blood cell.

TABLE 12.25 ■ Considerations in Initiating ESA in ESKD Patients

- Determine hemoglobin goal.
- Monitor hemoglobin level.
- Adjust ESA dose every 2–4 weeks.
- Consider contraindications to ESA administration.

ESA, Erythropoiesis stimulating agent; *ESKD*, end-stage kidney disease.

TABLE 12.26 ■ ESA Options for Treating Anemia in ESKD

EPO alfa (epoetin): genetically engineered recombinant human EPO. Initial dose 50 units/kg three times weekly.
EPO alfa-epbx: biosimilar of EPO alfa, identical in amino acid sequence and comparable in carbohydrate composition. Shown to be equivalent to EPO alfa in efficacy and similar to EPO alfa in safety when given intravenously to HD patients, and more recently when each was administered subcutaneously.^{a,b}
Darbepoetin: extended dosing intervals. Can be given less often than EPO alfa; has a half-life three times longer than EPO alfa. Studies show noninferiority to EPO alfa, with potential cost savings.^c
HIF PHIs: a new class of oral agents able to increase Hgb levels in patients naïve to ESAs or converted from ESA treatment.^{d-f} Roxadustat is approved in a number of countries for use in adult CKD and ESKD patients. In the United States, an FDA advisory committee has recommended not approving roxadustat because of safety concerns, primarily CVEs.
CERA: longer-acting ESA, lower frequency of administration; recent meta-analysis suggested CERA is similar to EPO alfa and darbepoetin in outcomes such as mortality, major adverse CVEs, or need for transfusions/iron therapy.^g

^aFishbane S, Spinowitz BS, Wisemandle WA, Martin NE. Randomized controlled trial of subcutaneous epoetin alfa-epbx versus epoetin alfa in end-stage renal disease. *Kidney Int Rep.* 2019 May 22;4(9):1235–1247. doi:10.1016/j.ekir.2019.05.010.

^bFishbane S, Singh B, Kumbhat S, Wisemandle WA, Martin NE. Intravenous epoetin alfa-epbx versus epoetin alfa for treatment of anemia in end-stage kidney disease. *Clin J Am Soc Nephrol.* 2018 Aug 7;13(8):1204–1214. doi:10.2215/CJN.11631017.

^cSinha SD, Bandi VK, Bheemareddy BR, et al. Efficacy, tolerability and safety of darbepoetin alfa injection for the treatment of anemia associated with chronic kidney disease (CKD) undergoing dialysis: a randomized, phase III trial. *BMC Nephrol.* 2019 Mar 13;20(1):90. doi:10.1186/s12882-019-1209-1.

^dChen N, Hao C, Liu B-C, et al. Roxadustat treatment for anemia in patients undergoing long-term dialysis. *N Engl J Med.* 2019 Sep 12;381(11):1011–1022. doi:10.1056/NEJMoa1901713.

^eAkizawa T, Iwasaki M, Yamaguchi Y, Majikawa Y, Reusch M. Phase 3, randomized, double-blind, active-comparator (darbepoetin alfa) study of oral roxadustat in CKD patients with anemia on hemodialysis in Japan. *J Am Soc Nephrol.* 2020 Jul;31(7):1628–1639. doi:10.1681/ASN.2019060623.

^fSingh AK, Carroll K, Perkovic V, et al. Daprodustat for the treatment of anemia in patients undergoing dialysis. *N Engl J Med.* 2021;385(25):2325–2335. doi:10.1056/NEJMoa2113379.

^gSaglimbene VM, Palmer SC, Ruospo M, Natale P, Craig JC, Strippoli GFM. Continuous erythropoiesis receptor activator (CERA) for the anaemia of chronic kidney disease. *Cochrane Database Syst Rev.* 2017 Aug 7;8(8):CD009904. doi:10.1002/14651858.CD009904.pub2.

CERA, Continuous erythropoiesis receptor activator; CKD, chronic kidney disease; CVEs, cardiovascular events; EPO, erythropoietin; ESA, erythropoiesis stimulating agent; ESKD, end-stage kidney disease; HIF, hypoxia-inducible factor; PHIs, prolyl hydroxylase inhibitors.

12.11 Management of Hyperkalemia

Hyperkalemia, defined as serum potassium greater than 5.0 to 5.5 mEq/L, is the most frequent electrolyte disturbance requiring urgent treatment in kidney failure and dialysis patients. Dietary noncompliance and insufficient dialytic removal are the main contributing factors. Other factors may include internal shifts of potassium due to acidosis or medications; tissue breakdown; decreased colonic potassium loss due to constipation; and loss of residual renal function.

Pseudohyperkalemia is when an elevated serum potassium does not reflect the *in vivo* plasma potassium levels. It may result from an improper collection process, hemolysis (within the extra-corporeal dialysis circuit or after collection), or thrombocytosis. This can be confirmed with a normal plasma (whole blood) potassium level.¹²²

Reverse pseudohyperkalemia, a rare occurrence in which the serum potassium is normal but the heparinized plasma potassium is falsely elevated, occurs with hematological malignancies and extreme leukocytosis from heparin-induced potassium loss from white blood cells (WBCs) elevating the plasma potassium after whole blood collection.¹²²

An elevated risk of adverse events has been demonstrated in dialysis patients with serum potassium levels >5.5 or 6.0 mEq/L, although an even lower threshold for risk of hospitalization and

TABLE 12.27 ■ Medical Management of Hyperkalemia

Purpose of Treatment	Agent	Dose	Comments/Side Effects
Stabilization of cardiac membrane	Calcium gluconate	1–2 g IV over minutes, can be given peripherally	Rapid cardiac effect in <5 min
	Calcium chloride	10 mL of 10% solution; preferably given centrally	
Shift K into cells	Regular insulin	10 U IV	K ⁺ decreases in 15–30 min, Given with 25–50 g IV dextrose to avoid hypoglycemia
	Beta-2 adrenergics	Nebulized; 5–20 mg in 4 mL over 10 min	K ⁺ decreases in 30 min,
	Sodium bicarbonate	50–150mEq IV	K ⁺ decreases in 1–4 hr especially if severe acidemia; sodium load
Removal from body	Loop diuretic	Furosemide 80 mg IV	Effective if diuretic-responsive
	Cation exchange resin	SPS 15 g orally 1–4 times daily	Associated with colonic necrosis; avoid if history of GI problems; less effective per rectum
		Patiromer 8.4 g 1–2 times daily with food; separate from other meds by 3 hr	Hypomagnesemia, constipation, impaired absorption of some drugs, slow onset of action
		SZC 10 g three times daily, then reduce to daily	Edema, fluid overload; onset in 1 hr; separate from other meds by 2 hr; 400-mg sodium/5-g dose
	Mineralocorticoid agonist	Fludrocortisone	Limited kaliuretic effect in ESKD

ESKD, End-stage kidney disease; *GI*, gastrointestinal; *IV*, intravenously; *SPS*, sodium polystyrene sulfonate; *SZC*, sodium zirconium cyclosilicate.

mortality may exist for some patients.¹²³ Hyperkalemia may lead to symptomatic muscle weakness, cardiac conduction delay, and cardiac arrhythmias. The main consequence of severe hyperkalemia is decreased myocardial conduction velocity, acceleration of repolarization, and possible cardiac arrest.

Mineralocorticoid receptor antagonists, of proven benefit in heart failure patients, have attracted growing attention in the ESKD population. A meta-analysis noted improvements in CV and all-cause mortality.¹²⁴ Their application of HD patients has been limited by concern about hyperkalemia. One U.S. data analysis indicated a higher risk of death with MRA use in ESKD.¹²⁵ A recent trial of spironolactone showed that 50 mg but not lower doses increased the risk of hyperkalemia.¹²⁶

Medical management of hyperkalemia, added to dietary potassium restriction and dialysis (the gold standard), may be urgently or emergently required (Table 12.27).

Polystyrene resins such as sodium polystyrene sulfonate (SPS, Kayexalate), which act by exchanging a cation for potassium within the intestinal lumen and are administered orally or rectally, have long been used to treat hyperkalemia. However, there have been concerns about weak evidence regarding efficacy, and well-known concerns about severe but rare adverse events such as bowel necrosis. A black box FDA warning from 2009 has warned against its use, particularly against coadministration with sorbitol to offset its constipating effects, as sorbitol appears to increase the risk of bowel toxicity. A recent systematic review reported a relative risk of 2.1 and a hazard ratio of 1.25 to 1.94, with a low absolute risk because of event rates of about 20/1000 person-years.¹²⁷ SPS crystals may be visible in rectal ulcer pathology.

Patiomer and sodium zirconium cyclosilicate (SZC) are now important alternatives to SPS.¹²⁸ In patients in need of acute management of hyperkalemia, SZC is preferred because of a more rapid effect, although the onset of action may be several hours and the effect modest.¹²⁹ Edema may result from its sodium content. Patiomer may be the drug of choice for chronic management, but may cause hypomagnesemia and constipation.

12.12 Bone and Mineral Disorders

ESKD-mineral bone disorder is a systemic disorder involving laboratory and skeletal abnormalities, vascular calcifications, or calcific uremic arteriolopathy (calciphylaxis). The clinical consequences are increased risk of bone fractures, CVEs, and mortality. The therapeutic approach must primarily address hyperparathyroidism and hyperphosphatemia.¹³⁰

12.12.1 MANAGEMENT OF SECONDARY HYPERPARATHYROIDISM

The related bone and mineral abnormalities of ESKD commonly manifest as progressive secondary hyperparathyroidism. Because of uremic resistance to the skeletal actions of parathyroid hormone (PTH), target PTH levels are 2 to 9 times the normal range. Secondary hyperparathyroidism may lead to progressive bone pain and fractures, refractory pruritus, accelerated vascular and cardiac valvular calcification, and myopathy. In tertiary hyperparathyroidism, progressive autonomous parathyroid production of PTH results in hypercalcemia.

Treatment options involve use of synthetic vitamin D analogues and calcimimetics. With the publication of the 2017 KDIGO guidelines,¹³⁰ the approach to controlling secondary hyperparathyroidism has evolved toward the first-line use of calcimimetic agents in addition to, or even instead of, starting vitamin D analogues. However, in most ESKD patients, a vitamin D analogue will also be required to correct vitamin D deficiency. Calcimimetics reduce parathyroid levels by increasing the sensitivity of the parathyroid calcium-sensing receptor to calcium. While effective at lowering PTH (especially if PTH is not very elevated), activated vitamin D administration may increase intestinal absorption and thereby serum concentrations of calcium and phosphorus; both are associated with increased cardiovascular risk. In contrast, calcimimetics are known to reduce both serum calcium and phosphorus levels, while lowering PTH.

Available calcimimetics include oral cinacalcet and IV etelcalcetide. Benefits of calcimimetics include lowering of serum PTH levels, but long-term effectiveness on cardiovascular and bone complications are uncertain. Both oral and IV forms are associated with nausea, vomiting, and hypocalcemia. Pooled data from initial regulatory trials indicated that oral cinacalcet in HD patients could reduce CVEs, fractures, and the need for surgical parathyroidectomy, compared to standard care with vitamin D analogues.¹³¹ Subsequently, in the Evaluation of Cinacalcet Hydrochloride Therapy to Lower Cardiovascular Events (EVOLVE) study, patient outcomes such as death or major CVEs were not demonstrated in an unadjusted analysis,¹³² although a beneficial effect in older patients appeared to occur. In a secondary analysis which included changes in the statistical analysis, the rate of clinical fractures was shown to be reduced.¹³³ Significant lowering of serum calcium with either calcimimetic may cause significant QT prolongation and ventricular arrhythmias.

Etelcalcetide IV is an effective alternative to oral cinacalcet.¹³⁴ It has been associated with hypersensitivity reactions, hypotension, congestive heart failure, prolonged QT interval, and decreased myocardial function. It may produce GI side effects similar to oral cinacalcet.

While fewer patients now appear to require parathyroidectomy, those with refractory hyperparathyroidism may nonetheless be best managed with surgical subtotal parathyroidectomy.

Activated vitamin D analogues are known to decrease parathyroid levels in dialysis patients, and may have other “pleotropic” effects.¹³⁵ Vitamin D analogues may increase both calcium and phosphorus levels, promoting soft tissue and vascular calcification. Commonly available analogues include calcitriol, paricalcitol, and doxercalciferol. No data indicate superiority of any

single agent. Some,¹³⁶ but not all, observational studies have indicated a reduction of mortality in ESKD patients on active vitamin D treatment.

12.12.2 MANAGEMENT OF HYPERPHOSPHATEMIA

Hyperphosphatemia is almost ubiquitous in patients on maintenance dialysis. Reduced adherence to phosphate-binder therapy, reported in over half of treated patients, contributes to poor control. The serum phosphorus goals are based on the guidelines of KDOQI (≤ 5.5 mg/dL)¹³⁷ and KDIGO (≤ 4.5 mg/dL).¹³⁰ The three key elements in the management of phosphate balance in ESKD are dietary restriction, oral phosphate binders, and adequate dialysis. Dietary phosphate absorption in the intestine is now known to involve not only active transcellular transport, but also passive paracellular absorption through tight junctions between enterocytes.

Basic research data on hyperphosphatemia suggest calcification toxicity to vasculature, and observational studies have indicated a positive association between elevated serum phosphate levels and mortality. Hyperphosphatemia has been associated with severity of coronary artery calcification, which is itself a predictor of mortality in dialysis patients. The recent EPISODE trial indicated that strict phosphate control toward the normal range may mitigate this effect.¹³⁸

The standard dietary phosphate restriction for ESKD patients is 900 to 1000 mg/day. However, there are very limited data on the benefit of restricted dietary phosphate intake on ESKD clinical outcomes.¹³⁹ A large observational study indicated treatment with phosphate binders was associated with improved survival among incident HD patients.¹⁴⁰

Conventional HD does not remove sufficient phosphate to achieve goal levels, so patients are managed with dietary phosphate restriction and, in most cases, with phosphate binders.¹⁴¹

Contemporary guidelines, although opinion based, strongly support the use of phosphate binders for serum phosphate lowering, to less than 5.5 mg/dL. For all prescribed binders, the lowest effective dose should be used.

In addition, the 2017 update of the KDIGO clinical practice guideline suggested restricting use of calcium-based phosphate binders in patients with ESKD, regardless of baseline serum calcium levels.¹³⁰ However, a subsequent trial did not suggest increased cardiovascular safety of the costlier sevelamer compared with calcium acetate.¹⁴²

It may be underappreciated that phosphate binders differ significantly in features, such as their phosphate-binding capacity and the range of pHs in which they are effective. For example, the binder sucroferic oxyhydroxide binds about 260 mg of phosphorus in the gut, compared to binding of 21 mg by an equivalent amount of sevelamer.¹⁴³ A reduced pill burden may result. Characteristics of oral phosphate binders are shown in [Table 12.28](#).

Current clinical approaches involve use of non-calcium-containing phosphate binders as the preferred choice, when possible, with an optimal clinical target goal of normalization of serum phosphate levels (3.5–4.5 mg/dL or at least <5.5 mg/dL). A recent study was the first randomized trial to demonstrate that normalization of serum phosphate levels was associated with improvement in cardiovascular health.¹³⁸ Analysis of Dialysis Outcomes and Practice Patterns Study (DOPPS) data indicates that worse phosphorus control may be strongly associated with cardiovascular mortality.¹⁴⁴

The fundamental clinical benefit of phosphate control in HD patients, however, remains unanswered, with a large clinical trial underway to look at the beneficial effects of higher or lower serum phosphate goals in the U.S. dialysis population.¹⁴⁵

12.12.3 CALCIFIC UREMIC ARTERIOLOPATHY (CALCIPHYLAXIS)

Calcific uremic arteriolopathy is more commonly referred to as calciphylaxis. The incidence in ESKD patients is in the range of 1% to 4%; it is far less commonly described in predialysis patients. While not fully understood, the fundamental pathology is progressive arterial calcification

TABLE 12.28 ■ Management of Hyperphosphatemia¹⁴¹

Phosphate Binder	Mechanism of Action	Dosage Strength	Initial/Maximum Recommended Dose (With Meals)	Advantages	Disadvantages
Aluminum hydroxide	Forms insoluble phosphate complexes	600-mg tablets	1–3 tablets three times/day	Inexpensive; calcium-free; effective at wide range of pH	Constipation; osteomalacia; CNS toxicity; anemia. For short-term use or with monitoring of aluminum levels.
Sevelamer carbonate	Cationic polymer/anion exchange resin	800-mg tablet, 800- or 2400-mg powder packet	800–2400 mg three times/day	No calcium loading; may lower lipid levels	GI intolerance
Calcium acetate	Binds with PO ₄ to form insoluble calcium phosphate in stool	667-mg tablet, 667-mg/5-mL solution; 25% elemental calcium	1–4 tablets three times/day (considered advisable to keep total elemental calcium dose to 1500 mg/day = 3-tablet dose)	Inexpensive; generally well tolerated	GI intolerance; positive calcium balance; possible risk of calcification
Lanthanum carbonate	Binds PO ₄ to form lanthanum phosphate in stool	500-, 750-, 1000-mg chewable tablet; 750- and 1000-mg oral powder	500–1000 mg three times/day	Potent; smaller pill burden; binds at lower pH	More costly; must be chewed; GI intolerance; tasteless
Ferric citrate	Iron-based phosphate binder	210-mg tablet	2–4 tablets three times/day	Provides bioavailable iron	More costly; avoid in iron overload; GI intolerance; dark stool
Sucroferric oxyhydroxide	Iron-based ligand exchange	500-mg chewable tablet	2–4 tablets three times/day	Potent; works in wide range of pH; smaller pill burden	More costly; flavored; may stain teeth; dark stool

CNS, Central nervous system; GI, gastrointestinal.

involving arterioles and capillaries in the dermis and subcutaneous adipose tissue, and affecting not only skin but at times other organs such as muscle, lungs, and brain.¹⁴⁶ In cutaneous calciphylaxis, painful and nonhealing violaceous skin ulcers may be preceded by nonulcerative subcutaneous plaques/nodules, and typically progress to tissue necrosis and gangrene.

Risk factors include female sex, obesity, elevated calcium/phosphorus levels, hyperparathyroidism, and hypercoagulable states.¹⁴⁷ Warfarin-associated calciphylaxis is a variant that is well described in the absence of CKD/ESKD. The drug promotes vascular calcification by inhibiting vitamin K-dependent matrix Gla protein, which normally prevents calcium deposition in arterioles.¹⁴⁸

The differential diagnosis of calciphylaxis may include ulcers related to peripheral vascular disease, cellulitis, atheroembolic disease, vasculitis, cryoglobulinemia, bacterial endocarditis, disseminated intravascular coagulation, and hypercoagulable states. The diagnosis is best established by skin biopsy of the peripheral area of the lesion, revealing the characteristic finding of diffuse calcification of small capillaries in adipose tissue, often with fibrin thrombi and dermal/epidermal necrosis, and inflammatory cell infiltration. However, the biopsy procedure itself may create a painful new lesion with bleeding and necrosis. Noninvasive support indicating the likelihood of the diagnosis may be found when a bone scan reveals increased technetium uptake in areas of involvement. Management considerations for calcific uremic arteriopathy are shown in [Table 12.29](#).

Calciphylaxis is associated with a mortality rate of 40% or more, but there have been encouraging reports of intravenous sodium thiosulfate producing improvement or even symptomatic resolution of lesions.¹⁴⁹ There are reports of intradermal injections and even intraperitoneal administration of sodium thiosulfate as well. Multiple clinical trials of other agents are currently underway.¹⁴⁶ Treatment options utilized for the management of calcific uremic arteriopathy are shown in [Table 12.29](#).

TABLE 12.29 ■ Management of Calcific Uremic Arteriopathy (Calciphylaxis)

- Sodium thiosulfate
- Stop vitamin D analogues
- Stop calcium supplements
- Lower dialysate calcium concentration
- Change warfarin to an NOAC (also called DOAC), if possible
- Increase dialysis intensity
- Treat hyperparathyroidism (PTH >800 pg/mL despite medical therapy)
- Consider vitamin K 10 mg orally (if not on warfarin)

DOAC, Direct oral anticoagulant; NOAC, new oral anticoagulant; PTH, parathyroid hormone.

12.13 Management of the ESKD Patient With Diabetes Mellitus

ESKD problems special to the patient with diabetes are shown in [Table 12.30](#).

12.13.1 GLYCEMIC MANAGEMENT

In fundamental ways, glycemic management in the ESKD patient is different and more complex.¹⁵⁰ Insufficient glycemic monitoring has been identified as one of the gaps in care for HD patients with diabetes mellitus. However, the role of tight glycemic control in ESKD patients remains controversial. The 2020 KDIGO guidelines on diabetes indicate that glycemic control must be individualized, based on comorbidities, risk of hypoglycemia, and other factors.^{151,152} The optimal target hemoglobin A1c (HbA1c) range for patients with diabetes on dialysis differs from that of the general population.¹⁵³ Suggested factors to consider in the spectrum of tighter (HbA1c <6.5%) or looser (HbA1c <8%) control in ESKD patients are shown in [Table 12.31](#).¹⁵²

TABLE 12.30 ■ ESKD Problems in Patients With Diabetes

Problem	Evaluation
Glycemic control	Hemoglobin A1c, continuous glucose monitoring, home fingerstick monitoring
Vascular access	Preservation of arterial and venous vasculature and early assessment for native fistula options
Coronary ischemia	Exercise tolerance test, P-thallium nuclear imaging, echocardiogram, heart catheterization, if indicated
Visual impairment	Ophthalmology evaluation and follow-up
Foot ulcers	Podiatry evaluation and follow-up
Peripheral vascular disease	Regular foot exams, podiatry care, Doppler flow studies of lower extremities
Gastroparesis/dysmotility	Gastric emptying study for symptoms
Neuropathy	Neurology evaluation, electromyography, if indicated
Malnutrition	Serum albumin, dietary counseling
<i>ESKD</i> , End-stage kidney disease.	

TABLE 12.31 ■ Factors to Consider in Determining Hemoglobin A1c Goal in ESKD Patients With Diabetes and CKD

Factors	GOAL HEMOGLOBIN A1c FOR BETTER OUTCOMES	
	<6.5%	<8.0%
CKD severity	Lower (e.g., eGFR stage 1)	Higher (e.g., eGFR stage 5)
Macrovascular complications	None or mild	Severe
Comorbidities	None or few	Many
Life expectancy	Long	Short
Hypoglycemia awareness	Good	Diminished
Resources for hypoglycemia management	Available	Limited
Likelihood of treatments causing hypoglycemia	Low	High
<i>CKD</i> , Chronic kidney disease; <i>eGFR</i> , estimated glomerular filtration rate; <i>ESKD</i> , end-stage kidney disease. Based on information in Chen TK, Sperati CJ, Thavarajah S, Grams ME. Reducing kidney function decline in patients with CKD: core curriculum 2021. <i>Am J Kidney Dis</i> . 2021 Jun;77(6):969–983. doi:10.1053/j.ajkd.2020.12.022.		

12.13.2 METHODS OF MEASURING GLYCEMIA

Continuous glucose monitoring, the most effective method to determine ongoing glucose exposure in patients with diabetes, appears to be accurate in CKD,¹⁵⁴ but chronic data in the ESKD population are not available. Levels of HbA1c, generally the gold standard, are affected by factors unrelated to glycemic control in ESKD, leading to a potential overestimation of glycemic control. Reduction in glycosylation of erythrocyte Hgb in anemic ESKD patients treated with ESAs accounts for the lowering of HbA1c. Diabetic HD patients with HbA1c measured below 6.5% and over 8.0% appear to have higher mortality risk.^{155–157} Guidelines recommend a target HbA1c of 7.0% to 8.0%.¹⁵¹ Glycated albumin has been suggested as the preferred alternative glycemic

marker in HD patients. The normal reference range for glycated albumin is 11.9% to 15.8%. There is limited evidence that the target range for glycated albumin should also exceed the reference range.¹⁵⁸

12.13.3 MANAGEMENT OF HYPERGLYCEMIA

The high prevalence of diabetes mellitus in the ESKD population means that the nephrologist should have some expertise in glycemic management.¹⁵⁹ A summary of drug selection would be as follows (Table 12.32). Note that sodium-glucose cotransporter-2 (SGLT2) inhibitors would not be effective in ESKD patients and should not be used.

A detailed accounting of diabetes treatments is shown in Tables 12.33 and 12.34.¹⁶⁰

Of note, a recent small study evaluating the benefit of a fully automated closed-loop insulin delivery system simulating the artificial pancreas (continuous glucose monitoring, insulin pump, diabetes control algorithm) demonstrated safe and effective glucose control over 24 hours.¹⁶¹

12.13.4 RISK OF HYPOGLYCEMIA

Hypoglycemia is a cause of morbidity and mortality in patients with diabetes, and affects the approach to treatment of diabetes in ESKD. Several factors, including diminished clearance of insulin, failure of renal gluconeogenesis, and inadequate nutrition, increase the risk of hypoglycemia in ESKD.¹⁶² Periods of hypoglycemia may be severe and prolonged.

The consequences of hypoglycemia are shown in Table 12.35.

Hypoglycemia may also be associated with stress-induced myocardial ischemia, thrombosis, proinflammatory events, and prolonged QT interval. With ESKD, more intense glucose monitoring is indicated in order to avoid hypoglycemic episodes.¹⁶³

12.14 Seizures

Acute seizures may occur as a complication of ESKD, or occur in patients with epilepsy that is exacerbated by uremia. The association with kidney failure may relate to many factors, such as hyponatremia, hypernatremia, hypocalcemia, acid-base disorders, azotemia, dialysis disequilibrium, uncontrolled HTN, and, rarely, ESAs.

For acute seizures, IV benzodiazepines can be used to halt seizure activity. Laboratory evaluation should include glucose, calcium, potassium, sodium, magnesium, blood urea nitrogen (BUN), and Cr. Options for antiepileptic drug treatment of seizures in ESKD patients are shown in Table 12.36.¹⁶⁴

Table 12.36 lists the properties of antiepileptic drugs used as therapy for chronic seizure control in the ESKD patient. Commonly used medications in ESKD are levetiracetam and lacosamide.¹⁶⁴

TABLE 12.32 ■ Oral Hypoglycemic Agents and ESKD

- Avoid metformin
- Glipizide is the preferred oral sulfonylurea
- No dose adjustment for repaglinide
- No dose adjustment for Pioglitazone or Rosiglitazone
- No dose adjustment for exenatide
- Need decreased dose of sitagliptin

ESKD, End-stage kidney disease.

TABLE 12.33 ■ Detailed Characteristics of Oral Hypoglycemic Agents

Class	Compound	Drug Name	Formulations	Mechanism of Action	Primary Physiological Action	Advantages	Disadvantages	Renal Metabolism/ Elimination	Eskd Dose Range
Sulfonylureas	Glipizide	Glucotrol	5, 10 mg	Close KATP channels on β -cell membrane	• $\uparrow$ Insulin secretion	• Common use	• Hypoglycemia • Weight gain • Low durability • ?Cardiovascular risk • Drug interactions	10% excreted unchanged in urine	2.5–10 mg daily
		Glucotrol XL	2.5, 5, 10 mg					Weekly active and inactive metabolites excreted in urine	1–4 mg daily
	Glimepiride	Amaryl	1, 2, 4 mg						
Meglitinides	Repaglinide	Prandin	0.5, 1, 2 mg	Close KATP channels on β -cell membrane	• $\uparrow$ Insulin secretion	• $\downarrow$Postprandial glucose excursions • Dosing flexibility • Relative safety	• Hypoglycemia • Weight gain • Frequent dosing • High cost	Limited excretion of inactive metabolites	0.5–1.0 mg three times daily w/meals
Thiazolidinediones	Pioglitazone	Actos	15, 30, 45 mg	Activate the nuclear transcription factor PPAR- γ	• $\uparrow$ Insulin sensitivity	• No hypoglycemia • Durability of effect • Favorable effect on lipid profile	• Weight gain • Possible fluid retention • ?Bladder cancer • Fractures • High cost	Limited excretion of active metabolites	15–30 mg daily

DPP-4 inhibitors	Sitagliptin	Januvia	25, 50, 100 mg	Inhibit the action of endogenous incretins by DPP-4	• ↑Insulin secretion • ↓Glucagon secretion	• Well tolerated • No weight gain • Less hypoglycemia	• Effect is modest • ?Pancreatitis • Urticaria/angioedema	Excreted mostly unchanged in urine	25 mg daily
	Saxagliptin	Onglyza	2.5, 5 mg					Renal excretion of drug and metabolites	2.5 mg daily
	Linagliptin	Tradjenta	5 mg					Very limited urinary excretion	5 mg daily
	Vildagliptin	Galvus	50 mg					Some excretion of drug by kidneys	50 mg daily
Incretin mimetics	Liraglutide	Victoza	18 mg/3 mL	Activates GLP-1 receptor	• ↑Insulin secretion • ↓Glucagon Secretion • ↓Gastric emptying • ↑Satiety	• Weight loss • No hypoglycemia	• GI side effects • No experience in renal failure • Injection required • ?Medullary thyroid cancer • ?Pancreatitis • High cost • Weight loss	Endogenous enzymes throughout the body	0.6–1.8 mg SQ daily

GI, Gastrointestinal; SQ, subcutaneously.

Adapted from Garg R, Williams M. Diabetes management in the kidney patient. *Med Clin North Am*. 2013 Jan;97(1):135–156. doi:10.1016/j.mcna.2012.11.001.

TABLE 12.34 ■ Characteristics of Insulin Analogues

Insulin Type	Name	Example	Onset of Actions	Peak Effect	Duration
Basal	NPH	Humulin N Novolin N	1–2 hr	4–8 hr	12–18 hr
	Glargine	Lantus	2 hr	No peak	20–24 hr
	Detemir	Levemir	2 hr	3–9 hr	16–24 hr
Prandial	Regular	Humulin R Novolin R	30 min	2–4 hr	6–8 hr
	Lispro	Humalog	5–15 min	1–2 hr	4–6 hr
	Aspart	NovoLog	5–15 min	1–2 hr	4–6 hr
	Glulisine	Apidra	5–15 min	1–2 hr	4–6 hr

Adapted from Garg R, Williams M. Diabetes management in the kidney patient. *Med Clin North Am.* 2013 Jan;97(1):135–156. doi:10.1016/j.mcna.2012.11.001.

TABLE 12.35 ■ Consequences of Hypoglycemia

Neurogenic	Neuroglycopenic
Tremulousness	Confusion/odd behavior
Palpitations	Drowsiness
Anxiety	Weakness
Sweating	Warmth
Tingling	Clumsiness
Hunger	Seizure
	Coma

TABLE 12.36 ■ Antiepileptic Drug Use in ESKD

Drug	Daily Dose in ESKD
Levetiracetam	500–1000 mg; 50% of daily dose as a post-HD supplement
Valproic acid	30–60 mg/kg/day as BID or TID
Lamotrigine	100–500 mg as QD or BID
Topiramate	100–200 mg BID; 50% supplement of BID dose as a post-HD
Zonisamide	100–600 mg/day as QD or BID; give post-HD
Phenytoin	150–200 mg BID or TID; follow free phenytoin levels
Phenobarbital	60–100 mg BID or TID
Carbamazepine	200 mg BID, maximum 800 mg BID
Oxcarbazepine	300 mg BID, maximum 1200 mg BID
Gabapentin	300 mg TID, maximum 1800 mg; give 200–300 mg post-HD dose
Pregabalin	150–600 mg/day as BID or TID, give 25–150 mg post-HD dose
Lacosamide	50 mg BID, maximum 200 mg BID; give 50% of BID dose as post-HD supplement

BID, Twice daily; ESKD, end-stage kidney disease; HD, hemodialysis; QD, daily; TID, three times daily.

12.15 Gout

Although the incidence of gout may be low in the ESKD patient, it is associated with a 1.5-fold increased risk of mortality.¹⁶⁵ Its incidence is increased with advancing age, obesity, alcohol use, female sex, and in the Black population. Hyperuricemia is almost universal in the patient with gout. Uric acid accumulates in kidney failure, but has a low molecular weight and is water soluble and not protein bound, so clearance with HD is excellent (and superior when HD is compared with PD). Nonetheless, hyperuricemia occurs in a large percentage of patients on dialysis. The mechanisms for the association with mortality risk are not well understood.

Gout should be considered in the differential diagnosis of acute joint complaints, although it may be mimicked by conditions such as pseudogout, septic monoarthritis, or localized cellulitis. Gout may resolve on its own, but it usually requires treatment, the sooner, the better.

The management of gout flares in the ESKD patient is challenging due to nephrotoxicity or contraindications of available agents.¹⁶⁶ The usual duration of therapy is 5 to 7 days. As described by Vargas-Santos et al.,¹⁶⁶ the treatment for gout can be separated into management of acute flares and prophylactic lowering of serum urate levels.

12.15.1 FOR ACUTE GOUT FLARES

Glucocorticoids: Treatment of choice in most ESKD cases; usually oral prednisone. Avoid if septic arthritis needs to first be ruled out. Initial dose is 30 to 40 mg. Avoid excessive dosing if gout occurs frequently. Will affect glycemic control in patients with diabetes. Intraarticular glucocorticoids may be preferred if oral agents are contraindicated.

Colchicine: Narrow therapeutic margin; should generally be avoided in ESKD. Poorly tolerated, dose must be adjusted; not removed by dialysis; increased risk of myo/neurotoxicity. Note that medications that inhibit the cytochrome P450 system component CYP3A4 (HIV protease inhibitors, clarithromycin, others) or the membrane P-glycoprotein drug efflux pump (cyclosporine, tacrolimus, amiodarone, verapamil, others) may dangerously raise colchicine levels and cause toxicity. FDA label limits dosing (in both HD and PD) to 0.3 mg twice weekly or a single 0.6-mg dose, not to be repeated in 2 weeks. Commonly causes diarrhea and abdominal cramping. ESKD patients should be monitored closely for colchicine toxicity if the drug is used.

NSAIDs: Relatively contraindicated in ESKD; available without a prescription; risk of acute gastritis and GI bleeding; systemic bleeding tendency due to impaired thromboxane-dependent platelet aggregation; may result in loss of any residual renal function. Aspirin, especially at low doses (<2–3 g/day), is contraindicated in treating gout flares because of elevation of serum uric acid levels that may result from the medication.

12.15.2 FOR URATE-LOWERING THERAPY

Allopurinol: Dose adjusted so that serum levels of oxypurinol do not rise; do not give during active flare; significant toxicity such as allopurinol hypersensitivity syndrome (related to the initial dose and whether the patient is a carrier of the HLA-B*5801 allele); risk is greatest in first 6 months of use; initial dose 50 mg/day to reduce risk of gout flare and allergic reaction with up-titration to maximum of 100 mg daily or 300 mg twice weekly. Should be discontinued at first appearance of skin rash or other signs of allergy. Dose limitation may reduce efficacy. Continue during flare without interruption. Do not initiate or reintroduce during flare as might predispose to another flare. If patient is on azathioprine, reduce its dose to one-fourth of preazathioprine allopurinol dose.

Febuxostat: Xanthine oxidase inhibitor; prohibitive cost; few case reports in ESKD showing reduction in uric acid levels; risk of severe adverse cutaneous reactions such as drug rash with

eosinophilia and systemic symptoms (DRESS) syndrome, Stevens-Johnson syndrome, and toxic epidermal necrolysis may be increased in CKD.

Pegloticase: Recombinant PEGylated uricase, converts uric acid to the more soluble allantoin; dosing IV every 2 weeks; no dose adjustment; G6PD deficiency is contraindication; anaphylaxis and infusion reactions have been reported.¹⁶⁷

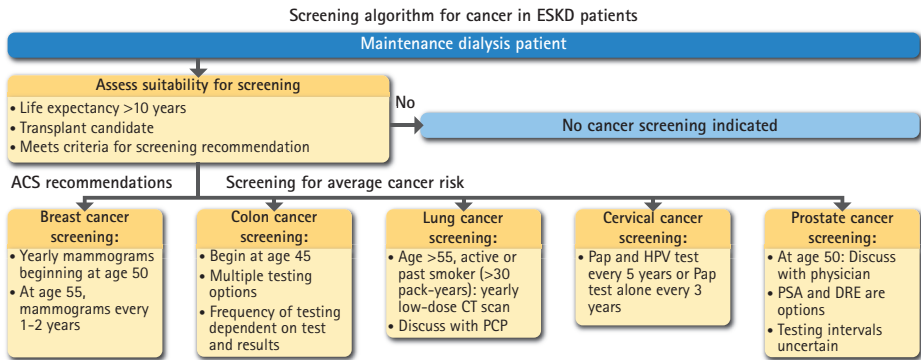
Rasburicase: Recombinant urate oxidase that can be used for acute hyperuricemia in patients with hematological and solid tumor malignancies who are receiving anticancer therapy to prevent or treat tumor lysis syndrome¹⁶⁸; [on line] can cause anaphylaxis, hemolysis (G6PD deficiency is a contraindication), or methemoglobinemia.

Dialysis: Increasing dialysis frequency in order to reduce uric acid levels.

12.16 Cancer Screening in ESKD

The overall cancer risk in ESKD patients is increased (standardized incidence ratio = 1.42), particularly kidney/bladder, breast, lymphoma, lung, and colon cancer.¹⁶⁹ Cancer screening guidelines developed for the general population may not apply to ESKD patients, related to the performance characteristics of the screening tests and the potential benefits related to reduced life expectancy. A screening algorithm for cancer in ESKD is shown in Figure 12.12.¹⁶⁹

A specific challenge is related to the increased incidence of acquired renal cystic disease with time on dialysis (over 50% by 10 years), and the association of the acquired cysts with renal cell carcinoma. Periodic screening by initial ultrasound (US), followed by CT scan when indicated, is recommended for ESKD patients who are transplant candidates or who have longer expected survival.^{169,170}



ACS, American Cancer Society; DRE, digital rectal exam; ESKD, end-stage kidney disease; HPV, human papillomavirus; PCP, primary care physician; PSA, prostate-specific antigen.

Figure 12.12 Screening algorithm for cancer in ESKD patients. (From Rosner MH. Cancer screening in patients undergoing maintenance dialysis: who, what, and when. *Am J Kidney Dis*. 2020 Oct;76[4]:558–566. doi:10.1053/j.ajkd.2019.12.018.)

12.17 COVID-19 and ESKD

Patients with CKD/ESKD have been severely affected by the COVID-19 pandemic.^{171,172} Chronic dialysis patients have specific risk factors for COVID-19, including older age, obesity, diabetes, HTN, and Black/Hispanic demographics, as well as risk of person-to-person exposure during transportation and in the crowded dialysis venue. Outbreaks in dialysis units and clustering of cases on specific dialysis shifts have been reported.

ESKD patients are particularly susceptible to COVID-19,¹⁷³ only in part related to overrepresentation of the elderly and Blacks relative to the general population. Universal screening

indicates a high prevalence of positive RT-PCR results from nasopharyngeal swabs in both patients and staff. The overall dialysis seropositivity rates in patients have been 2 to 3 times those of the general population. Despite implementation of stringent policies and procedures to reduce risk in the HD setting, overall seropositivity rates averaged around 6% during the pandemic,¹⁷⁴ with geographic variation ranging between 2% and 24%. Black and Hispanic patients had rates over twice as high as White patients. The seropositivity rate was surprisingly lower in older patients, possibly due to their higher mortality risk. The overall rate of patients clinically diagnosed with COVID-19 was 3.3%, making the seropositivity rate about 1.7 times the clinical diagnosis rate.

Patients on dialysis appeared to have fewer unknown COVID-19 infections than the general population, although many cases are subclinical or asymptomatic.¹⁷⁵ Reports from around the world have indicated a fatality rate of 20% to 30% among patients with ESKD.¹⁷⁶ All-cause mortality among dialysis patients during the 2020 COVID-19 surge was about 50% higher than the same time in preceding years. Maintenance dialysis patients needing ICU care for COVID-19 have a hospital mortality rate of 50%. In addition, 20% to 40% of COVID-19 patients in the general population have developed AKI, many of them intensive care patients requiring dialysis. Patients who recover from symptomatic episodes have prolonged viral shedding, although they do develop an IgG antibody response to the virus.

During the 2020 crisis phase of the pandemic, one result of shortages of supplies and trained staff was a reduction for some acute patients in the number of HD treatments to twice weekly. Also, data indicate that a reduction in CKD care during the peak months of the pandemic occurred that may lead to a downstream impact of reduced RRT preparedness.

By comparison, the incidence of symptomatic disease is reduced by at least half in home hemodialysis (HHD) patients and is only slightly above that of the general population in PD patients.¹⁷² A transition toward greater utilization of home dialysis modalities may also be a legacy of the COVID-19 pandemic.

Importantly, prevention of transmission in the HD unit setting has been shown to be achievable using universal COVID-19 testing along with universal droplet and contact precautions.¹⁷⁷ Strategies to reduce COVID-19 risk in chronic dialysis patients¹⁷⁸ are shown in Table 12.37.

A variety of treatment options for COVID-19 have been tested (chloroquine, IL-6 blockade, interferon, corticosteroids, convalescent plasma, remdesivir, favipiravir, therapeutic plasma exchange, monoclonal antibodies, and new oral antiviral drugs), but data in the ESKD population are sparse in part due to exclusion of patients from clinical trials.¹⁷⁹ Remdesivir pharmacokinetics predict an accumulation of the parent compound, an active metabolite, and the vehicle compound with sustained use in the setting of kidney failure.¹⁸⁰ Nirmatrelvir-ritonavir (Paxlovid) is significantly renally excreted, but can be utilized at reduced dosage for patients with CKD. It is generally not recommended for those with advanced kidney failure, but has been increasingly used in symptomatic dialysis patients at reduced dosages. For dialysis, heightened anticoagulation necessary to

TABLE 12.37 ■ Strategies to Reduce Risk of COVID-19 Infection in ESKD Patients¹⁷⁸

Risk reduction through patient and family education
Expanded testing in high-risk community settings
Universal mask use among dialysis patients and staff
Reduced waiting-room exposure
Routine primary screening before entry to the facility, for signs and symptoms or known exposure to COVID-19
Cohorting of patients with symptomatic or asymptomatic COVID-19 status
Mandatory screening of nursing home residents and new hospital admissions
<i>ESKD</i> , End-stage kidney disease.

prevent filter clotting has been well demonstrated in those on CRRT.¹⁸¹ Acute PD has emerged as an alternative emergency therapy due in part to HD machine/staff shortages in surges. For chronic dialysis therapy, PD has been promoted when possible, to avoid hospitalization time and reduce the risk of hospital-acquired infections.¹⁸²

The ability of current vaccines to provide protective capacity in ESKD patients, who are known to have a suboptimal acquired immunity from other vaccines, remains to be determined.¹⁸³ Dialysis patients appear to develop lower titers of protective antibody than control groups.¹⁸⁴ However, evidence suggests that over 80% will respond to 2 doses of an mRNA vaccine. Nonetheless, elderly ESKD patients are especially advised to receive a third vaccine dose. Effects on the severity of the disease and long-term safety remain to be determined. Seroconversion, which occurs after infection, approaches 100% in the dialysis population, but titers may decline after a few months; natural immunity to COVID-19 among chronic HD patients occurs, but is incomplete and is of reduced magnitude compared to healthy individuals.¹⁸⁵ However, acquired immunity significantly reduces risk of subsequent COVID-19 infection among ESKD patients.

12.18 Symptom Management in ESKD Patients

Even less objectively assessable than dialysis adequacy, but increasingly considered a relevant measure of it, are patient-reported outcomes such as symptom control (Table 12.38¹⁸⁶). Patients with ESKD experience many physical and psychological symptoms, and their physical state typically deteriorates over time. This may be underrecognized by the physician. Regular symptom management should be assessed through available survey tools.¹⁸⁶

Fatigue: Fatigue is a common complaint, diminishes quality of life, and may not improve with the dialysis procedure.¹⁸⁷ Little is actually understood about the pathogenesis and treatment of fatigue.¹⁸⁸ Almost three-quarters of patients report some degree of fatigue, and it is severe in one-third of them. Postdialysis fatigue may be the most intolerable part of dialysis dependence. A new fatigue measure (tiredness, energy level, limits on usual activities) may be useful.¹⁸⁹ Empiric treatment includes increasing physical activity, maximizing the target Hgb level, correcting acidemia, and evaluating/treating depression. More frequent dialysis sessions may actually diminish fatigue.

TABLE 12.38 ■ Symptoms Reported in ESKD Patients

Fatigue/weakness: 68%
Dry skin: 72%
Pruritus: 54%
Pain: 50%
Dry mouth: 45%
Insomnia: 44%
Muscle cramps: 43%
Diarrhea: 17%
Anxiety: 28%
Shortness of breath: 19%
Poor appetite: 29%
Depression: 24%
Restless legs: 29%
Nausea: 26%
Constipation: 21%
Vomiting: 11%

ESKD, End-stage kidney disease.
Adapted from Gelfand SL, Scherer JS, Koncicki HM. Kidney supportive care: core curriculum 2020. *Am J Kidney Dis*. 2020 May;75(5):793–806. doi:10.1053/j.ajkd.2019.10.016.

Pain: Dialysis patients experience a high pain symptom burden.¹⁹⁰ Avoidance of NSAIDs in ESKD has led to increased administration of opioids and other adjuvant therapies to manage pain. Dialysis patients with residual renal function may face the same NSAID risks as other stage 5 CKD patients, including AKI, loss of residual kidney function, hypervolemia, and hyperkalemia. If NSAIDs are to be used in ESKD, low doses of short half-life agents restricted to 5 or fewer days are appropriate.¹⁹¹ Opioid prescribing in ESKD patients has increased,¹⁹² but has been associated with increased hospitalizations, death risk, and other adverse outcomes. Dosing for preferred opioids in advanced CKD is shown in Table 12.39.¹⁹²

Pharmacokinetics and clearance vary among parent opioid compounds. Hydromorphone (0.5 mg every 6 hr), fentanyl (12.5-mcg patch every 72 hr), methadone, and buprenorphine (5-mcg patch every 7 days) are preferred; not recommended are morphine, hydrocodone, codeine, and extended-release tramadol.

Pruritus: Uremic pruritus is common, and is associated with poorer quality of life as well as higher mortality.¹⁹³ Its prevalence is underestimated by physicians, and many patients remain untreated. It frequently exists with dry skin, and often is precipitated by the dialysis procedure. Other dermatological conditions, such as contact dermatitis or scabies infestation, and systemic conditions such as cholestasis and postherpetic neuralgia, should be ruled out.

Treatment generally targets high phosphate levels, hyperparathyroidism, or inadequate dialysis, based on weak evidence. In addition, all sufferers should receive a topical emollient (such as 1% pramoxine hydrochloride lotion) and reapply after bathing; in addition, sufferers should avoid hot-water bathing, use moisturizers regularly, and take low-dose gabapentin or pregabalin. Use of

TABLE 12.39 ■ Suggested Starting Doses of Preferred Opioids for Use in Advanced CKD¹⁹²

Opioid	Normal Kidney Function (eGFR > 100)	CKD Stage 4 (eGFR 15–30)	CKD Stage 5 (eGFR <15)
Hydromorphone (short-acting)	2- to 4-mg oral tab every 4–6 hr	1-mg oral tab every 6 hr	0.5-mg oral tab every 6 hr
Fentanyl (long-acting)	Not recommended in opioid-naïve patients; dose may be variable based on oral opioid equivalent dose	12.5- or 25-mcg patch transdermally every 72 hr (decrease to 50%-75% of normal dose)	12.5-mcg patch transdermally every 72 hr (decrease to 50% of normal dose)
Methadone	Recommend referral to specialist; requires pretreatment and follow-up ECG monitoring for QT interval prolongation		
Buprenorphine	5-mcg patch transdermally every 7 days; may precipitate withdrawal in patients already receiving opioids; should discontinue other long-acting opioids; may need to continue short-acting analgesics until adequate analgesia from buprenorphine is achieved	5-mcg patch transdermally every 7 days (no clear evidence for dosage adjustments; use with caution)	5-mcg patch transdermally every 7 days (no clear evidence for dosage adjustments; use with caution)
Oxycodone	10–30 mg every 4–6 hr; use with caution	5 mg every 6–8 hr; use with caution	2.5–5 mg every 8–12 hr; use with caution
CKD, Chronic kidney disease; ECG, electrocardiogram; eGFR, estimated glomerular filtration rate (in mL/min/1.73 m ²).			

TABLE 12.40 ■ Treatment of Uremic Pruritus

1. Optimization of dialysis: maximize Kt/V; use of high-flux dialyzer
2. Treatment of hyperphosphatemia
3. Parathyroidectomy for hyperparathyroidism if calcimimetic treatment fails
4. Topical emollients, capsaicin, and pramoxine
5. Antihistamines and gabapentin
6. Opioid receptor modulators (naltrexone or naloxone; difelikefalin or nalfurafine)
7. Ultraviolet B phototherapy for resistant cases

antihistamines such as hydroxyzine is based on weak evidence and may result in sedation. With the exception of gabapentin, the evidence for benefit is weak.¹⁹⁴

Stimulators of the μ -opioid receptor (morphine) in the brain lead to pruritus, and κ -opioid receptor stimulation leads to inhibition of μ -receptors, making modulation an attractive therapeutic target. Opioid receptor antagonists include naltrexone, nalfurafine, and difelikefalin. A recent regulatory trial using the Worst Itching Intensity Numerical Rating Scale indicated superiority over placebo of the intravenous agent difelikefalin, a selective κ -opioid receptor agonist.¹⁹⁵ In select patients, narrow-band ultraviolet B phototherapy, parathyroidectomy, or acupuncture may be appropriate. Treatment for uremic pruritus is indicated in Table 12.40.

12.18.1 DEPRESSION

Depression affects at least one-quarter of ESKD patients and affects their outcomes.¹⁹⁶ The Institute for Health and Care Excellence (NICE) guidance for depression in adults with chronic health problems recommends a stepped care approach to treatment, including medications for mild to moderate depression.

Management of dialysis patients is mandated to include annual screening and follow-up by the dialysis team for depression. Screening tools have been validated in ESKD. Treatment is similar to the general population, with evidence suggesting the benefit of psychosocial interventions in addition to medications. However, in many instances, renal providers are unwilling to prescribe antidepressants, or patients do not accept treatment.

Selective serotonin reuptake inhibitors (SSRIs) are first-line therapy in ESKD patients. Paroxetine clearance is significantly reduced in renal failure. No antidepressants are cleared by dialysis. Therefore, no dose adjustment is required with sertraline and fluoxetine in dialysis patients, while the dose of paroxetine is reduced by at least half. Evidence indicates that cognitive behavioral treatment as well as exercise therapy are beneficial.

12.18.2 SLEEP DISORDERS

Up to 80% of ESKD patients experience sleep disturbances, including insomnia, restless leg syndrome, sleep-disordered breathing, and periodic limb movement disorder. Furthermore, there are associations between sleep disorders and cognitive impairment.¹⁹⁷ Parasomnias refer to sleep states of partial arousal with abnormal movements and dreams. Obesity and neck circumference (>40 cm) are risk factors. Mechanisms in uremia include fluid accumulation, systemic inflammation, and disturbed respiratory drive. The normal evening surge of melatonin is inadequate in ESKD patients. Sleep apnea improves in patients with intensive dialysis such as nocturnal HD. In addition to optimization of the dialysis regimen, treatment of insomnia in ESKD is the same as in the general population. Benzodiazepines should be avoided in older patients or those with a history of substance use disorder. Contributing factors such as dementia, obstructive sleep apnea, and heart failure should be ruled out.

12.19 Home Dialysis Options

Much effort has been put into correcting a perceived underutilization of home dialysis therapy, including both HHD and PD, in recent years.¹⁹⁸ Incentivized by health policy changes and regulatory efforts, the home dialysis option has recently increased to its highest point in 2 decades.¹⁹⁹ Close to 90% of patients undergoing dialysis use in-center HD, and the probability of switching to home dialysis decreases from 7% in the first month to <1% by the fourth month of dialysis. If home dialysis is likely to expand further in the future to meet current goals, predialysis planning will need to fundamentally change.

There are no randomized clinical trials comparing home and in-center dialysis. Home modalities do appear to offer the potential for improved clinical outcomes, increased patient satisfaction and autonomy, and increased employment. Regarding health-related quality of life (HRQoL), a recent analysis showed better physical HRQoL with home dialysis than in-center dialysis, while mental HRQoL was comparable.²⁰⁰ Barriers include reduced patient referrals, workforce shortages, lack of professional expertise, patient costs, and patient/caregiver burdens. A recent NKF-KDOQI conference concluded that policy and reimbursement changes to better empower patients to make informed choices about home dialysis are necessary.²⁰¹ Potential advantages of a home dialysis regimen are shown in Table 12.41.²⁰¹

12.20 Home Hemodialysis

Growth in home dialysis is particularly evident in HHD, which is being given high priority by dialysis organizations. While for many nephrologists, a barrier to HHD is their lack of familiarity with the modality, intensive efforts are being made to educate both physicians and patients regarding the full scope of this treatment option and whether the patient can manage it. Patient-perceived barriers include need for self-cannulation of AVFs or AVGs and fear of a catastrophic event. More user-friendly machine technology has contributed to the growth in HHD.

Recent practice guidelines suggest an individualized prescription for HHD, taking into account lifestyle goals, with the same dose and time target considerations each week as for in-center-based patients.²⁰² Current clinical practice guidelines for weekly HD adequacy suggest a target standard (stdKt/V_{urea}) of 2.3, with a minimum delivered dose of 2.1. This stdKt/V formula is used for higher home dialysis frequencies than the single-pool (spKt/V) formula used for thrice-weekly in-center HD.²⁰³

Potential benefits of HHD include regression in left ventricular (LV) mass, reduced depressive symptoms, improved volume control with liberalized diet, superior bone/mineral parameters, decreased postdialysis recovery time, improved quality of life, and reduced mortality risk. One

TABLE 12.41 ■ Potential Advantages of Home Dialysis

- Improved patient satisfaction
- Flexibility in scheduling
- Employment more likely
- Superior physical health-related quality of life
- Intensified dialysis regimen
- Shorter interdialytic period
- Fewer hospital visits
- Regression of left ventricular hypertrophy
- Longer life expectancy

Adapted from Chan CT, Collins K, Ditschman EP, et al. Overcoming barriers for uptake and continued use of home dialysis: an NKF-KDOQI conference report. *Am J Kidney Dis*. 2020 Jun;75(6):926–934. doi:10.1053/j.ajkd.2019.11.007.

landmark study comparing three conventional weekly HD sessions to somewhat shorter, 6-weekly sessions reported a 40% mortality reduction in the frequent HD group.²⁰⁴ In a randomized controlled trial, patients who switched from conventional HD to HHD performed overnight 5 to 6 times weekly showed improvements in LV mass, BP, and mineral metabolism.²⁰⁵

12.21 Hemodialysis Adequacy

According to the most recent KDOQI clinical practice guideline,²⁰⁶ Kt/V monitoring is done on a monthly basis for the majority of chronic HD patients. The current convention is a minimum delivered single-pool Kt/V of 1.2, three times per week for chronic HD patients, with a target Kt/V of 1.4^{202,206}; the contribution from residual kidney function is no longer generally measured or included. Targeting a Kt/V over 1.4 does not appear to improve outcomes.²⁰⁷ In a simpler but less tested alternative fashion, the dialysis dose may also be expressed as a percent reduction in plasma urea concentration (urea reduction ratio or URR). Recommendations for dialysis adequacy with schedules other than conventional 3 times weekly HD (such as HHD or nocturnal dialysis) are less certain. KDOQI guidelines¹⁹⁹ suggest a minimum delivered Kt/V dose of 2.1 with a method that includes residual kidney function and ultrafiltration volume.²⁰⁸

Virtually all randomized controlled trials on HD outcomes have used Kt/V or URR to reflect dialysis adequacy. Several observational studies^{209,210} have indicated a correlation between Kt/V and mortality outcomes. A major shortcoming of Kt/V (urea) is its reliance on the clinical importance of urea despite the presence of numerous other uremic toxins. In addition, data to indicate that “patient-relevant” outcomes relate to better Kt/V results are largely lacking.²¹¹ As a result, determining the adequate dose of dialysis for an individual patient remains somewhat uncertain, and there is growing consensus that the totality of adequacy in dialysis involves more than just small solute removal.²¹² The assessment of dialysis adequacy for ESKD is seen in [Figure 12.13](#).

12.22 Peritoneal Dialysis

INTRODUCTION

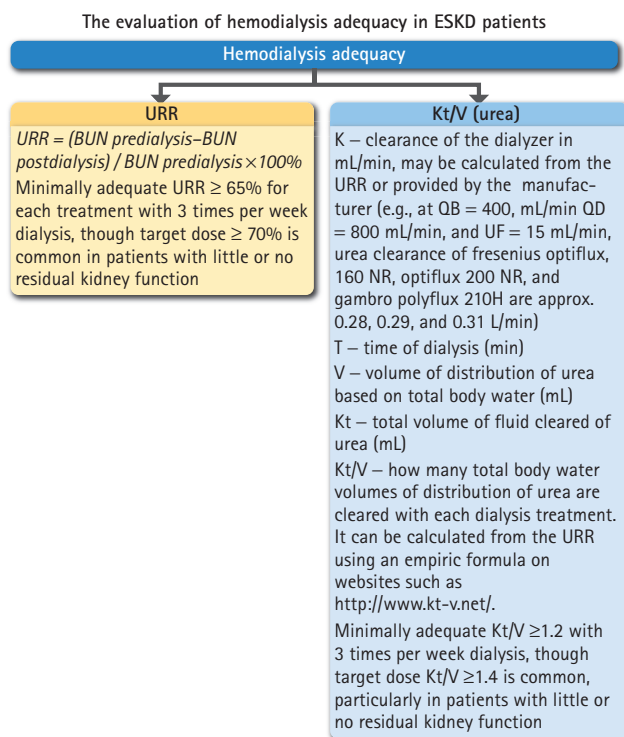
PD is the major alternative modality to HD, and is the most common form of home dialysis. Modality choice is typically determined by patient preference. Although associated with similar mortality, higher quality of life, lower rates of complications, and lower costs compared to HD, PD has been underutilized.²¹³ Use has increased in the United States since Medicare payment reform in 2011.²¹⁴

A disadvantage of PD is a number of infectious and mechanical complications specific to the modality, such as peritonitis, catheter malfunction, ultrafiltration failure, hydrothorax, and others.

Factors to be overcome in expanding the use of PD have included an incorrect perception that clinical outcomes are inferior, that peritonitis rates are high, that small-solute clearances are inadequate, and that ultrafiltration failure rates are high. Note that PD has emerged as an attractive alternative to HD for some ESKD patients with COVID-19 infection, by minimizing healthcare facility visits, reducing exposure in congregate settings, and improving social distancing. PD can be delivered at lower costs than the costs for HD.²¹⁵ “Assisted” PD utilizes a paid caregiver model and is being developed to allow this home modality for those with physical or cognitive barriers to self-care PD at home.

Recent publications have suggested a role for PD in non-ESKD patients with cardiorenal syndrome, although the benefit of hemofiltration itself when compared to diuretic therapy in that setting was inconsistent in three trials (UNLOAD, RAPID-CHF, and CARRESS-HF).²¹⁶ PD should be considered in patients with volume overload related to chronic cardiorenal syndrome.²¹⁷

Recent guidelines emphasize incorporating quality of life into the assessment of care; preserving residual renal function; utilizing incremental PD; and prescribing PD for older people with frailty.²¹⁸



BUN, Blood urea nitrogen; ESKD, end-stage kidney disease; URR, urea reduction ratio.

Figure 12.13 The evaluation of hemodialysis adequacy in ESKD patients. (From Windpessl M, Prischl FC, Prenner A, Vychytil A. Managing hospitalized peritoneal dialysis patients: ten practical points for the non-nephrologist. *Am J Med.* 2021 Jul;134[7]:833–839. doi:10.1016/j.amjmed.2021.02.007.)

12.22.1 PERITONEAL DIALYSIS INITIATION

PD takes advantage of the peritoneum to act as a dialysis “membrane” for solute and fluid removal. Location of the peritoneal catheter is depicted in [Figure 12.14](#).²¹³

The catheter consists of a visible external portion, the mid-portion of which is tunneled through the abdominal wall and ultimately penetrates the peritoneal cavity, and the intraabdominal portion. Optimal catheter tip location is in the pelvis. Connected to the external end of the catheter is a transfer set (not shown), to which the dialysis bag’s tubing is connected.

It is ideal to have the PD catheter placed at least 2 weeks before the anticipated need for initiation of PD, mainly to prevent fluid leakage from the peritoneal cavity.²¹⁹ Alternatively, PD may need to be initiated on an “urgent” basis (see below).

12.22.2 PERITONEAL DIALYSIS SOLUTIONS

Commercially available peritoneal dialysate solutions are composed of an osmotic agent, electrolytes, and a buffer ([Table 12.42](#)).

The glucose-based solutions are hypertonic-containing dextrose, a monosaccharide, as the primary osmotic agent in a concentration of 1.5%, 2.5%, and 4.25%. The solutions contain the electrolytes sodium, chloride, calcium, and magnesium, but no potassium. As a buffer, they

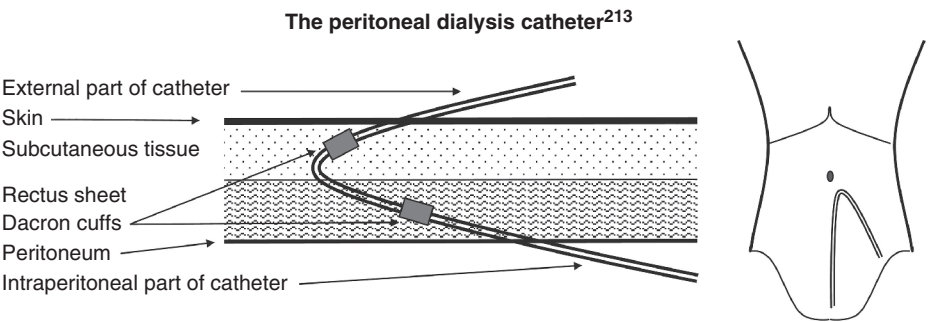


Figure 12.14 The peritoneal dialysis catheter.

TABLE 12.42 ■ Comparison of Peritoneal Dialysis Solutions

Solution	1.5% Dextrose	2.5% Dextrose	4.25% Dextrose	7.5% Icodextrin
Dextrose	15 g/L	25 g/L	42.5 g/L	0 (75 g/L icodextrin)
Sodium	132 mEq/L	132 mEq/L	132 mEq/L	132 mEq/L
Chloride	95 mEq/L	95 mEq/L	95 mEq/L	96 mEq/L
Calcium	2.5 mEq/L	2.5 mEq/L	2.5 mEq/L	3.5 mEq/L
Magnesium	0.5 mEq/L	0.5 mEq/L	0.5 mEq/L	0.5 mEq/L
Lactate	40 mEq/L	40 mEq/L	40 mEq/L	40 mEq/L
Osmolarity	344 mOsm/L	395 mOsm/L	485 mOsm/L	285 mOsm/L
pH	4.0–6.5	4.0–6.5	4.0–6.5	5.0–6.0

contain significant concentrations of lactate. The glucose exposure has been linked to adverse metabolic consequences related to significant systemic glucose absorption.

Icodextrin is an isosmotic glucose-sparing solution derived from maltodextrin and is used in the long dwell to improve fluid removal, particularly in patients with fast peritoneal solute transfer rates (“rapid transporters”).²²⁰ Icodextrin-containing solutions enable the long dwell and promote more sustained ultrafiltration capacity, for up to 8 to 12 hours. Compared to crystalloid solutions which induce ultrafiltration through aquaporin and large/small pores (the former producing water-only ultrafiltration), icodextrin induces more sodium removal through colloid osmosis, acting predominantly through large/small pores, with minimal absorption. It also prevents nonfluid (presumed fat) weight gain and improves insulin resistance. In PD therapy for diabetic kidney disease patients, the use of icodextrin-containing solutions has a beneficial effect on technique survival.²²¹ Note that only glucose-specific monitors and test strips to monitor blood glucose should be used in patients on icodextrin; other devices may show falsely elevated glucose levels. Icodextrin is contraindicated in patients with lactic acidosis or in patients with a known allergy to cornstarch or icodextrin itself. Serious reactions such as toxic epidermal necrolysis, angioedema, or erythema multiforme may result. Rarely, encapsulating peritoneal sclerosis, which may be fatal, has been reported with icodextrin.

Icodextrin may improve metabolic control in patients with diabetes, and may also lower serum insulin levels in patients without diabetes, although patients should be monitored for volume overload.²²²

12.22.3 PERITONEAL DIALYSIS TECHNIQUES

PD exchanges may be done manually or with an automated cycler, or in combination usually performed 6 to 7 days per week:

Manual technique

- CAPD: Continuous ambulatory peritoneal dialysis (CAPD); continuous technique with about 4 sequential 2- to 3-liter exchanges per day, 4- to 6-hour dwell cycles

Techniques requiring a cycler machine or automated PD (APD)

- CCPD: Continuous cycling peritoneal dialysis using a cycler machine, usually during the night; with shorter dwell times than CAPD but more exchanges and commonly with an additional long dwell, or even a manual exchange, during the day
- CCPD with abdomen dry during the day may be called intermittent PD (IPD) or nocturnal intermittent PD (NIPD) performed as frequent, short cycles during the night with no daytime dwell
- TPD: Tidal peritoneal dialysis (series of quick fills with incomplete drains so that some residual volume (15%–25% of usual dwell volume) remains in the peritoneal space; used infrequently as a peritoneal “conditioning” regimen or for patients with abdominal/rectal/pelvic pain when completely drained

12.22.4 PERITONEAL DIALYSIS PRESCRIPTION

The fill volume and dwell time of PD should be tailored to the individual patient's needs.²²³ The initial PD prescription is based on clinical assessment of the patient. An adequate “prescription” of PD should be designed to provide adequate clearance and volume removal, while fitting the patient's lifestyle/schedule. The components of the PD prescription include PD modality, number of exchanges, volume of exchanges, and dialysate solution osmolarity (“strength” of PD dialysate to ultrafilter off fluid is usually determined by dextrose concentration, though icodextrin is available). In “incremental” PD, the initial prescription takes advantage of residual kidney function to limit the solute transfer required to meet treatment goals.

A key tool for assessing peritoneal membrane function, and therefore the dialysis prescription, is the peritoneal equilibration test (PET).²²⁴ The PET is typically performed several weeks after initiation of chronic PD. The dialysis prescription is then modified, based on the transport characteristics of the peritoneal membrane determined by the PET results.

Note that, in addition to forming the basis for the initial PD prescription, the PET may be used later in the course of PD when poor fluid removal or inadequate dialysis suggests membrane dysfunction.

Importantly, when membrane dysfunction subsequently occurs, additional PET evaluation may be an important tool in providing an explanation for the dysfunction.

12.22.5 PET INTERPRETATION AND DIALYSIS MODALITY SELECTION

The purpose of the PET is to characterize the rate of transfer of solute and water across the peritoneal membrane.²²⁵ The test yields data on Cr and glucose levels and ultrafiltration volume. The results of the PET are used to maximize solute clearances and fluid removal for individual patients. Limited data are available to validate the impact of the PET on outcomes such as hospitalization and mortality.

Details of PET administration are standardized.²²⁶ After morning drain, 2L of dialysate containing 2.5% dextrose (2500 mg/dL) are infused and dwelled for 4 hours, and samples of dialysate and serum are taken at time 0, 2, and 4 hours. Dialysate (D) and plasma (P) Cr and dialysate glucose are measured to calculate the Cr D/P ratio and glucose Dt/D0 ratio. The results can be

TABLE 12.43 ■ Peritoneal Equilibration Test Results Used to Determine PD Transport Type

Transport Type	D/P Creatinine 4 hr	D _{4hr} /D ₀ Glucose	PD Modality
High	0.82–1.03	<0.26	CAPD or CCPD with rapid short exchanges is needed
High average	0.66–0.81	0.26–0.38	Standard CAPD or CCPD will be effective
Low average	0.5–0.64	0.38–0.49	If residual renal CrCl >2 mL/min: standard CAPD or CCPD will be effective If residual renal CrCl <2 mL/min: high-dose CAPD or CCPD with day dwell is needed
Low	0.34–0.49	>0.49	High-dose CAPD, consider HD

CAPD, Continuous ambulatory peritoneal dialysis; CCPD, continuous cycling peritoneal dialysis; CrCl, creatinine clearance; D/P, dialysate/plasma; HD, hemodialysis; PD, peritoneal dialysis.
Adapted from Mehrotra R, Ravel V, Streja E, et al. Peritoneal equilibration test and patient outcomes. *Clin J Am Soc Nephrol*. 2015 Nov 6;10(11):1990–2001. doi:10.2215/CJN.03470315.

plotted on a graph to assess the peritoneal transport characteristics of the patient as a high or rapid transporter, average transporter, or low or slow transporter. Recent guidelines have also suggested incorporating a “sodium dip” or sodium sieving ratio to evaluate ultrafiltration adequacy.^{224,227}

The 4-hour values and suggested PD modality based membrane transport characteristics are shown in Table 12.43.²²⁵

12.23 Peritoneal Dialysis Adequacy

As with forms of HD, the minimum effective dose of PD is determined by measurements of urea clearance. Both the KDOQI guidelines²²⁸ and the International Society for Peritoneal Dialysis (ISPD) guideline²²⁹ recommend small-solute clearance targets (total weekly Kt/V urea) as the primary measure of adequacy of dialysis. It should be noted that residual kidney function is included in the total urea clearance for the PD patient; this is no longer typically done in the in-center HD population. Targets for PD adequacy are shown in Table 12.44.

However, recent practice recommendations for dialysis adequacy in PD also place far greater emphasis on the principles of a patient-centered approach with shared decision-making,²³⁰ and decreased emphasis on a specific target for small-solute clearance.²³¹ Increased emphasis is also being given to

TABLE 12.44 ■ Criteria for Peritoneal Dialysis Adequacy

Peritoneal Dialysis Adequacy
Removal of toxins (dialysis dose) can be estimated by using: 1. CrCl ≥60 L per week per 1.73 m ² body surface area (calculate using serum and dialysate creatinine measurements from a 24-hr collection of dialysate) OR 2. Weekly Kt/V urea ≥1.7–2.0 (either just dialysis clearance or a combination of dialysis clearance and residual kidney function). Calculate from http://www.kt-v.net using addition of residual renal function if UO is >100 mL/day. To calculate weekly Kt/V for peritoneal dialysis: K = clearance of the urea (not creatinine)—measured by timed urine or dialysate collection t = 10080 min (per week) V = 0.6 (for males) or 0.5 (for females) × body weight in kg
CrCl, Creatinine clearance; UO, urine output.

regular assessment of fluid status and avoidance of fluid overload. PD prescriptions may therefore need to be highly individualized. The current Kt/V target of 1.7 remains the standard, however.

Other standard targets for high-quality PD would include management of HTN and fluid overload, control of metabolic acidemia ($\text{HCO}_3^- \geq 24 \text{ mEq/L}$), and maintenance of serum levels of albumin, sodium, potassium, calcium, phosphate, and magnesium as close to normal as possible.

Nutritional status is known to have a positive impact on the treatment outcomes of patients dependent upon PD,²³² although nutritional intake and status may be challenging to assess. Inadequate protein intake may result in protein malnutrition and protein-energy wasting, a condition diagnosed by low dietary protein intake, hypoalbuminemia, unintentional weight loss, and muscle wasting. An important complication of malnutrition in PD patients includes increased susceptibility to infection. Protein is lost into the peritoneal dialysate (averaging 6–7 g/day), and peritoneal protein clearance may be a marker for inflammation, endothelial dysfunction, and mortality.²³³ Daily nutritional recommendations for the PD patient are based on international society guidelines as follows: protein 1.2 to 1.4 g/kg body weight, sodium 2 to 3 g, calcium 1000 to 1200 mg, phosphorus 1.2 to 4.0 g, and no potassium restriction.²³²

12.24 Urgent-Start Peritoneal Dialysis

Unfortunately, over half of new ESKD patients are lacking a dialysis plan at the time of dialysis initiation, with little progress in the past two decades. “Urgent-start” dialysis refers to the urgent initiation of dialysis in an ESKD patient with no preestablished functional vascular access for HD and no catheter in place for PD. Urgent-start PD in unplanned ESKD patients refers to the initiation of PD much earlier (24–72 hr after PD catheter placement) than is done with conventional PD.²³⁴ Urgent-start PD is available in many centers for late-referred patients in urgent need of dialysis but with no modality plan, who are considered otherwise to be good PD candidates. Barriers to urgent-start PD have been cited as contributing to the relatively low utilization of PD overall. It involves prompt patient education, timely catheter placement, and provision of PD with lower dwell volumes (1 L) in the recumbent position, until the patient can be trained to perform dialysis at home. Compared with non-urgent-start PD, urgent-start PD appears to be associated with similar risks of infection, hospitalization, and short-term technique and patient survival, and a slight increase in peritoneal fluid leaks not associated with adverse outcomes.²³⁴ Data from experienced urgent-start programs indicate a minor increased risk of mechanical complications such as dialysate leaks, but no impact on technical survival compared to conventional PD.²³⁵ Comparisons of PD and HD in urgent-start patients are limited, but PD is at least as good, and is associated with a lower risk of bacteremia.²³⁶

A key feature is the timely placement of the PD catheter, either by interventional radiology or surgery. A structured urgent-start program (Table 12.45.) is required to ensure implementation once the catheter is placed. It provides an alternative to urgent initiation of HD using a central venous dialysis catheter.²³⁶

12.25 Complications of Peritoneal Dialysis

Complications of acute/chronic PD include weight gain, formation of hernias, leakage of dialysis fluid into the subcutaneous tissue, catheter exit site or pleura, abdominal pain, worsening blood sugars, and low potassium levels. A common worrisome complication is the appearance of turbid PD fluid, likely related to peritonitis (Table 12.46).

12.25.1 PD PERITONITIS

PD-associated peritonitis is associated with substantial morbidity, hospitalizations, transition to HD, and mortality risk.²³⁷ The evaluation of possible peritonitis is shown in Figure 12.15.

TABLE 12.45 ■ Features of Urgent Start Peritoneal Dialysis

A. Patient education about dialysis options by the nephrologist/team
B. Evaluation of candidate patient:
• Does not need emergency dialysis
• Functional status/trainability determined by nephrologist
• No history of abdominal surgeries that would complicate PD
• No psychiatric or memory problems
• Adequate visual/hearing capability
• Satisfactory home assessment
• Family support is helpful for certain patients
C. Timely catheter placement by surgeon or interventional radiology
D. Clinical course:
• Break-in period of less than 2 weeks
• Dwell in recumbent position
• Dwell volume limited to maximum of 1 L
• Dextrose concentration determined by ultrafiltration need

PD, Peritoneal dialysis.

Adapted from Htay H, Johnson DW, Craig JC, et al. Urgent-start peritoneal dialysis versus conventional-start peritoneal dialysis for people with chronic kidney disease. *Cochrane Database Syst Rev*. 020 Dec 15;12(12): CD012913. doi:10.1002/14651858.CD012913.pub2.

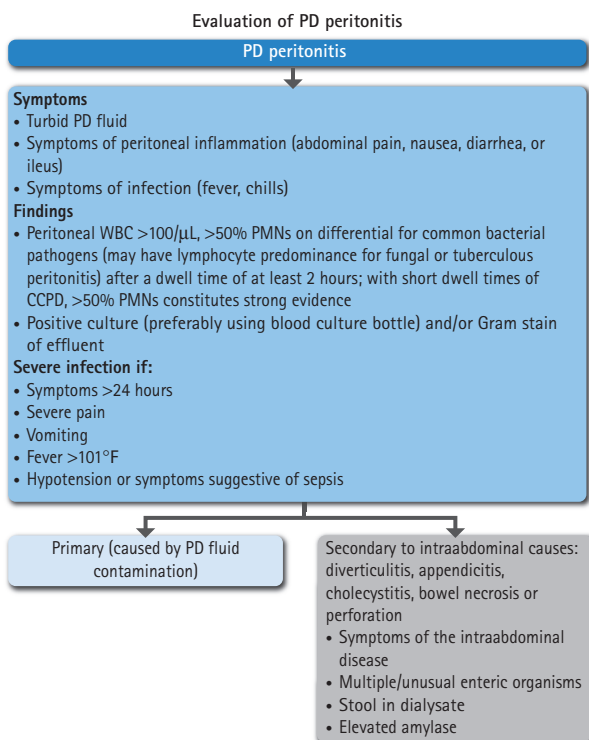
TABLE 12.46 ■ Causes of Turbid PD Fluid

• Peritonitis (culture-positive or culture-negative)
• Hemoperitoneum (hemorrhage, ovulation, menses)
• Severe constipation
• Fibrin or other proteins
• Lipids (chylous ascites)
• Prolonged dwell time
• Pancreatitis
• Dihydropyridine calcium channel blockers
• Eosinophilic peritonitis (10%–30%); may be allergic reaction to PD solutions or tubing; IP antibiotics
• Specimen taken from “dry” abdomen

IP, intraperitoneal; PD, peritoneal dialysis.

The most common cause of peritonitis is “touch contamination” of the PD catheter connection leading to infection of the dwelled peritoneal fluid. In addition, an important clinical risk factor for peritonitis is use of high peritoneal glucose loads, perhaps by impairing host defenses or damaging the peritoneal membrane itself.²³⁸

- Key measures to reduce PD-associated peritonitis include^{230,239,240}:
- prophylactic systemic antibiotics immediately before catheter insertion;
 - for CAPD treatment, a disconnect system with a “flush before fill” design;
 - adequate training of PD patients and their caregivers;
 - daily topical gentamicin 0.1% at exit site for reducing both Gram-positive and Gram-negative organisms (do not apply either to PD catheter itself) and coprescription of oral nystatin or fluconazole during treatment for PD peritonitis;
 - prompt treatment of exit-site infection;
 - antibiotic prophylaxis before dental procedures, colonoscopy, or invasive gynecological procedures; and
 - following an episode of peritonitis, retraining for PD patients.²⁴¹



CCPD, Continuous cycling peritoneal dialysis; PD, peritoneal dialysis;
PMNs, polymorphonuclear cells; WBCs, white blood cells.

Figure 12.15 Evaluation of PD peritonitis.

Whenever peritonitis is suspected, the PD effluent should be tested for cell count, differential, Gram stain, and culture. Diagnosis of peritonitis relies upon: (1) clinical features consistent with peritonitis, such as abdominal pain and/or a cloudy dialysis effluent; (2) a dialysis effluent total white cell count >100/μL, along with >50% polymorphonuclear WBCs (sample taken after a dwell time of at least 2 hr); and (3) a positive dialysis effluent culture.

Note that the WBC count in the effluent may depend on the time PD fluid has been in the abdomen. For CCPD patients with a short dwell time, more reliance should be placed on the percentage of PMNs. Patients with abdominal pain in the absence of a PD dwell may have no fluid to drain, in which case 1 L of PD solution should be instilled, dwelled for 2 hours, and then drained for sampling/testing. A blood culture bottle is the preferred technique for bacterial culture.

Once culture and sensitivity results are known, antibiotic selection should be narrowed accordingly.

Fungal peritonitis is a relatively rare but serious complication in PD patients. Risk factors include preceding bacterial peritonitis and exposure to antimicrobial agents. *Candida* species are the most frequent cause.²⁴² Several routes of contamination causing infectious peritonitis should be considered, including touch contamination, exit-site/tunnel infection, GI infection, and hematogenous infection. Mortality is unusual, the most common cause being fungal infection. Management recommendations are indicated in [Figure 12.16](#).²⁴⁰

In some cases, antibiotics for suspected and mild peritonitis cases can be initially administered at home, pending subsequent culture results of dialysate effluent. There is no widely accepted regimen. Recommendations are shown in [Table 12.47](#).^{240,243}

Evaluation and management recommendations for PD peritonitis

- Send cell count and differential, culture, and Gram stain on initial drainage; note that labs may only report neutrophil count and no total WBC, or may report all WBCs so that they must be added to determine total WBC count in order to establish a diagnosis (along with % PMNs) of peritonitis
- Start empiric antibiotics after sample is obtained:
 - Cefazolin or cephalothin 15 mg/kg of body weight or 1 g/2–3-L bag QD (in the long dwell) AND EITHER
 - Ceftazidime 10–15 mg/kg of body weight or 1 g/2–3-L bag QD (in the long dwell) OR
 - Aminoglycoside (not recommended if residual urine output >100 mL/day): tobramycin or gentamicin, 0.6 mg/kg body weight in only 1 exchange per day or amikacin 2.0 mg/kg body weight, in only 1 exchange per day (or gentamicin 3 mg/kg IV)
 - Or oral quinolone (500 mg levofloxacin $\times$ 1, then 250 mg QD for 2 weeks or ciprofloxacin 750 mg BID)
 - If secondary to an intraabdominal bowel cause, treat with metronidazole plus vancomycin, in combination with ceftazidime or an aminoglycoside
 - Note that antibiotic-containing PD solution should be allowed to dwell for at least 6 hours
 - If effluent is bloody or has fibrin strands, add heparin 500 units/L IP
 - Vancomycin and gentamicin, if used in combination, should be injected into different PD dwells
 - Concurrent prophylactic antifungal oral nystatin or fluconazole may be indicated to prevent secondary fungal peritonitis related to enteric fungal overgrowth and reduced defense due to peritonitis
 - Repeat PD fluid cell count and culture after 3 days
 - Refer patient to the ER for symptoms of severe infection
 - IV antibiotics are appropriate if patient has features of systemic sepsis; otherwise, IP route is preferred.
 - Note that vancomycin may be preferable if program has a high rate of MRSA or patient has a prior history of MRSA
 - Gram-positive cocci should be treated for 2 weeks (*Staphylococcus aureus* and Enterococci for 3 weeks); Gram-negative rods for 3 weeks

Culture results return in 24–48 hours

Culture positive:

- Gram (+) (40%–60%)
 - Enterococci: IP vancomycin for 3 weeks; add IP aminoglycoside for severe cases
 - MSSA: Cephalosporin IP + rifampin 600 mg PO for 3 weeks
 - MRSA: Vancomycin 15–30 mg/kg in 1 bag per day, repeat in 3–7 days to keep serum level >15 mcg/mL; treat 3 weeks
 - Coagulase-negative *S. aureus*: Assume methicillin resistant, so IP vancomycin for 2 weeks
 - VRE: Removal of catheter may be necessary, daptomycin 4–6 mg/kg every 48 hr IV or oral linezolid 200–300 mg once daily
 - Other: Stop aminoglycoside, continue cephalosporin for 2 weeks
- Gram (–) (15%–40%)
 - Single gram (–) non-*Xanthomonas*: Adjust ABx to sensitivity for 3 weeks
 - *Pseudomonas*: IP aminoglycoside or oral ciprofloxacin + ceftazidime or cefepime for 3 weeks, removal of catheter may be necessary
 - Multiple organisms/anaerobes: Add metronidazole 500 every 8 hr IV/PO for 3 weeks, consider evaluation for bowel perforation if not better in 72 hours
 - Note high failure rate with extended-spectrum beta-lactamase organisms

Culture negative (15%–20%)

- No improvement: Repeat culture, continue antibiotics for 2 weeks
- Improvement: Discontinue aminoglycoside, continue cephalosporin for 2 weeks
- Often caused by coagulase-negative *S. aureus*
- Evaluate for other disease: TB, other mycobacterial, fungal, or parasitic peritonitis, eosinophilic peritonitis, intraabdominal disease, renal cell or ovarian cancer, leukemia, lymphoma
- Reculture can identify the cause in some patients

Fungi (2%–5%)

- Immediate catheter removal is often necessary
- Fluconazole 150–200 mg every other day IP or amphotericin 25 mg/day IV or amphotericin 25 mg/d IV and
- Flucytosine load 2000 mg PO, maintenance 1000 mg PO for 4–6 weeks
- Change to amphotericin if not better in 72 hours
- IP route not preferred over systemic treatment
- Aspergillus: Use caspofungin
- Non-albicans candida: Use caspofungin
- Filamentous fungi: Use posaconazole or voriconazole

ABx, antibiotics; BID, twice daily; IP, intraperitoneal injection; MRSA, methicillin-resistant *Staphylococcus aureus*; MSSA, methicillin-sensitive *Staphylococcus aureus*; PD, peritoneal dialysis; PMNs, polymorphonuclear neutrophils; PO, by mouth; QD, daily; TB, tuberculosis; VRE, vancomycin-resistant enterococci; WBC, white blood cell.

Figure 12.16 Evaluation and management recommendation for PD peritonitis. (Adapted in part from Li, PK, Szeto CC, Piraino B, et al. ISPD peritonitis recommendations: 2016 update on prevention and treatment. *Perit Dial Int*. 2016 Sep 10;36[5]:481–508. doi:10.3747/pdi.2016.00078.)

TABLE 12.47 ■ “After-Hours” Initial Treatment of Suspected Peritonitis

1. Only for mild episodes (mild pain, low-grade fever, no hypotension, duration <24 hr, able to drain)
2. Patient to place cloudy effluent bag in refrigerator for culture
3. Cephalexin 500 mg orally every 12 hr with initial dose of 1000 mg OR Cefdinir 300 mg orally daily OR trimethoprim 160 mg/sulfamethoxazole 800 mg twice daily if allergic to cephalosporins
4. Report to regular dialysis facility next business day for evaluation with effluent bag

Adapted from Li P K-T, Szeto CC, Piraino B, et al. ISPD peritonitis recommendations: 2016 update on prevention and treatment. *Perit Dial Int.* 2016 Sep 10;36(5):481–508. doi:10.3747/pdi.2016.00078.

TABLE 12.48 ■ Indications for Catheter Removal

- Refractory peritonitis—failure of effluent to clear after 5 days of treatment
- Relapsing or recurrent peritonitis
- Refractory exit-site and tunnel infection
- Concomitant pseudomonal or staphylococcal aureus exit-site infection
- Unresponsive pseudomonas or fungal peritonitis
- Nontuberculous mycobacterial peritonitis
- Fecal peritonitis or other significant intraabdominal pathology
- Mechanical failure of catheter

In some cases, removal of the PD catheter is recommended (Table 12.48). Note that refractory peritonitis is now defined as failure of the effluent to clear after 5 days of appropriate antibiotics. Reinsertion of the catheter and resumption of PD should only be attempted after complete resolution of symptoms and a holiday period of at least 2 weeks after catheter removal.

12.25.2 PD TECHNIQUE FAILURE

One reason for relatively low PD utilization is attrition off of PD due to technique failure, related to either patient factors or dialysis center factors.²⁴⁴ Technique failure rates have improved, decreasing to roughly 25% of patients treated per year in death-censored analyses. Smaller centers with less cumulative PD experience may have higher rates. Technique failure may take the form of inadequate solute clearance (weekly Kt/V urea of <1.7) or of ultrafiltration insufficiency. Insufficient ultrafiltration may be suspected when the net ultrafiltration from a 4-hour PET using 4.25% dextrose is <400 mL, or with 2.5% dextrose <100 mL.²²⁴ Icodextrin has been associated with improved technique survival, including in those with diabetes.²²¹ Causes of ultrafiltration failure are shown in Table 12.49.

Whether at home or in the hospital, the patient’s PD catheter may require evaluation for catheter malfunction. Following are common types of catheter malfunction requiring assessment:

- Mechanical failure of either dialysate inflow or outflow is not uncommon. Inflow failure is uncommon and may be due to obstruction within the catheter (fibrin strands or kinking of

TABLE 12.49 ■ Causes of Ultrafiltration Failure

- Peritonitis
- Adhesions/loculation
- Malpositioned catheter
- Poor peritoneal blood flow

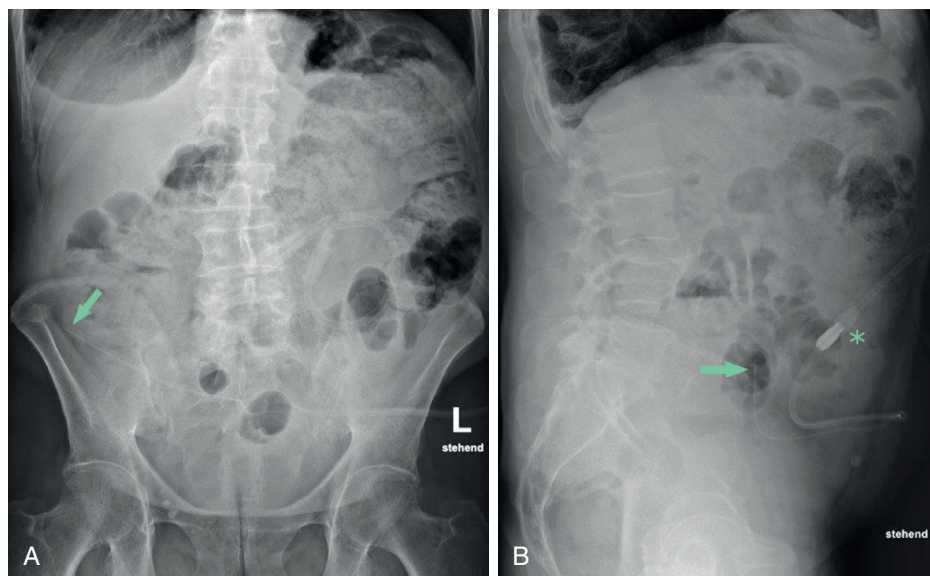


Figure 12.17 Location of peritoneal dialysis catheter on plain film of the abdomen. (A) Anteroposterior view demonstrates malposition of the catheter with its tip located in the mid-abdomen, close to the right iliac crest (arrow). (B) Lateral view depicts the external part of the catheter with the titanium adaptor (asterisk) and the intraabdominal portion turned up (arrow). The radio-opaque stripe is clearly visible and allows localization of the catheter. (From Windpessl M, Prischl FC, Prenner A, Vychytil A. Managing hospitalized peritoneal dialysis patients: ten practical points for the non-nephrologist. *Am J Med.* 2021 Jul;134[7]:833–839. doi:10.1016/j.amjmed.2021.02.007.)

the catheter) or outside the catheter (externally by a clamp inadvertently left closed) or internally (such as entrapment in omentum). Outflow failure is relatively common and may be defined as failure to drain the dwelling fluid within 45 minutes. It may be preceded by sluggish outflow, or may occur without warning.

- The most common cause of catheter malfunction is constipation. Other causes are intraperitoneal adhesions, omental wrapping, catheter tip migration, luminal obstruction due to fibrin/clot, and catheter kinking. The latter two may be accompanied by two-way obstruction.
- Evaluation should proceed from conservative to more aggressive interventions.
 - a. Treat constipation aggressively.
 - b. Rule out bladder distension causing extrinsic catheter compression.
 - c. Evaluate catheter for location and kinking. On plain film, catheter location may be identified as a radiopaque stripe, preferentially located in the pelvic portion of the peritoneal cavity, as shown in [Figure 12.17](#). Note that some cases may require a CT scan for visualization.
 - d. Peritonitis should be ruled out.
 - e. An empiric attempt to restore patency should be made, by a trained nurse using aseptic technique. Initially, attempt to repeatedly flush the catheter with 20 to 50 mL of heparinized 0.9% saline. If flushing is unsuccessful, fibrinolytic therapy with tPA may be attempted^{218,245}; 8 mL of tPA (1 mg/mL) is slowly injected into the catheter. After a 1- to 2-hour dwell, attempt to drain PD fluid by gentle aspiration and flushing of the

- catheter. If partially successful, this can be repeated. If flow is restored, 500 units of heparin should be added to each liter of PD fluid subsequently.
- f. Laparoscopy to establish a diagnosis and provide a rescue procedure to salvage the catheter may ultimately be required.
 - g. When these steps have failed, catheter replacement may be done simultaneously with removal of the nonfunctioning peritoneal catheter.

12.25.3 ACID-BASE AND ELECTROLYTE DISTURBANCES IN PERITONEAL DIALYSIS

Multiple abnormalities in acid-base balance and electrolytes may occur in the patient treated with PD, and may be associated with adverse outcomes.²⁴⁶ They sometimes provide limitations to the use of PD and may be associated with morbidity and even mortality.^{247–249} Common laboratory abnormalities are shown in [Table 12.50](#).

12.25.4 HEMOPERITONEUM

New onset of bloody peritoneal fluid is relatively uncommon and has multiple etiologies. Causes of intraperitoneal hemorrhage may be divided based on the severity and the underlying cause ([Figure 12.18](#)). In some cases, the cause is uncertain.²⁵⁰ The diagnostic approach includes peritoneal fluid analysis and culture, coagulation studies, abdominal US, CT, diagnostic paracentesis, and laparoscopy/laparotomy. The method of management depends on the source of blood loss.²⁵¹

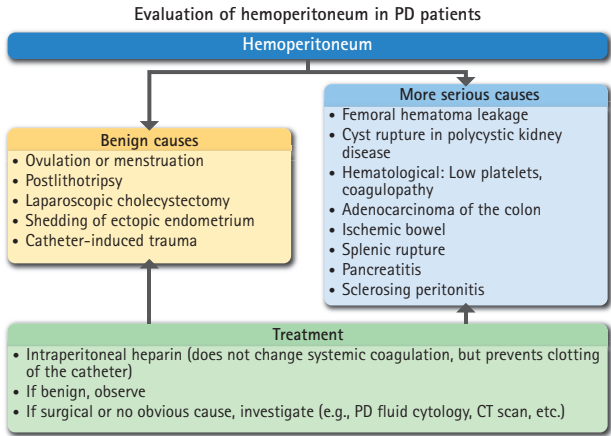
12.25.5 OTHER COMPLICATIONS OF PD

With the decline in PD peritonitis rates, other complications have become relatively more common ([Table 12.51](#)). They may account for cases of PD technique failure.²⁵² Some require surgical management.

TABLE 12.50 ■ Laboratory Disturbances in the PD Patient

	Mechanism	Correction
Hypnatremia	Hypertonic dialysate ⇒ water shift into peritoneal space (removing water in excess of Na)	Water orally or D5W IV or lower glucose or Na level in dialysate
Hyponatremia	Low Na intake, excessive thirst, renal or stool losses, inadequate UF	Salt intake must be proportional to volume loss induced by dialysis, increase hypertonic dialysate if water overload
Hyperkalemia	High K ⁺ intake with low renal excretion and PD clearance, extracellular shift of K ⁺ due to low insulin or drugs	Higher dialysis clearance, standard treatment of acute hyperkalemia
Hypokalemia	Excessive K ⁺ clearance by PD, occurs in 60% of ESKD patients on PD	10%–30% of patients need K ⁺ supplement
Lactic acidosis	Conversion of lactate to bicarbonate can be affected in sepsis or by metformin	Replace lactate buffer with bicarbonate in dialysate

ESKD, End-stage kidney disease; *PD*, peritoneal dialysis; *UF*, ultrafiltration.



PD, Peritoneal dialysis.

Figure 12.18 Evaluation of hemoperitoneum in PD patient.

TABLE 12.51 ■ Other Complications in the PD Patient

Problem	Diagnosis	Treatment
Hernia (↑ intraabdominal pressure)	Clinical exam or CT scan	Surgical repair; corsets; dialysis with lower intraabdominal pressure (APD, eliminate daytime dwell or decrease volume)
Genital edema (<10% CAPD patients); tracking of PD fluid to scrotum/labia	Clinical exam, decreased PD fluid effluent return, US/CT scan	Stop PD and use temporary HD; low-volume APD at bed rest; further treatment depends on source of leak
Abdominal wall leak	Clinical exam, decreased PD fluid effluent return, US/CT scan	Stop PD, or convert to low-volume APD at bed rest; avoid sitting position; if severe, convert to temporary HD; consider catheter replacement or injecting fibrin glue (1 mL of a solution of fibrinogen and thrombin)
Hydrothorax/pleural effusion (incidence <5%)	Dyspnea, no improvement with hypertonic exchange, decreased PD fluid effluent return, diagnostic CXR with pleural effusion, thoracentesis with pleural fluid analysis showing high glucose, scan with isotope in abdomen	Stop PD and use temporary HD; thoracentesis; low-volume PD (after 2 weeks of HD may return to PD); pleurodesis (fibrin glue, autologous blood, talc, tetracycline); surgical repair
Sclerosing encapsulating peritonitis	Recurrent abdominal pain with fills, repeated peritonitis predisposes to it, may cause bowel obstruction, or hemoperitoneum, decreased solute and water transport, characteristic appearance with CT; rarely, reported with use of icodextrin	Careful attention to nutrition and bowel function; laparoscopy; surgical intervention; antiinflammatory or immunosuppressive meds (controversial); tamoxifen

TABLE 12.51 ■ Other Complications in the PD Patient (Continued)

Problem	Diagnosis	Treatment
Glucose intolerance	Hemoglobin A1c	Minimize use of high-dextrose concentration PD solutions; use of icodextrin for glucose-sparing
Pneumoperitoneum	Radiology imaging	Consider benign incidental cause, allowing air entry into PD catheter; assess for pathological abdominal cause (i.e., bowel perforation)

APD, ambulatory peritoneal dialysis; *CAPD*, continuous ambulatory peritoneal dialysis; *CXR*, chest X-ray; *HD*, hemodialysis; *PD*, peritoneal dialysis; *US*, ultrasound.

12.26 ESKD Peritoneal Dialysis Outcomes

PD outcomes in ESKD patients from randomized controlled trials are sparse, measured inconsistently, and limited by surrogate endpoints that have not been validated.²⁵³ High risk of infections, inadequate clearance of small solutes, and eventual ultrafiltration failure were the basis for past concerns that PD was an inferior modality compared to HD. However, virtually all recent studies indicate that PD and in-center HD have similar short (1–2 years) or medium-term (up to 5 years) outcomes.²⁵⁴ Metabolic factors potentially exacerbated by glucose-based dialysate solutions include dyslipidemia, insulin resistance/metabolic syndrome, and weight gain, but the effect on overall outcomes is unclear. PD is associated with better preservation of residual kidney function, which is associated with improved survival.²⁵⁵

Improvement in patient-reported outcomes is especially important. Hard outcomes such as mortality and technique failure are underreported in trials, and patient-important outcomes are even more limited. The Standardized Outcomes in Nephrology-Peritoneal Dialysis (SONG-PD) and other initiatives are seeking to validate relevant, meaningful PD outcomes in future trials with a goal of improving shared decision-making²⁵⁶ (i.e., outcomes including PD-related infection, cardiovascular disease, mortality, technique survival, and quality of life).

Medication Management in ESKD

Both the effectiveness and the toxicity of drugs are commonly related to their concentrations in the body. Kidney dysfunction influences the pharmacokinetic parameters of at least 50% of drugs, and is also important in drug pharmacodynamics (the biochemical and physiological effects of the agent). Not only renal excretion, but also drug bioavailability and absorption, protein binding, and metabolism may be altered with impaired kidney function.

Changes in pharmacokinetics in ESKD can predispose the patient to either over- or underdosing. Dose-related drug toxicity or therapeutic failure may result.²⁵⁷ For example, when the plasma half-life of a drug is prolonged in the setting of ESKD, the time to reach the steady-state concentration may increase proportionately, so administration of a loading dose may achieve therapeutic levels more quickly.²⁵⁷

The required change in dosing regimen for drugs in ESKD can in some cases be established through the application of pharmacokinetic principles.²⁵⁸ In this case, drugs that are largely eliminated by the kidneys with normal renal function commonly require a dose adjustment (either by decreased dose [with more consistent but possibly toxic levels] or increased interval [with risk of subtherapeutic trough levels]) in patients with impaired kidney function or dialysis dependence. In general, dosage reduction is indicated when over one-third of the drug or active

metabolite appears unchanged in the urine, although a normal loading dose may be given even in CKD or ESKD patients. Drug elimination tends to correlate with GFR, although in many cases proximal tubular secretion is the primary mechanism of drug elimination.

Clearance of drugs and pharmacokinetics during HD are complex. Both dialysis properties (Table 12.52) and drug properties (Table 12.53) affect dosing regimens for many drugs in ESKD patients.

Attributes that would reduce the likelihood of clearance through dialysis include large molecular weight, large volume of distribution, high level of protein binding, and clearance through nonrenal mechanisms. It is important to estimate dialysis clearance of medications to adequately adjust the dose. That is especially true in drugs with a relatively narrow therapeutic window (e.g., lithium, chemotherapy, antibiotics). The complexity of the process has to do with multiple factors affecting drug clearance in dialysis; specifically, the characteristics of the dialyzer, dialysis procedure, dialysate, and properties of the drug itself. Clearance mechanisms may involve both diffusion and convection, and, rarely, adsorption to the dialysis membrane. Many drugs are small enough to be removed by dialysis so that the major determinant of drug removal is actually protein binding, since only unbound drug is removed. The term *sieving coefficient* describes how freely a solute crosses the dialysis membrane. Using these considerations, only very crude approximations of the clearance is possible in the absence of experimental data. Such experimental data are obtained for some, but not all, drugs (e.g., vancomycin²⁵⁹). Clearance of drugs by PD is generally less than with HD, but continues on a daily basis, making weekly drug dosing often comparable between the two modalities.²⁶⁰

Therefore, monitoring of drug levels is important, especially when the therapeutic target range is known.²⁶¹ During CRRT, for example, vancomycin levels are known to be frequently subtherapeutic.²⁵⁹ Drug removal by dialysis may necessitate a supplemental dose following the procedure. For some specific drugs and toxins, dialysis is valuable in treating

TABLE 12.52 ■ Dialysis Properties That Affect Drug Clearance

• Dialyzer properties (pore size, surface area, type of membrane might affect binding) • Dialysis procedure properties (blood flow rate, dialysate flow rate, ultrafiltration rate) • Dialysate properties (solute concentration, pH, temperature) • Duration of dialysis treatment

TABLE 12.53 ■ Drug Properties That Affect Dialytic Clearance

• Molecular size (Da) • <1000 Da—small molecules—diffusion-dependent transport. Most drugs are <500 Da. • 1000–2000 Da—only convective transport. Example: vancomycin (1448 Da). • >2000 Da—partially reflected by membrane even during UF, so clearance is low. • Increased protein binding reduces clearance; heparin increases free fraction of many drugs (free fraction = 1-Protein-bound fraction). Kidney failure tends to decrease protein binding of drugs. • VOD—the greater the VOD, the less the dialyzability. An estimate of all the compartments in which the drug may be found. VOD increased with lipid solubility and low plasma protein binding. • 1 L/kg BW—distribution volume—likely removed by dialysis. • 1–2 L/kg BW—distribution volume—marginal clearance by dialysis. • >2 L/kg BW—unlikely to be effectively removed by dialysis. • Multicompartmental distribution leads to dramatic rebound after dialysis. • Molecular charge decreases dialyzability. • Water or lipid solubility: poor dialyzability of lipid-soluble compounds. • Dialyzer membrane binding increases clearance of the compound.

BW, Body weight; UF, ultrafiltration; VOD, volume of distribution.

TABLE 12.54 ■ Reference Sources for Dosing Adjustments in Patients With ESKD

MICROMEDEX Healthcare Series	www.micromedexsolutions.com
UPTODATE	www.uptodate.com/online/index.do
EPOCRATES	www.epocrates.com
National Kidney Disease Education Program	www.nkdep.nih.gov
FDA MedWatch	http://fda.gov/medwatch
The Medical Letter on Drugs and Therapeutics	http://secure.medicalletter.org
Drug Prescribing in Renal Failure: Dosing Guidelines for Adults, 5th Edition	www.acponline.org
<i>ESKD</i> , End-stage kidney disease.	

overdoses/intoxications, particularly IHD, which may exceed the clearance by normal kidney function in some cases.

Recent guidelines have addressed vancomycin dosing in dialysis patients, with important caveats.²⁶² Instead of trough-only dosing (based on the premise that the trough serum level is an adequate efficacy target), new recommendations make use of the serum concentration-time curve/minimum inhibitory concentration ratio. However, data justifying a specified area under the serum concentration curve (AUC)/minimum inhibitory concentration (MIC) ratio in the dialysis patient are lacking, and the approach presents important logistical issues. It is well recognized that numerous medications require dosage adjustment, or even cessation, in patients with renal failure, so the use of drug references is essential.

Because of the panoply of medications used in the management of the ESKD patient, it is important to have available reference sources to guide such decision-making. Several are listed in Table 12.54.

12.27 Extracorporeal Treatments for Intoxication

About 20% of reported poison exposures are intentional, and they are far more serious than unintentional exposures. The Extracorporeal Treatments In Poisoning (EXTRIP) workgroup has developed consensus guidelines on the role of extracorporeal treatment in patients suffering from accidental or intentional intoxications.²⁶³ Additional considerations are supportive care, antidotes (such as naloxone for acute opioid intoxication), and corporeal elimination by activated charcoal or urine alkalinization. Over 50 poison control centers in the United States provide telephone or online guidance for human poison exposures.

When indicated, rapid introduction of effective dialysis is critical in improving outcomes. The major determinants of the benefit of extracorporeal treatments include the toxin's molecular weight, protein binding, volume of distribution, and endogenous clearance.²⁶⁴ The utility of extracorporeal modalities in poisoning is shown in Table 12.55.²⁶⁵

Removal by dialysis is most efficient for agents with a low volume of distribution, limited protein binding, and low molecular weight. In the majority of indications, IHD is preferred. If dialysis is used to treat an acute intoxication and AKI is not present, replacement of potassium and phosphorus will likely be necessary.

The role for charcoal hemoperfusion has been markedly reduced, particularly when drug removal can be achieved by dialysis. A list of when extracorporeal therapy for various toxins is advised can be found at <https://www.extrip-workgroup.org/recommendations>.

TABLE 12.55 ■ Role of Extracorporeal Modalities in Poisoning

Modality	Toxin Molecular Mass (Da)	Toxin VOD (L/kg)	Protein Binding of Toxin	Examples of Toxins Amenable to Therapy	Primary Limitations of Therapy
Hemodialysis	Up to 10,000–15,000	≤1.5–2	≤80%	Salicylates, toxic alcohols, lithium, metformin	Hemodynamic stability
HCO filter HD	Up to 50,000	≤1.5–2	≤80%	Small-peptide therapeutics; any therapy amenable to HD	Limited availability Limited role in poisoning
CRRT	Up to 15,000–25,000	≤1.5–2	≤80%	Lithium	Slow toxin clearance (except for toxins with slow redistribution)
Hemoperfusion	Unclear, but high	≤1 L/kg	Any	Valproic acid, carbamazepine	Limited availability Clotting Hypocalcemia
Plasma exchange	No limit	≤1 L/kg	Any	Monoclonal antibodies, arsine	Limited availability Very slow clearance

HCO, High-molecular-mass cutoff; *HD*, hemodialysis; *CRRT*, continuous renal replacement therapy; *VOD*, volume of distribution
Adapted from King JD, Kern MH, Jaar BG. Extracorporeal removal of poisons and toxins. *Clin J Am Soc Nephrol*. 2019 Sep 6;14(9):1408–1415. doi:10.2215/CJN.02560319.

If the intoxicant has a large volume of distribution, HD can be used to reduce the initial intravascular burden of the agent. CRRT can be especially useful for intoxications for drugs with pharmacokinetics that lead to a “rebound” increase in blood levels when rapidly removed by HD, such as lithium and N-acetyl procainamide. CRRT can avoid the rebound that follows discontinuation of dialysis, while continuing the drug removal process.

Lithium remains one of the poisons where extracorporeal removal is most often needed. Dialysis is recommended when kidney function is impaired and the serum Li+ level is >4.0 mEq/L, or in the presence of decreased consciousness, seizures, or life-threatening dysrhythmias regardless of the serum level.²⁶⁶ While CRRT may be a useful alternative, HD is the most effective way to rapidly reduce toxic serum levels and brain exposure in poisoned patients.

For metformin-associated lactic acidosis, HD is recommended in cases of marked serum lactate elevation, severe acidemia, and decreased level of consciousness.²⁶⁷ Alternatively, SLED has been shown to improve outcomes.

12.28 Therapeutic Apheresis

The nephrology service may be consulted to determine the value of therapeutic apheresis (TA), or in some programs be asked to perform TA for various indications.²⁶⁸ TA is a process that removes a circulating pathogenic solute substance from the plasma and replaces it with allogeneic plasma (therapeutic plasma exchange) or with albumin. Examples may include autoantibodies, paraproteins, Ag-Ab complexes, nonimmunological proteins, endotoxin, or exogenous toxins. The nephrologist may be involved in patient consultation, or in some cases may manage the TA process itself.⁶⁹ The 2019 *Journal of Clinical Apheresis* guidelines address several diseases managed by the nephrologist and provide ratings of clinical efficacy of TA.²⁶⁹

TABLE 12.56 ■ Potential Renal Indications for Therapeutic Apheresis

1. Anti-GBM diseases (dialysis independence or diffuse alveolar hemorrhage)
2. Thrombotic thrombocytopenic purpura
3. Kidney transplant antibody-mediated rejection
4. Kidney transplant desensitization (ABO incompatible)
5. Symptomatic, severe cryoglobulinemia
6. Prevention of recurrent focal segmental glomerulosclerosis in transplant
7. HUS—Factor H antibodies
8. Drug-associated thrombotic microangiopathy—Ticlopidine, quinine, quetiapine, and gemcitabine, but effect of TA in drug-associated TMA is questionable.^a

^aGeorge JN, Nester CM. Syndromes of thrombotic microangiopathy. *N Engl J Med*. 2014 Aug 14;371(7):654–666. doi:10.1056/NEJMr1312353.

GBM, Glomerular basement membrane; HUS, hemolytic uremic syndrome; TA, therapeutic apheresis; TMA, thrombotic microangiopathy.

Adapted from Williams ME, Balogan RA. Principles of separation: indications and therapeutic targets for plasma exchange. *Clin J Am Soc Nephrol*. 2014 Jan;9(1):181–190. doi:10.2215/CJN.04680513.

Plasma separation may be achieved through either centrifugation or a filter membrane. Nephrologists are more likely to be familiar with membrane-based filter technology, which may be adapted to conventional HD or hemofiltration. The TA membrane's pores are large enough to allow removal of large circulating molecules while retaining cellular components.²⁶⁸ A typical TA schedule is every other day, with a volume exchange of 1.0 to 1.5 plasma volumes. In anti-glomerular basement membrane (anti-GBM) disease, daily treatment is required. Albumin is the most common nonplasma replacement fluid and has a superior side-effect profile. Allogeneic plasma may be needed to provide proteins or other factors, such as a source of ADAMTS13 in thrombotic thrombocytopenic purpura (TTP).

Potential renal indications for therapeutic apheresis are shown in Table 12.56.^{69,269–270}

However, note that while the benefit of TA in antineutrophilic cytoplasmic autoantibody (ANCA) vasculitis was suggested by the MEPEX trial (if serum Cr >5.7 mg/dL) and by a subsequent meta-analysis, the recent PEVIXAS randomized trial showed no renal benefit over steroids, and TA was also not beneficial for pulmonary hemorrhage.²⁷¹ A recent report suggested a role for TA in patients with septic shock/multiple organ failure.²⁷²

References

1. Mehrotra R. Choice of dialysis modality. *Kidney Int*. 2011;80(9):909–911. doi:10.1038/ki.2011.262.
2. Holley JL. We offer renal replacement therapy to patients who are not benefitted by it. *Semin Dial*. 2016;29(4):306–308. doi:10.1111/sdi.12492.
3. Da Silva-Gane M, Wellsted D, Greenshields H, Norton S, Chandna SM, Farrington K. Quality of life and survival in patients with advanced chronic kidney failure managed conservatively or by dialysis. *Clin J Am Soc Nephrol*. 2012;7(12):2002–2009. doi:10.2215/CJN.01130112.
4. Moss AH. Ethical principles and processes guiding dialysis decision-making. *Clin J Am Soc Nephrol*. 2011;6(9):2313–2317. doi:10.2215/CJN.03960411.
5. Granado R C-D, Mehta RL. Withholding and withdrawing renal support in acute kidney injury. *Semin Dial*. 2011;24(2):208–214. doi:10.1111/j.1525-139X.2011.00832.x.
6. Patel SS, Holley JL. Withholding and withdrawing dialysis in the intensive care unit: benefits derived from consulting the Renal Physicians Association/American Society of Nephrology clinical practice guideline, shared decision-making in the appropriate initiation of and withdrawal from dialysis. *Clin J Am Soc Nephrol*. 2008;3(2):587–593. doi:10.2215/CJN.04040907.

7. Nash DM, Przech S, Wald R, O'Reilly D. Systematic review and meta-analysis of renal replacement therapy modalities for acute kidney injury in the intensive care unit. *J Crit Care*. 2017;41:138–144. doi:10.1016/j.jcrc.2017.05.002.
8. Schefold JC, von Haehling S, Pischowski R, et al. The effect of continuous versus intermittent renal replacement therapy on the outcome of critically ill patients with acute renal failure (CONVINT): a prospective randomized controlled trial. *Crit Care*. 2014;18(1):R11. doi:10.1186/cc13188.
9. Liang KV et al. Modality of RRT and recovery of kidney function after AKI in patients surviving to hospital discharge. *Clin J Am Soc Nephrol*. 2016;11(1):30–38. doi:10.2215/CJN.01290215.
10. Cullis B, Al-Hwiesh A, Kilonzo K, et al. ISPD guidelines for peritoneal dialysis in acute kidney injury: 2020 update (adults). *Perit Dial Int*. 2021;41(1):15–31. doi:10.1177/0896860820970834.
11. Gabriel DP, Caramori JT, Martim LE, Barretti P, Balbi AL. High volume peritoneal dialysis vs daily hemodialysis: a randomized, controlled trial in patients with acute kidney injury. *Kidney Int Suppl*. 2008;(108):S87–S93. doi:10.1038/sj.ki.5002608.
12. Al-Hwiesh A, Abdul-Rahman I, Finkelstein F, et al. Acute kidney injury in critically ill patients: a prospective randomized study of tidal peritoneal dialysis versus continuous renal replacement therapy. *Ther Apher Dial*. 2018;22(4):371–379. doi:10.1111/1744-9987.12660.
13. Granado R C-D, Mehta RL. Withholding and withdrawing renal support in acute kidney injury. *Semin Dial*. 2011;24(2):208–214. doi:10.1111/j.1525-139X.2011.00832.x.
14. Patel SS, Holley JL. Withholding and withdrawing dialysis in the intensive care unit: benefits derived from consulting the renal physicians association/American Society of Nephrology clinical practice guideline, shared decision-making in the appropriate initiation of and withdrawal from dialysis. *Clin J Am Soc Nephrol*. 2008;3(2):587–593. doi:10.2215/CJN.04040907.
15. Zarbock A, Mehta RL. Timing of kidney replacement therapy in acute kidney injury. *Clin J Am Soc Nephrol*. 2019;14(1):147–149. doi:10.2215/CJN.08810718.
16. Sohaney R, Yessayan LT, Heung M. Towards consensus in timing of kidney replacement therapy for acute kidney injury. *Am J Kidney Dis*. 2021;77(4):542–545. doi:10.1053/j.ajkd.2020.08.004.
17. Gaudry S, Hajage D, Schortgen F, et al. Initiation strategies for renal-replacement therapy in the intensive care unit. *N Engl J Med*. 2016;375(2):122–133. doi:10.1056/NEJMoa1603017.
18. STARRT-AKI Investigators, Canadian Critical Care Trials Group, Australian and New Zealand Intensive Care Society Clinical Trials Group, et al. Timing of initiation of renal-replacement therapy in acute kidney injury. *N Engl J Med*. 2020;383(3):240–251. doi:10.1056/NEJMoa2000741.
19. Gaudry S, Hajage D, Martin-Lefevre L, et al. Comparison of two delayed strategies for renal replacement therapy initiation for severe acute kidney injury (AKIKI-2): a multicentre, open-label, randomized, controlled trial. *Lancet*. 2021;397(10281):1293–1300. doi:10.1016/S0140-6736(21)00350-0.
20. Zarbock A, Kellum JA, Schmidt C, et al. Effect of early vs delayed initiation of renal replacement therapy on mortality in critically ill patients with acute kidney injury: the ELAIN randomized clinical trial. *JAMA*. 2016;315(20):2190–2199. doi:10.1001/jama.2016.5828.
21. Shiao C-C, Wu V-C, Li WY, et al. Late initiation of renal replacement therapy is associated with worse outcomes in acute kidney injury after major abdominal surgery. *Crit Care*. 2009;13(5):R171. doi:10.1186/cc8147.
22. Gaudry S, Hajage D, Benichou N, et al. Delayed versus early initiation of renal replacement therapy for severe acute kidney injury: a systematic review and individual patient data meta-analysis of randomized clinical trials. *Lancet*. 2020;395(10235):1506–1515. doi:10.1016/S0140-6736(20)30531-6.
23. Viallet N, Brunot V, Kuster N, et al. Daily urinary creatinine predicts the weaning of renal replacement therapy in ICU acute kidney injury patients. *Ann Intensive Care*. 2016;6(1):71. doi:10.1186/s13613-016-0176-y.
24. KDIGO Clinical Practice Guideline for Acute Kidney Injury. *Kidney Internat. Suppl*. 2012;2(1):1–138. <https://kdigo.org/wp-content/uploads/2016/10/KDIGO-2012-AKI-Guideline-English.pdf>.
25. The VA/NIH Acute Renal Failure Trial Network, Palevsky PM, Hongyuan Zhang J, et al. Intensity of renal support in critically ill patients with acute kidney injury. *N Engl J Med*. 2008;359(1):7–20. doi:10.1056/NEJMoa0802639.
26. Liang KV, Zhang JH, Palevsky PM. Urea reduction ratio may be a simpler approach for measurement of adequacy of intermittent hemodialysis in acute kidney injury. *BMC Nephrol*. 2019;20(1):82. doi:10.1186/s12882-019-1272-7.
27. Macedo E, Mehta RL. Continuous dialysis therapies: core curriculum 2016. *Am J Kidney Dis*. 2016; 68(4):645–657. doi:10.1053/j.ajkd.2016.03.427.

28. Tolwani A. Continuous renal-replacement therapy for acute kidney injury. *N Engl J Med*. 2012; 367(26):2505–2514. doi:10.1056/NEJMct1206045.
29. Ghoshal S, Parikh A. Dialysis-associated neurovascular injury (DANI) in acute brain injury. Practical considerations for intermittent dialysis in the Neuro-ICU. *Clin J Am Soc Nephrol*. 2021;16(7): 1110–1112. doi:10.2215/CJN.15000920.
30. Rosner MH, Connor MJ Jr. Management of severe hyponatremia with continuous renal replacement therapies. *Clin J Am Soc Nephrol*. 2018;13(5):787–789. doi:10.2215/CJN.13281117.
31. Neyra JA, Yessayan L, Thompson Bastin ML, Wille KM, Tolwani AJ. How to prescribe and troubleshoot continuous renal replacement therapy: a case-based review. *Kidney360*. 2020;2(2):371–384. doi:10.34067/KID.0004912020.
32. Ting S, Chua H-R, Cove ME. Euglycemic ketosis during continuous kidney replacement therapy with glucose-free solution: a report of 8 cases. *Am J Kidney Dis*. 2021;78(2):305–308. doi:10.1053/j.ajkd.2020.10.014.
33. Hatamizadeh P, Tolwani A, Palevsky P. Revisiting filtration fraction as an index of the risk of hemofilter clotting in continuous venovenous hemofiltration. *Clin J Am Soc Nephrol*. 2020;15(11):1660–1662. doi:10.2215/CJN.02410220.
34. Thompson AJ. Drug dosing during continuous renal replacement therapies. *J Pediatr Pharmacol Ther*. 2008;13(2):99–113. doi:10.5863/1551-6776-13.2.99.
35. Zarbock A, Küllmar M, Kindgen-Milles D, et al. Effect of regional citrate anticoagulation vs systemic heparin anticoagulation during continuous kidney replacement therapy on dialysis filter life span and mortality among critically ill patients with acute kidney injury: a randomized clinical trial. *JAMA*. 2020;324(16):1629–1639. doi:10.1001/jama.2020.18618.
36. Szamosvalvi B, Yessayan LT, Heung M. Citrate anticoagulation for continuous kidney replacement therapy: an embarrassment of RICH-es. *Am J Kidney Dis*. 2021;78(1):146–150. doi:10.1053/j.ajkd.2021.01.005.
37. Cervantes CE, Menez S, Hanounch M. CKRT clotting and cerebrovascular accident in a critically ill patient. *Kidney360*. 2020;1(7):718–719. doi:10.34067/KID.0003112020.
38. Dickie H, Tovey L, Berry W, Ostermann M. Revised algorithm for heparin anticoagulation during continuous renal replacement therapy. *Crit Care*. 2015;19:376. doi:10.1186/s13054-015-1099-y.
39. Mottes TA, Goldstein SL, Basu RK. Process based quality improvement using a continuous renal replacement therapy dashboard. *BMC Nephrol*. 2019;20(1):17. doi:10.1186/s12882-018-1195-8.
40. Gautam SC, Srialluri N, Jaar BG. Strategies for continuous renal replacement therapy de-escalation. *Kidney 360*. 2021;2(7):1166–1169. doi:10.34067/KID.0000912021
41. Ricci Z, Romagnoli S, Ronco C. Acute kidney injury: to dialyse or to filter? *Nephrol Dial Transplant*. 2020;35(1):44–46. doi:10.1093/ndt/gfz022.
42. Yoshida T, Komaru Y, Matsuura R, et al. Findings from two large randomized controlled trials on renal replacement therapy in acute kidney injury. *Ren Replace Ther*. 2016;2:13–20. doi:10.1186/s41100-016-0027-1.
43. Bellomo R, Cass A, Cole L, et al. Intensity of continuous renal-replacement therapy in critically ill patients. *N Engl J Med*. 2009;361(17):1627–1638. doi:10.1056/NEJMoa0902413.
44. Jun M, Lambers Heerspink HJ, Ninomiya T, et al. Intensities of renal replacement therapy in acute kidney injury: a systematic review and meta-analysis. *Clin J Am Soc Nephrol*. 2010;5(6):956–963. doi:10.2215/CJN.09111209.
45. Kovacs B, Sullivan KJ, Hiremath S, Patel RV. Effect of sustained low efficient dialysis versus continuous renal replacement therapy on renal recovery after acute kidney injury in the intensive care unit: a systematic review and meta-analysis. *Nephrology (Carlton)*. 2017;22(5):343–353. doi:10.1111/nep.13009.
46. Zhang L, Yang J, Eastwood GM, Zhu G, Tanaka A, Bellomo R. Extended daily dialysis versus continuous renal replacement therapy for acute kidney injury: a meta-analysis. *Am J Kidney Dis*. 2015;66(2):322–330. doi:10.1053/j.ajkd.2015.02.328.
47. Chionh CY, Ronco C, Finkelstein FO, Soni SS, Cruz DN. Acute peritoneal dialysis: what is the “adequate” dose for acute kidney injury? *Nephrol Dial Transplant*. 2010;25(10):3155–3160. doi:10.1093/ndt/gfq178.
48. Cullis B, Al-Hwiesh A, Kilonzo K, et al. ISPD guidelines for peritoneal dialysis in acute kidney injury: 2020 update (adults). *Perit Dial Int*. 2021;41(1):15–31. doi:10.1177/0896860820970834.
49. Srivatanav V, Aggarwal V, Finkelstein FO, Najlayan M, Crabtree JH, Perl J. Peritoneal dialysis for acute kidney injury treatment in the United States: brought to you by the COVID-19 pandemic. *Kidney360*. 2020;1(5):410–415. doi:10.34067/KID.0002152020.

50. Chen W, Caplin N, El Shamy O, et al. NYC-PD Consortium. Use of peritoneal dialysis for acute kidney injury during the COVID-19 pandemic in New York City: a multicenter observational study. *Kidney Int.* 2021;100(1):2–5. doi:10.1016/j.kint.2021.04.017.
51. Rastogi A, Lerma EV. Anemia management for home dialysis including the new US public policy initiative. *Kidney Int Suppl* (2011). 2021;11(1):59–69. doi:10.1016/j.kisu.2020.12.005.
52. Chertow GM, Levin NW, Beck GJ, et al. Long-term effects of frequent in-center hemodialysis. *J Am Soc Nephrol.* 2016;27(6):1830–1836. doi:10.1681/ASN.2015040426.
53. Tentori F, Zhang J, Li Y, et al. Longer dialysis session length is associated with better intermediate outcomes and survival among patients on in-center three times per week hemodialysis: results from the Dialysis Outcomes and Practice Patterns Study (DOPPS). *Nephrol Dial Transplant.* 2012;27(11):4180–4188. doi:10.1093/ndt/gfs021.
54. Kalim S, Wald R, Yan AT, et al. Extended duration nocturnal hemodialysis and changes in plasma metabolite profiles. *Clin J Am Soc Nephrol.* 2018;13(3):436–444. doi:10.2215/CJN.08790817.
55. Chazot C, Ok E, Lacson E Jr, Kerr PG, Jean G, Misra M. Thrice-weekly nocturnal hemodialysis: the overlooked alternative to improve patient outcomes. *Nephrol Dial Transplant.* 2013;28(10):2447–2455. doi:10.1093/ndt/gft078.
56. Foley RN, Gilbertson DT, Murray T, Collins, AJ. Long interdialytic interval and mortality among patients receiving hemodialysis. *N Engl J Med.* 2011;365(12):1099–1107. doi:10.1056/NEJMoa1103313.
57. McIntyre CW, Rosansky SJ. Starting dialysis is dangerous: how do we balance the risk? *Kidney Int.* 2012;82(4):382–387. doi:10.1038/ki.2012.133.
58. Cooper BA, Branley P, Bulfone L, et al. A randomized, controlled trial of early versus late initiation of dialysis. *N Engl J Med.* 2010;363(7):609–619. doi:10.1056/NEJMoa1000552.
59. Wright D, Klausner B, Baird ME, et al. Timing of dialysis initiation and survival in ESRD. *Clin J Am Soc Nephrol.* 2010;5(10):1828–1835. doi:10.2215/CJN.06230909.
60. Chan CT, Blankestijn PJ, Dember LM, et al. Dialysis initiation, modality choice, access, and prescription: conclusions from a Kidney Disease: Improving Global Outcomes (KDIGO) controversies conference. *Kidney Int.* 2019;96(1):37–47. doi:10.1016/j.kint.2019.01.017.
61. Moss AH. Ethical principles and processes guiding dialysis decision-making. *Clin J Am Soc Nephrol.* 2011;6(9):2313–2317. doi:10.2215/CJN.03960411.
62. Kalantar-Zadeh K, Wightman A, Liao S. Ensuring choice for people with kidney failure—dialysis, supportive care, and hope. *N Engl J Med.* 2020;383(2):99–101. doi:10.1056/NEJMp2001794.
63. Brennan F, Stewart C, Burgess H, et al. Time to improve informed consent for dialysis: an international perspective. *Clin J Am Soc Nephrol.* 2017;12(6):1001–1009. doi:10.2215/CJN.09740916.
64. Brown MA, Collett GK, Josland EA, Foote C, Li Q, Brennan FP. CKD in elderly patients managed without dialysis: survival, symptoms, and quality of life. *Clin J Am Soc Nephrol.* 2015;10(2):260–268. doi:10.2215/CJN.03330414.
65. Murtagh FEM, Burns A, Moranne O, Morton RL, Naicker S. Supportive care: comprehensive conservative care in end-stage kidney disease. *Clin J Am Soc Nephrol.* 2016;11(10):1909–1914. doi:10.2215/CJN.04840516.
66. Kalantar-Zadeh K, Jafar TH, Nitsch D, Neuen BL, Perkovic V. Chronic kidney disease. *Lancet.* 2021;398(10302):786–802. doi:10.1016/S0140-6736(21)00519-5.
67. Ward RA, Ronco C. Dialyzer and machine technologies: application of recent advances to clinical practice. *Blood Purif.* 2006;24(1):6–10. doi:10.1159/000089429.
68. Storr M, Ward RA. Membrane innovation: closer to native kidneys. *Nephrol Dial Transplant.* 2018;33(suppl_3):iii22–iii27. doi:10.1093/ndt/gfy228.
69. Williams ME, Balogun RA. Principles of separation: indications and therapeutic targets for plasma exchange. *Clin J Am Soc Nephrol.* 2014;9(1):181–190. doi:10.2215/CJN.04680513.
70. Weiner DE, Falzon L, Skoufos L, et al. Efficacy and safety of expanded hemodialysis with the TheraNova 400 Dialyzer: a randomized controlled trial. *Clin J Am Soc Nephrol.* 2020;15(9):1310–1319. doi:10.2215/CJN.01210120.
71. Cozzolino M, Magagnoli L, Ciceri P, Conte F, Galassi A. Effects of a medium cut-off (TheraNova) dialyser on haemodialysis patients: a prospective, cross-over study. *Clin Kidney J.* 2019;14(1):382–389. doi:10.1093/ckj/sfz155.
72. Kokubo K, Kurihara Y, Kobayashi K, Tsukao H, Kobayashi H. Evaluation of the biocompatibility of dialysis membranes. *Blood Purif.* 2015;40(4):293–297. doi:10.1159/000441576.
73. Lok CE, Huber TS, Lee T, et al. KDOQI clinical practice guideline for vascular access: 2019 update. *Am J Kidney Dis.* 2020;75(4 suppl 2):S1–S164. doi:10.1053/j.ajkd.2019.12.001.

74. Lok CE. Fistula first initiative: advantages and pitfalls. *Clin J Am Soc Nephrol*. 2007;2(5):1043–1053. doi:10.2215/CJN.01080307.
75. DeSilva RN, Patibandla BK, Vin Y, et al. Fistula first is not always the best strategy for the elderly. *J Am Soc Nephrol*. 2013;24(8):1297–1304. doi:10.1681/ASN.2012060632.
76. Brown RS, Patibandla BK, Goldfarb-Rumyantsev AS. The survival benefit of “fistula first, catheter last” in hemodialysis is primarily due to patient factors. *J Am Soc Nephrol*. 2017;28(2):645–652. doi:10.1681/ASN.2016010019.
77. Shah S, Chan MR, Lee T. Perspectives in individualizing solutions for dialysis access. *Adv Chronic Kidney Dis*. 2020;27(3):183–190. doi:10.1053/j.ackd.2020.03.004.
78. Brown RS. Barriers to optimal vascular access for hemodialysis. *Semin Dial*. 2020;33(6):457–463. doi:10.1111/sdi.12922.
79. Pisoni RL, Zepel L, Zhao J, et al. International comparisons of native arteriovenous fistula patency and time to becoming catheter-free: findings from the Dialysis Outcomes and Practice Patterns Study (DOPPS). *Am J Kidney Dis*. 2021;77(2):245–254. doi:10.1053/j.ajkd.2020.06.020.
80. Koirala N, Anvari E, McLennan G. Monitoring and surveillance of hemodialysis access. *Semin Intervent Radiol*. 2016;33(1):25–30. doi:10.1055/s-0036-1572548.
81. Boubes K, Shaikh A, Alsaukas Z, Dwyer A. New directions in ensuring catheter safety. *Adv Chronic Kidney Dis*. 2020;27(3):228–235. doi:10.1053/j.ackd.2020.02.004.
82. Vascular Access Work Group. Clinical practice guidelines for vascular access. *Am J Kidney Dis*. 2006;48 suppl 1:S248–S273. doi:10.1053/j.ajkd.2006.04.040.
83. Ravani P, Palmer SC, Oliver MJ, et al. Associations between hemodialysis access type and clinical outcomes: a systematic review. *J Am Soc Nephrol*. 2013;24(3):465–473. doi:10.1681/ASN.2012070643.
84. Allon M. Treatment guidelines for dialysis catheter-related bacteremia: an update. *Am J Kidney Dis*. 2009;54(1):13–17. doi:10.1053/j.ajkd.2009.04.006.
85. Ouseph R, Ward RA. Anticoagulation for intermittent hemodialysis. *Semin Dial*. 2000;13(3):181–187. doi:10.1046/j.1525-139x.2000.00052.x.
86. Jain N, Reilly RF. Clinical pharmacology of oral anticoagulants in patients with kidney disease. *Clin J Am Soc Nephrol*. 2019;14(2):278–287. doi:10.2215/CJN.02170218.
87. Randhawa MS, Vishwanath R, Rai MP, et al. Association between use of warfarin for atrial fibrillation and outcomes among patients with end-stage renal disease: a systematic review and meta-analysis. *JAMA Netw Open*. 2020;3(4):e202175. doi:10.1001/jamanetworkopen.2020.2175.
88. Feldberg J, Patel P, Farrell A, et al. A systematic review of direct oral anticoagulant use in chronic kidney disease and dialysis patients with atrial fibrillation. *Nephrol Dial Transplant*. 2019;34(2):265–277. doi:10.1093/ndt/gfy031.
89. DeVriese AS, Caluwé R, Van Der Meersch H, De Boeck K, De Bacquer D. Safety and efficacy of vitamin K antagonists versus rivaroxaban in hemodialysis patients with atrial fibrillation: a multicenter randomized controlled trial. *J Am Soc Nephrol*. 2021;32(6):1474–1483. doi:10.1681/ASN.2020111566.
90. Greenberg KI, Choi MJ. Hemodialysis emergencies: core curriculum 2021. *Am J Kidney Dis*. 2021;77(5):796–809. doi:10.1053/j.ajkd.2020.11.024.
91. Saha M, Allon M. Diagnosis, treatment, and prevention of hemodialysis emergencies. *Clin J Am Soc Nephrol*. 2017;12(2):357–369. doi:10.2215/CJN.05260516.
92. Dalia T, Tuffaha AM. Dialysis disequilibrium syndrome leading to sudden brain death in a chronic hemodialysis patient. *Hemodial Int*. 2018;22(3):E39–E44. doi:10.1111/hdi.12635.
93. Samanta R, Chan C, Chauhan VS. Arrhythmias and sudden cardiac death in end stage renal disease: epidemiology, risk factors, and management. *Can J Cardiol*. 2019;35(9):1228–1240. doi:10.1016/j.cjca.2019.05.005.
94. Loutradis C, Sarafidis PA, Papadopoulos CE, Papagianni A, Zoccali C. The ebb and flow of echocardiographic cardiac function parameters in relationship to hemodialysis treatment in patients with ESRD. *J Am Soc Nephrol*. 2018;29(5):1372–1381. doi:10.1681/ASN.2017101102.
95. Sarafidis PA, Persu A, Agarwal R, et al. Hypertension in dialysis patients: a consensus document by the European Renal and Cardiovascular Medicine (EURECA-m) working group of the European Renal Association—European Dialysis and Transplant Association (ERA-EDTA) and the Hypertension and the Kidney Working Group of the European Society of Hypertension. *Nephrol Dial Transplant*. 2017;32(4):620–640. doi:10.1093/ndt/gfw433.
96. Denker MG, Cohen DL. Antihypertensive medications in end-stage renal disease. *Semin Dial*. 2015;28(4):330–336. doi:10.1111/sdi.12369.

97. Zoccali C, Torino C, Tripepi R, et al. Pulmonary congestion predicts cardiac events and mortality in ESRD. *J Am Soc Nephrol*. 2013;24(4):639–646. doi:10.1681/ASN.2012100990.
98. Flythe JE, Kimmel SE, Brunelli SM. Rapid fluid removal during dialysis is associated with cardiovascular morbidity and mortality. *Kidney Int*. 2011;79(2):250–257. doi:10.1038/ki.2010.383.
99. Zhou H, Sim JJ, Shi J, et al. Beta-blocker use and risk of mortality in heart failure patients initiating maintenance dialysis. *Am J Kidney Dis*. 2021;77(5):704–712. doi:10.1053/j.ajkd.2020.07.023.
100. Navaneethan SD, Wehbe E, Heresi GA, et al. Presence and outcomes of kidney disease in patients with pulmonary hypertension. *Clin J Am Soc Nephrol*. 2014;9(5):855–863. doi:10.2215/CJN.10191013.
101. Bansal S, Prasad A, Linas S. Right heart failure—unrecognized cause of cardiorenal syndrome. *J Am Soc Nephrol*. 2018;29(7):1795–1798. doi:10.1681/ASN.2018020224.
102. Sise ME, Courtwright AM, Channick RN. Pulmonary hypertension in patients with chronic and end-stage kidney disease. *Kidney Int*. 2013;84(4):682–692. doi:10.1038/ki.2013.186.
103. Seong EY, Liu S, Song SH, et al. Intradialytic hypotension and newly recognized peripheral artery disease in patients receiving hemodialysis. *Am J Kidney Dis*. 2021;77(5):730–738. doi:10.1053/j.ajkd.2020.10.012.
104. Okamoto S, Iida O, Mano T. Current perspective on hemodialysis patients with peripheral artery disease. *Ann Vasc Dis*. 2017;10(2):88–91. doi:10.3400/avd.ra.17-00034.
105. Reeves PB, McCausland FR. Mechanisms, clinical implications, and treatment of intradialytic hypotension. *Clin J Am Soc Nephrol*. 2018;13(8):1297–1303. doi:10.2215/CJN.12141017.
106. McIntyre CW, Salerno FR. Diagnosis and treatment of intradialytic hypotension in maintenance hemodialysis patients. *Clin J Am Soc Nephrol*. 2018;13(3):486–489. doi:10.2215/CJN.11131017.
107. Kuipers J, Verboom LM, Ipema KJR, et al. The prevalence of intradialytic hypotension in patients on conventional hemodialysis: a systematic review with meta-analysis. *Am J Nephrol*. 2019;49(6):497–506. doi:10.1159/000500877.
108. Flythe JE, Xue H, Lynch KE, Curhan GC, Brunelli SM. Association of mortality risk with various definitions of intradialytic hypotension. *J Am Soc Nephrol*. 2015;26(3):724–734. doi:10.1681/ASN.2014020222.
109. Kanbay M, Ertuglu LA, Afsar B, et al. An update review of intradialytic hypotension: concept, risk factors, clinical implications and management. *Clin Kidney J*. 2020;13(6):981–993. doi:10.1093/ckj/sfaa078.
110. McIntyre CW, Burton JO, Selby NM, et al. Hemodialysis-induced cardiac dysfunction is associated with an acute reduction in global and segmental myocardial blood flow. *Clin J Am Soc Nephrol*. 2008;3(1):19–26. doi:10.2215/CJN.03170707.
111. Hryciw N, Joannidis M, Hiremath S, Callum J, Clark EG. Intravenous albumin for mitigating hypotension and augmenting ultrafiltration during kidney replacement therapy. *Clin J Am Soc Nephrol*. 2021;16(5):820–828. doi:10.2215/CJN.09670620.
112. Karaboyas A, Morgenstern H, Waechter S, et al. Low hemoglobin at hemodialysis initiation: an international study of anemia management and mortality in the early dialysis period. *Clin Kidney J*. 2019;13(3):425–433. doi:10.1093/ckj/sfz065.
113. Klinger AS, Foley RN, Goldfarb DS, et al. KDOQI US commentary on the 2012 KDIGO clinical practice guideline for anemia in CKD. *Am J Kidney Dis*. 2013;62(5):849–859. doi:10.1053/j.ajkd.2013.06.008.
114. Besarab A, Bolton WK, Browne JK, et al. The effects of normal as compared with low hematocrit values in patients with cardiac disease who are receiving hemodialysis and epoetin. *N Engl J Med*. 1998;339(9):584–590. doi:10.1056/NEJM199808273390903.
115. Fishbane S, Spinowitz BS, Wisemandle WA, Martin NE. Randomized controlled trial of subcutaneous epoetin alfa-epbx versus epoetin alfa in end-stage renal disease. *Kidney Int Rep*. 2019;4(9):1235–1247. doi:10.1016/j.ekir.2019.05.010.
116. Fishbane S, Singh B, Kumbhat S, Wisemandle WA, Martin NE. Intravenous epoetin alfa-epbx versus epoetin alfa for treatment of anemia in end-stage kidney disease. *Clin J Am Soc Nephrol*. 2018;13(8):1204–1214. doi:10.2215/CJN.11631017.
117. Sinha SD, Bandi VK, Bheemareddy BR, et al. Efficacy, tolerability and safety of darbepoetin alfa injection for the treatment of anemia associated with chronic kidney disease (CKD) undergoing dialysis: a randomized, phase III trial. *BMC Nephrol*. 2019;20(1):90. doi:10.1186/s12882-019-1209-1.
118. Chen N, Hao C, Liu B-C, et al. Roxadustat treatment for anemia in patients undergoing long-term dialysis. *N Engl J Med*. 2019;381(11):1011–1022. doi:10.1056/NEJMoa1901713.

119. Akizawa T, Iwasaki M, Yamaguchi Y, Majikawa Y, Reusch M. Phase 3, randomized, double-blind, active-comparator (darbepoetin alfa) study of oral roxadustat in CKD patients with anemia on hemodialysis in Japan. *J Am Soc Nephrol*. 2020;31(7):1628–1639. doi:10.1681/ASN.2019060623.
120. Singh AK, Carroll K, Perkovic V, et al. Daprodustat for the treatment of anemia in patients undergoing dialysis. *N Engl J Med*. 2021;385(25):2325–2335. doi:10.1056/NEJMoa2113379.
121. Saglimbene VM, Palmer SC, Ruooslahti P, Natale P, Craig JC, Strippoli GFM. Continuous erythropoiesis receptor activator (CERA) for the anaemia of chronic kidney disease. *Cochrane Database Syst Rev*. 2017;8(8):CD009904. doi:10.1002/14651858.CD009904.pub2.
122. Asirvatham JR, Moses V, Bjornson L. Errors in potassium measurement: a laboratory perspective for the clinician. *N Am J Med Sci*. 2013;5(4):255–259. doi:10.4103/1947-2714.110426.
123. Karaboyas A, Robinson BM, James G, et al. Hyperkalemia excursions are associated with an increased risk of mortality and hospitalizations in hemodialysis patients. *Clin Kidney J*. 2020;14(7):1760–1769. doi:10.1093/ckj/sfaa208.
124. Quach K, Ltvyn L, Baigent C, et al. The safety and efficacy of mineralocorticoid receptor antagonists in patients who require dialysis: a systematic review and meta-analysis. *Am J Kidney Dis*. 2016;68(4):591–598. doi:10.1053/j.ajkd.2016.04.011.
125. Lee R, Yin M, He Z, Gurm H, Heung M. Dialysis paradox: impact of aldosterone antagonism on survival in patients with HF and ESRD. *J Am Coll Cardiol*. 2019;73(9_suppl_1):924. doi:10.1016/S0735-1097(19)31531-1.
126. Charytan DM, Himmelfarb J, Ikizler TA, et al. Safety and cardiovascular efficacy of spironolactone in dialysis-dependent ESRD (Spin-D): a randomized, placebo-controlled, multiple dosage trial. *Kidney Int*. 2019;95(4):973–982. doi:10.1016/j.kint.2018.08.034.
127. Wong SWS, Zhang G, Norman P, Welihinda H, Thiwanaka Wijeratne D. Polysulfonate resins in hyperkalemia: a systematic review. *Can J Kidney Health Dis*. 2020;7:2054358120965838. doi:10.1177/2054358120965838.
128. Shrestha DB, Budhathoki P, Sedhai YR, et al. Patiromer and sodium zirconium cyclosilicate in treatment of hyperkalemia: a systematic review and meta-analysis. *Curr Ther Res Clin Exp*. 2021;95:100635. doi:10.1016/j.curtheres.2021.100635.
129. Bushinsky DA, Williams GH, Pitt B, et al. Patiromer induces rapid and sustained potassium lowering in patients with chronic kidney disease. *Kidney Int*. 2015;88(6):1427–1433. doi:10.1038/ki.2015.270.
130. Kidney Disease: Improving Global Outcomes CKD-MBD Working Group. KDIGO 2017 clinical practice guideline update for the diagnosis, evaluation, prevention, and treatment of chronic kidney disease-mineral and bone disorder (CKD-MBD). *Kidney Int Suppl* (2011). 2017;7(1):1–59. doi:10.1016/j.kisu.2017.04.001.
131. Cunningham J, Danese M, Olson K, Klassen P, Chertow GM. Effects of the calcimimetic cinacalcet HCl on cardiovascular disease, fracture, and health-related quality of life in secondary hyperparathyroidism. *Kidney Int*. 2005;68(4):1793–1800. doi:10.1111/j.1523-1755.2005.00596.x.
132. EVOLVE Trial Investigators. Effect of cinacalcet on cardiovascular disease in patients undergoing dialysis. *N Engl J Med*. 2012;367(26):2482–2494. doi:10.1056/NEJMoa1205624.
133. Moe SM, Abdalla S, Chertow GM, et al. Effects of cinacalcet on fracture events in patients receiving hemodialysis: the EVOLVE Study. *J Am Soc Nephrol*. 2015;26(6):1466–1475. doi:10.1681/ASN.2014040414.
134. Block GA, Bushinsky DA, Sunfa Cheng S, et al. Effect of etelcalcitide versus cinacalcet on serum hormone in patients receiving hemodialysis with secondary hyperparathyroidism: a randomized clinical trial. *JAMA*. 2017;317(2):156–164. doi:10.1001/jama.2016.19468.
135. Parikh C, Gutgarts V, Eisenberg E, Melamed ML. Vitamin D and clinical outcomes in dialysis. *Semin Dial*. 2015;28(6):604–609. doi:10.1111/sdi.12446.
136. Zheng Z, Shi H, Jia J, Li D, Lin S. Vitamin D supplementation and mortality risk in chronic kidney disease: a meta-analysis of 20 observational studies. *BMC Nephrol*. 2013;14:199. doi:10.1186/1471-2369-14-199.
137. Isakova T, Nickolas TL, Denburg M, et al. KDOQI US commentary on the 2017 KDIGO Clinical Practice Guideline Update for the Diagnosis, Evaluation, Prevention, and Treatment of Chronic Kidney Disease—Mineral and Bone Disorder. *Am J Kidney Dis*;70(6):737–751. doi:10.1053/j.ajkd.2017.07.019.
138. Isaka Y, Hamano T, Fujii H, et al. Optimal phosphate control related to coronary artery calcification in dialysis patients. *J Am Soc Nephrol*. 2021;32(3):723–735. doi:10.1681/ASN.2020050598.
139. Hutchison AJ. Oral phosphate binders. *Kidney Int*. 2009;75(9):906–914. doi:10.1038/ki.2009.60.

140. Isakova T, Gutiérrez OM, Chang Y, et al. Phosphorus binders and survival on hemodialysis. *J Am Soc Nephrol*. 2009;20(2):388–396. doi:10.1681/ASN.2008060609.
141. Chan S, Au K, Francis RS, Mudge DW, Johnson DW, Pillans PI. Phosphate binders in patients with chronic kidney disease. *Aust Prescr*. 2017;40(1):10–14. doi:10.18773/austprescr.2017.002.
142. Spoendlin J, Paik JM, Tsacogianis T, Kim SC, Schneeweiss S, Desai RJ. Cardiovascular outcomes of calcium-free vs calcium-based phosphate binders in patients 65 years or older with end-stage renal disease requiring hemodialysis. *JAMA Intern Med*. 2019;179(6):741–749. doi:10.1001/jamainternmed.2019.0045.
143. Gutekunst L. An update on phosphate binders: a dietician's perspective. *J Ren Nutr*. 2016;26(4):209–218. doi:10.1053/j.jrn.2016.01.009.
144. Lopes MB, Karaboyas A, Bieber B, et al. Impact of longer term phosphorus control on cardiovascular mortality in hemodialysis patients using an area under the curve approach: results from the DOPPS. *Nephrol Dial Transplant*. 2020;35(10):1794–1801. doi:10.1093/ndt/gfaa054.
145. Edmonston DL, Isakova T, Dember LM, et al. Design and rationale of HiLo: a pragmatic, randomized trial of phosphate management for patients receiving maintenance hemodialysis. *Am J Kidney Dis*. 2021;77(6):920–930.e1. doi:10.1053/j.ajkd.2020.10.008.
146. Kodumudi V, Jeha GM, Mydlo N, Kaye AD. Management of cutaneous calciphylaxis. *Adv Ther*. 2020;37(12):4797–4807. doi:10.1007/s12325-020-01504-w.
147. Sprague SM. Painful skin ulcers in a hemodialysis patient. *Clin J Am Soc Nephrol*. 2014;9(1):166–173. doi:10.2215/CJN.00320113.
148. Yu WY-H, Bhutani T, Kornik R, et al. Warfarin-associated nonuremic calciphylaxis. *JAMA Dermatol*. 2017;153(3):309–314. doi:10.1001/jamadermatol.2016.4821.
149. Singh RP, Derendorf H, Ross EA. Simulation-based sodium thiosulfate dosing strategies for the treatment of calciphylaxis. *Clin J Am Soc Nephrol*. 2011;6(5):1155–1159. doi:10.2215/CJN.09671010.
150. Coelho S. Is the management of diabetes different in dialysis patients? *Semin Dial*. 2018;31(4):367–376. doi:10.1111/sdi.12698.
151. DeBoer I, Caramori ML, Chan JCN, et al. Executive summary of the 2020 KDIGO diabetes management in CKD guideline: evidence-based advances in monitoring and treatment. *Kidney Int*. 2020;98(4):839–848. doi:10.1016/j.kint.2020.06.024.
152. Chen TK, Sperati CJ, Thavarajah S, Grams ME. Reducing kidney function decline in patients with CKD: core curriculum 2021. *Am J Kidney Dis*. 2021;77(6):969–983. doi:10.1053/j.ajkd.2020.12.022.
153. Hill CJ, Maxwell AP, Cardwell CR, et al. Glycated hemoglobin and risk of death in diabetic patients treated with hemodialysis: a meta-analysis. *Am J Kidney Dis*. 2014;63(1):84–94. doi:10.1053/j.ajkd.2013.06.020.
154. Presswala L, Hong S, Harris Y, et al. Continuous glucose monitoring and glycemic control in patients with type 2 diabetes mellitus and CKD. *Kidney Med*. 2019;1(5):281–287. doi:10.1016/j.xkme.2019.07.006.
155. Williams ME, Lacson E Jr, Wang W, Lazarus JM, Hakim R. Glycemic control and extended hemodialysis survival in patients with diabetes mellitus: comparative results of traditional and time-dependent Cox model analyses. *Clin J Am Soc Nephrol*. 2010;5(9):1595–1601. doi:10.2215/CJN.09301209.
156. Hoshino J, Larkina M, Karaboyas A, et al. Unique hemoglobin A1c level distribution and its relationship with mortality in diabetic hemodialysis patients. *Kidney Int*. 2017;92(2):497–503. doi:10.1016/j.kint.2017.02.008.
157. Ricks J, Molnar MZ, Kovesdy CP, et al. Glycemic control and cardiovascular mortality in hemodialysis patients with diabetes: a 6-year cohort study. *Diabetes*. 2012;61(3):708–715. doi:10.2337/db11-1015.
158. Hoshino J, Hamano T, Abe M, et al. Glycated albumin versus hemoglobin A1c and mortality in diabetic hemodialysis patients: a cohort study. *Nephrol Dial Transplant*. 2018;33(7):1150–1158. doi:10.1093/ndt/gfy014.
159. American Diabetes Association. Pharmacological approaches to glycemic treatment: standards of medical care in diabetes—2021. *Diabetes Care*. 2021;44(suppl 1):S111–S124. doi:10.2337/dc21-S009.
160. Garg R, Williams M. Diabetes management in the kidney patient. *Med Clin North Am*. 2013;97(1):135–156. doi:10.1016/j.mcna.2012.11.001.
161. Bally L, Gubler P, Thabit H, et al. Fully closed-loop insulin delivery improves glucose control of inpatients with type 2 diabetes receiving hemodialysis. *Kidney Int*. 2019;96(3):593–596. doi:10.1016/j.kint.2019.03.006.
162. Hong S, Presswala L, Harris YT, et al. Hypoglycemia in patients with type 2 diabetes mellitus and chronic kidney disease: a prospective observational study. *Kidney360*. 2020;1(9):897–903. doi:10.34067/KID.0001272020.

163. Abe M, Kalantar-Zadeh K. Haemodialysis-induced hypoglycaemia and glycaemic disarrays. *Nat Rev Nephrol.* 2015;11(5):302–313. doi:10.1038/nrneph.2015.38.
164. Bansal AD, Hill CE, Berns JS. Use of antiepileptic drugs in patients with chronic kidney disease in end-stage renal disease. *Semin Dial.* 2015;28(4):404–412. doi:10.1111/sdi.12385.
165. Cohen SD, Kimmel PL, Neff R, Agodoa L, Abbott KC. Association of incident gout and mortality in dialysis patients. *J Am Soc Nephrol.* 2008;19(11):2204–2210. doi:10.1681/ASN.2007111256.
166. Vargas-Santos AB, Neogi T. Management of gout and hyperuricemia in CKD. *Am J Kidney Dis.* 2017;70(3):422–439. doi:10.1053/j.ajkd.2017.01.055.
167. Schlesinger N, Lipsky PE. Pegloticase treatment of chronic refractory gout: update on efficacy and safety. *Semin Arthritis Rheum.* 2020;50(3S):S31–S38. doi:10.1016/j.semarthrit.2020.04.011.
168. Alakel N, Middeke JM, Schetelig J, Bornhäuser M. Prevention and treatment of tumor lysis syndrome, and the efficacy and role of rasburicase. *Onco Targets Ther.* 2017;10:597–605. doi:10.2147/OTT.S103864.
169. Rosner MH. Cancer screening in patients undergoing maintenance dialysis: who, what, and when. *Am J Kidney Dis.* 2020;76(4):558–566. doi:10.1053/j.ajkd.2019.12.018.
170. Farivar-Mohseni H, Perlmuter AE, Wilson S, Shingleton WB, Bigler SA, Fowler JE Jr. Renal cell carcinoma and end stage renal disease. *J Urol.* 2006;175(6):2018–2020; discussion 2021. doi:10.1016/S0022-5347(06)00340-5.
171. De Meester J, De Bacquer D, Naesens M, et al. Incidence, characteristics, and outcome of COVID-19 in adults on kidney replacement therapy: a regionwide registry study. *J Am Soc Nephrol.* 2021;32(2):385–396. doi:10.1681/ASN.2020060875.
172. Jiang H-J, Tang H, Xiong F, et al. COVID-19 in peritoneal dialysis patients. *Clin J Am Soc Nephrol.* 2020;16(1):121–123. doi:10.2215/CJN.07200520.
173. Flythe JE, Assimon MM, Tugman MJ, et al. Characteristics and outcomes of individuals with pre-existing kidney disease and COVID-19 admitted to intensive care units in the United States. *Am J Kidney Dis.* 2021;77(2):190–203.e1. doi:10.1053/j.ajkd.2020.09.003.
174. Walker AG, Sibbel S, Wade C, et al. SARS-CoV-2 antibody seroprevalence among maintenance dialysis patients in the United States. *Kidney Med.* 2021;3(2):216–222.e1. doi:10.1016/j.xkme.2021.01.002.
175. Tang H, Tian J-B, Dong J-W, et al. Serologic detection of SARS-CoV-2 infections in hemodialysis centers: a multicenter retrospective study in Wuhan, China. *Am J Kidney Dis.* 2020;76(4):490–499.e1. doi:10.1053/j.ajkd.2020.06.008.
176. Weinhandl ED, Wetmore JB, Peng Y, Liu J, Gilbertson DT, Johansen KL. Initial effects of COVID-19 on patients with ESKD. *J Am Soc Nephrol.* 2021;32(6):1444–1453. doi:10.1681/ASN.2021010009.
177. Yau K, Muller MP, Lin M, et al. COVID-19 outbreak in an urban hemodialysis unit. *Am J Kidney Dis.* 2020;76(5):690–695.e1. doi:10.1053/j.ajkd.2020.07.001.
178. Hsu CM, Weiner DE. COVID-19 in dialysis patients: outlasting and outsmarting a pandemic. *Kidney Int.* 2020;98(6):1402–1404. doi:10.1016/j.kint.2020.10.005.
179. Miller-Handley H, Luckett K, Govil A. Treatment options for coronavirus disease 2019 in patients with reduced or absent kidney function. *Adv Chronic Kidney Dis.* 2020;27(5):434–441. doi:10.1053/j.ackd.2020.09.001.
180. Adamsick ML, Gandhi RG, Bidell MR, et al. Remdesivir in patients with acute or chronic kidney disease and COVID-19. *J Am Soc Nephrol.* 2020;31(7):1384–1386. doi:10.1681/ASN.2020050589.
181. Shankaranarayanan D, Muthukumar T, Barbar T, et al. Anticoagulation strategies and filter life in COVID-19 patients receiving continuous renal replacement therapy: a single-center experience. *Clin J Am Soc Nephrol.* 2020;16(1):124–126. doi:10.2215/CJN.08430520.
182. Alfano G, Fontana F, Ferrari A, et al. Peritoneal dialysis in the time of coronavirus disease 2019. *Clin Kidney J.* 2020;13(3):265–268. doi:10.1093/ckj/sfaa093.
183. Windpessl M, Bruchfeld A, Anders H-J, et al. COVID-19 vaccines and kidney disease. *Nat Rev Nephrol.* 2021;17(5):291–293. doi:10.1038/s41581-021-00406-6.
184. Grupper A, Sharon N, Finn T, et al. Humoral response to the Pfizer BNT162b2 vaccine in patients undergoing maintenance hemodialysis. *Clin J Am Soc Nephrol.* 2021;16(7):1037–1042. doi:10.2215/CJN.03500321.
185. Cohen DE, Sibbel S, Marlowe G, et al. Antibody status, disease history, and incidence of SARS-CoV-2 infection among patients on chronic dialysis. *J Am Soc Nephrol.* 2021;32(8):1880–1886. doi:10.1681/ASN.2021030387.

186. Gelfand SL, Scherer JS, Koncicki HM. Kidney supportive care: core curriculum 2020. *Am J Kidney Dis.* 2020;75(5):793–806. doi:10.1053/j.ajkd.2019.10.016.
187. Sondergaard H. Fatigue while undergoing long-term hemodialysis. *Clin J Am Soc Nephrol.* 2020;15(11):1539–1540. doi:10.2215/CJN.14870920.
188. Gregg LP, Bossola M, Ostrosky-Frid M, Hedayati SS. Fatigue in CKD: epidemiology, pathophysiology, and treatment. *Clin J Am Soc Nephrol.* 2021;16(9):1445–1455. doi:10.2215/CJN.19891220.
189. Ju A, Teixeira-Pinto A, Tong A, et al. Validation of a core patient-reported outcome measure for fatigue in patients receiving hemodialysis: the SONG-HD Fatigue instrument. *Clin J Am Soc Nephrol.* 2020;15(11):1614–1621. doi:10.2215/CJN.05880420.
190. Moist L. Pain management in a patient with kidney failure. *Clin J Am Soc Nephrol.* 2020;15(11):1657–1659. doi:10.2215/CJN.01440220.
191. Baker M, Perazella MA. NSAIDs in CKD: are they safe? *Am J Kidney Dis.* 2020;76(4):546–557. doi:10.1053/j.ajkd.2020.03.023.
192. Lu E, Schell JO, Koncicki HM. Opioid management in CKD. *Am J Kidney Dis.* 2021;77(5):786–795. doi:10.1053/j.ajkd.2020.08.018.
193. Pisoni RL, Wikström B, Elder SJ, et al. Pruritus in hemodialysis patients: international results from the Dialysis Outcomes and Practice Patterns Study (DOPPS). *Nephrol Dial Transplant.* 2006;21(12):3495–3505. doi:10.1093/ndt/gff461.
194. Simonsen E, Komenda P, Lerner B, et al. Treatment of uremic pruritus: a systematic review. *Am J Kidney Dis.* 2017;70(5):638–655. doi:10.1053/j.ajkd.2017.05.018.
195. Fishbane S, Jamal A, Munera C, et al. A phase 3 trial of difelikefalin in hemodialysis patients with pruritus. *N Engl J Med.* 2020;382(3):222–232. doi:10.1056/NEJMoa1912770.
196. Friedli K, Guirguis A, Almond M, et al. Sertraline versus placebo in patients with major depressive disorder undergoing hemodialysis: a randomized, controlled feasibility trial. *Clin J Am Soc Nephrol.* 2017;12(2):280–286. doi:10.2215/CJN.02120216.
197. Zhao Y, Zhang Y, Yang Z, et al. Sleep disorders and cognitive impairment in peritoneal dialysis: a multicenter prospective cohort study. *Kidney Blood Press Res.* 2019;44(5):1115–1127. doi:10.1159/000502355.
198. Chan CT, Wallace E, Golper TA, et al. Exploring barriers and potential solutions in home dialysis: an NKF-KDOQI Conference Outcomes Report. *Am J Kidney Dis.* 2019;73(3):363–371. doi:10.1053/j.ajkd.2018.09.015.
199. Chan CT, Collins K, Ditschman EP, et al. Overcoming barriers for uptake and retention of home dialysis patients: an NKF KDOQI Home Dialysis Conference. *Am J Kidney Dis.* 2020;75(6):926–934. doi:10.1053/j.ajkd.2019.11.007.
200. Bonenkamp AA, van Eck van der Sluijs A, Hoekstra T, et al. Health-related quality of life in home dialysis patients compared to in-center hemodialysis patients: a systematic review and meta-analysis. *Kidney Med.* 2020;2(2):139–154. doi:10.1016/j.xkme.2019.11.005.
201. Chan CT, Collins K, Ditschman EP, et al. Overcoming barriers for uptake and continued use of home dialysis: an NKF-KDOQI conference report. *Am J Kidney Dis.* 2020;75(6):926–934. doi:10.1053/j.ajkd.2019.11.007.
202. Ashby D, Borman N, Burton J, et al. Renal Association clinical practice guideline on haemodialysis. *BMC Nephrol.* 2019;20(1):379. doi:10.1186/s12882-019-1527-3.
203. Rivara MB, Ravel V, Streja E, et al. Weekly standard Kt/V urea and clinical outcomes in home and in-center hemodialysis. *Clin J Am Soc Nephrol.* 2018;13(3):445–455. doi:10.2215/CJN.05680517.
204. Chertow GM, Levin NW, Beck GJ, et al. Long-term effects of frequent in-center hemodialysis. *J Am Soc Nephrol.* 2016;27(6):1830–1836. doi:10.1681/ASN.2015040426.
205. Culleton BF, Walsh M, Klarenbach SW, et al. Effect of frequent nocturnal hemodialysis versus conventional hemodialysis on left ventricular mass and quality of life: a randomized controlled trial. *JAMA.* 2007;298(11):1291–1299. doi:10.1001/jama.298.11.1291.
206. National Kidney Foundation. KDOQI clinical practice guideline for hemodialysis adequacy: 2015 update. *Am J Kidney Dis.* 2015;66(5):884–930. doi:10.1053/j.ajkd.2015.07.015.
207. Eknayan G, Beck GJ, Cheung AK, et al. Effect of dialysis dose and membrane flux in maintenance hemodialysis. *N Engl J Med.* 2002;347(25):2010–2019. doi:10.1056/NEJMoa021583.
208. Cherukuri S, Bajo M, Colussi G, et al. Home hemodialysis treatment and outcomes: retrospective analysis of the Knowledge to Improve Home Dialysis Network in Europe (KIHdNEy) cohort. *BMC Nephrol.* 2018;19(1):262. doi:10.1186/s12882-018-1059-2.

209. Lowrie EG, Laird NM, Parker TF, Sargent JA. Effect of the hemodialysis prescription on patient morbidity: report from the National Cooperative Dialysis Study. *N Engl J Med*. 1981;305(20):1176–1181. doi:10.1056/NEJM198111123052003.
210. Gotch FA, Sargent JA. A mechanistic analysis of the National Cooperative Dialysis Study (NCDS). *Kidney Int*. 1985;28(3):526–534. doi:10.1038/ki.1985.160.
211. Steyaert S, Holvoet E, Nagler E, Malfait S, Van Biesen W. Reporting of “dialysis adequacy” as an outcome in randomized trials conducted in adults on haemodialysis. *PLoS One*. 2019;14(2):e0207045. doi:10.1371/journal.pone.0207045.
212. Perl J, Dember LM, Bargman JM, et al. The use of a multidimensional measure of dialysis adequacy—moving beyond small solute kinetics. *Clin J Am Soc Nephrol*. 2017;12(5):839–847. doi:10.2215/CJN.08460816.
213. Windpessl M, Prischl FC, Prenner A, Vychytil A. Managing hospitalized peritoneal dialysis patients: ten practical points for the non-nephrologist. *Am J Med*. 2021;134(7):833–839. doi:10.1016/j.amjmed.2021.02.007.
214. Sloan CE, Coffman CJ, Sanders LL, et al. Trends in peritoneal dialysis use in the United States after Medicare Payment Reform. *Clin J Am Soc Nephrol*. 2019;14(12):1763–1772. doi:10.2215/CJN.05910519.
215. Klomjit N, Kattah AG, Cheungpasitporn W. The cost-effectiveness of peritoneal dialysis is superior to hemodialysis: updated evidence from a more precise model. *Kidney Med*. 2020;3(1):15–17. doi:10.1016/j.xkme.2020.12.003.
216. Francois K, Ronco C, Bargman JM. Peritoneal dialysis for chronic congestive heart failure. *Blood Purif*. 2015;40(1):45–52. doi:10.1159/000430084.
217. Kazory A. Peritoneal dialysis for chronic cardiorenal syndrome: lessons learned from ultrafiltration trials. *World J Cardiol*. 2015;7(7):392–396. doi:10.4330/wjc.v7.i7.392.
218. Cullis B, Al-Hwiesh A, Kilonzo K, et al. ISPD guidelines for peritoneal dialysis in acute kidney injury: 2020 update (adults). *Perit Dial Int*. 2021;41(1):15–31. doi:10.1177/0896860820970834.
219. Crabtree JH, Shrestha BM, Chow K-M, et al. ISPD guidelines/recommendations: creating and maintaining optimal peritoneal dialysis access in the adult patient: 2019 update. *Perit Dial Int*. 2019;39(5):414–436. doi:10.3747/pdi.2018.00232.
220. Wolfson M, Piraino B, Hamburger RJ, Morton AR, Icodextrin Study Group. A randomized controlled trial to evaluate the efficacy and safety of icodextrin in peritoneal dialysis. *Am J Kidney Dis*. 2002;40(5):1055–1065. doi:10.1053/ajkd.2002.36344.
221. Takatori Y, Akagi S, Sugiyama H, et al. Icodextrin increases technique survival rate in peritoneal dialysis patients with diabetic nephropathy by improving body fluid management: a randomized controlled trial. *Clin J Am Soc Nephrol*. 2011;6(6):1337–1344. doi:10.2215/CJN.10041110.
222. Li PKT, Culleton BF, Ariza A, et al. Randomized, controlled trial of glucose-sparing peritoneal dialysis in diabetic patients. *J Am Soc Nephrol*. 2013;24(11):1889–1900. doi:10.1681/ASN.2012100987.
223. Teitelbaum I. Crafting the prescription for patients starting peritoneal dialysis. *Clin J Am Soc Nephrol*. 2018;13(3):483–485. doi:10.2215/CJN.10770917.
224. Morelle J, Stachowska-Pietka J, Öberg C, et al. ISPD recommendations for the evaluation of peritoneal membrane dysfunction in adults: classification, measurement, interpretation and rationale for intervention. *Perit Dial Int*. 2021;41(4):352–372. doi:10.1177/0896860820982218.
225. Mehrotra R, Ravel V, Streja E, et al. Peritoneal equilibration test and patient outcomes. *Clin J Am Soc Nephrol*. 2015;10(11):1990–2001. doi:10.2215/CJN.03470315.
226. Twardowski ZJ, Nolph KD, Khanna R, et al. Peritoneal equilibration test. *Perit Dial Bull*. 1987;7:138.
227. Mujais S, Vonesh E. Profiling of peritoneal ultrafiltration. *Kidney Int Suppl*. 2002;(81):S17–S22. doi:10.1046/j.1523-1755.62.s81.4.x.
228. NKF-DOQI clinical practice guidelines for peritoneal dialysis adequacy. National Kidney Foundation. *Am J Kidney Dis*. 1997;30(3 suppl 2):S67–S136. doi:10.1016/s0272-6386(97)70028-3.
229. Lo WK, Bargman JM, Burkart J, et al. Guideline on targets for solute and fluid removal in adult patients on chronic peritoneal dialysis. *Perit Dial Int*. 2006;26(5):520–522. doi:10.1177/089686080602600502.
230. Brown EA, Blake PG, Boudville N, et al. International Society for Peritoneal Dialysis practice recommendations: prescribing high-quality goal-directed peritoneal dialysis. *Perit Dial Int*. 2020;40(3):244–253. doi:10.1177/0896860819895364.

231. Teitelbaum I, Glickman J, Neu A, et al. KDOQI US commentary on the 2020 ISPD practice recommendations for prescribing high-quality goal-directed peritoneal dialysis. *Am J Kidney Dis.* 2021; 77(2):157–171. doi:10.1053/j.ajkd.2020.09.010.
232. Kiebalo T, Holotka J, Habura I, Pawlaczyk K. Nutritional status in peritoneal dialysis: nutritional guidelines, adequacy and the management of malnutrition. *Nutrients.* 2020;12(6):1715. doi:10.3390/nu12061715.
233. Lu W, Pang WF, Jin L, et al. Peritoneal protein clearance predicts mortality in peritoneal dialysis patients. *Clin Exp Nephrol.* 2019;23(4):551–560. doi:10.1007/s10157-018-1677-9.
234. Ghaffari A. Urgent-start peritoneal dialysis: a quality improvement report. *Am J Kidney Dis.* 2012; 59(3):400–408. doi:10.1053/j.ajkd.2011.08.034.
235. Htay H, Johnson DW, Craig JC, et al. Urgent-start peritoneal dialysis versus conventional-start peritoneal dialysis for people with chronic kidney disease. *Cochrane Database Syst Rev.* 020;12(12):CD012913. doi:10.1002/14651858.CD012913.pub2.
236. Jin H, Fang W, Zhu M, et al. Urgent-start peritoneal dialysis and hemodialysis in ESRD patients: complications and outcomes. *PLoS One.* 2016;11(11):e0166181. doi:10.1371/journal.pone.0166181.
237. Al Sahlawi M, Bargman JM, Perl J. Peritoneal dialysis-associated peritonitis: suggestions for management and mistakes to avoid. *Kidney Med.* 2020;2(4):467–475. doi:10.1016/j.xkme.2020.04.010.
238. Uiterwijk H, Franssen CFM, Kuipers J, Westerhuis R, Nauta FL. Glucose exposure in peritoneal dialysis is a significant factor predicting peritonitis. *Am J Nephrol.* 2020;51(3):237–243. doi:10.1159/000506324.
239. Szeto C-C, Li P K-T. Peritoneal dialysis-associated peritonitis. *Clin J Am Soc Nephrol.* 2019;14(7): 1100–1105. doi:10.2215/CJN.14631218.
240. Li P K-T, Szeto CC, Piraino B, et al. ISPD peritonitis recommendations: 2016 update on prevention and treatment. *Perit Dial Int.* 2016;36(5):481–508. doi:10.3747/pdi.2016.00078.
241. Szeto CC. The new ISPD peritonitis guideline. *Renal Replacement Therapy.* 2018;4:7–12. doi:10.1186/s41100-018-0150-2.
242. Auricchio S, Giovenzana ME, Pozzi M, et al. Fungal peritonitis in peritoneal dialysis: a 34-year single centre evaluation. *Clin Kidney J.* 2018;11(6):874–880. doi:10.1093/ckj/sfy045.
243. Schreiber MJ, Pegues DA, van Hout B, et al. Analysis of after-hours oral antibiotic protocol for the treatment of suspected peritonitis in peritoneal dialysis patients (Abstract). Poster Presentation at American Society of Nephrology Kidney Week 2018. DaVita Clinical Research. Accessed Month, day, year. <https://www.davitaclinicalresearch.com/publication/analysis-of-after-hours-oral-antibiotic-protocol-for-the-treatment-of-suspected-peritonitis-in-peritoneal-dialysis-patients>.
244. Htay H, Cho Y, Pascoe EM, et al. Multicenter registry analysis of center characteristics associated with technique failure in patients on incident peritoneal dialysis. *Clin J Am Soc Nephrol.* 2017;12(7): 1090–1099. doi:10.2215/CJN.12321216.
245. Shani MM, Mukhtar KN, Boorgu R, Leehey DJ, Popli S, Ing TS. Tissue plasminogen activator can effectively declot peritoneal dialysis catheters (letter). *Am J Kidney Dis.* 2000;36(3):675. doi:10.1053/ajkd.2000.16212.
246. Ravel VA, Streja E, Mehrotra R, et al. Serum sodium and mortality in a national peritoneal dialysis cohort. *Nephrol Dial Transplant.* 2017;32(7):1224–1233. doi:10.1093/ndt/gfw254.
247. Torlen K, Kalantar-Zadeh K, Molnar MZ, Vashistha T, Mehrotra R. Serum potassium and cause-specific mortality in a large peritoneal dialysis cohort. *Clin J Am Soc Nephrol.* 2012;7(8):1272–1284. doi:10.2215/CJN.00960112.
248. Uribarri J, Prabhakar S, Kahn T. Hyponatremia in peritoneal dialysis patients. *Clin Nephrol.* 2004;61(1):54–58. doi:10.5414/cnp61054.
249. Zanger R et al. Hyponatremia and hypokalemia in patients on peritoneal dialysis. *Semin Dial.* 2010;23(6):575–580. doi:10.1111/j.1525-139X.2010.00789.x.
250. Greenberg A, Bernardini J, Piraino BM, Johnston JR, Perlmutter JA. Hemoperitoneum complicating chronic peritoneal dialysis: a single-center experience and literature review. *Am J Kidney Dis.* 1992; 19(3):252–256. doi:10.1016/s0272-6386(13)80006-6.
251. Lew SQ. Hemoperitoneum: bloody peritoneal dialysate in ESRD patients receiving peritoneal dialysis. *Perit Dial Int.* 2007;27(3):226–233.
252. McCormick BB, Bargman JM. Noninfectious complications of peritoneal dialysis: implications for patient and technique survival. *J Am Soc Nephrol.* 2007;18(12):3023–3025. doi:10.1681/ASN.2007070796.

253. Teitelbaum I. Delivering high-quality peritoneal dialysis. What really matters? *Clin J Am Soc Nephrol*. 2020;15(11):1663–1665. doi:10.2215/CJN.02930320.
254. Mehrotra R, Devuyst O, Davies SJ, Johnson DW. The current state of peritoneal dialysis. *J Am Soc Nephrol*. 2016;27(11):3238–3252. doi:10.1681/ASN.2016010112.
255. Selby NM, Kazmi I. Peritoneal dialysis has optimal intradialytic hemodynamics and preserves residual renal function: why isn't it better than hemodialysis? *Semin Dial*. 2019;32(1):3–8. doi:10.1111/sdi.12752.
256. Manera KE, Johnson DW, Craig JC, et al. Establishing a core outcome set for peritoneal dialysis: report of the SONG-PD (Standardized Outcomes in Nephrology-Peritoneal Dialysis) Consensus Workshop. *Am J Kidney Dis*. 2020;75(3):404–412. doi:10.1053/j.ajkd.2019.09.017.
257. Roberts DM, Sevastos J, Carland JE, Stocker SL, Lea-Henry TN. Clinical pharmacokinetics in kidney disease: application to rational design of dosing regimens. *Clin J Am Soc Nephrol*. 2018;13(8):1254–1263. doi:10.2215/CJN.05150418.
258. Lea-Henry TN, Carland JE, Stocker SL, Sevastos J, Roberts DM. Clinical pharmacokinetics in kidney disease. Fundamental principles. *Clin J Am Soc Nephrol*. 2018;13(7):1085–1095. doi:10.2215/CJN.00340118.
259. Wilson FP, Berns JS. Vancomycin levels are frequently subtherapeutic during continuous venovenous hemodialysis. *Clin Nephrol*. 2012;77(4):329–331. doi:10.5414/cn106993.
260. Hirata S, Kadowaki D. Appropriate drug dosing in patients receiving peritoneal dialysis. *Contrib Nephrol*. 2012;177:30–37. doi:10.1159/000336933.
261. Cimino C, Burnett Y, Vyas N, Norris AH. Post-dialysis parenteral antimicrobial therapy in patients receiving intermittent high-flux hemodialysis. *Drugs*. 2021;81(5):555–574. doi:10.1007/s40265-021-01469-2.
262. Lewis SL, Nolin TD. New vancomycin dosing guidelines for hemodialysis patients: rationale, caveats, limitations. *Kidney360*. 2021;2(8):1313–1315. doi:10.34067/KID.0000192021.
263. Lavergne V, Ouellet G, Bouchard J, et al. Guidelines for reporting case studies on extracorporeal treatments in poisonings: methodology. *Semin Dial*. 2014;27(4):407–414. doi:10.1111/sdi.12251.
264. Ghannoun M, Hoffman RS, Gosselin S, Nolin TD, Lavergne V, Roberts DM. Use of extracorporeal treatments in the management of poisonings. *Kidney Int*. 2018;94(4):682–688. doi:10.1016/j.kint.2018.03.026.
265. King JD, Kern MH, Jaar BG. Extracorporeal removal of poisons and toxins. *Clin J Am Soc Nephrol*. 2019;14(9):1408–1415. doi:10.2215/CJN.02560319.
266. Decker BS, Goldfarb DS, Dargan PI, et al. Extracorporeal treatment for lithium poisoning: systematic review and recommendations from the EXTRIP Workgroup. *Clin J Am Soc Nephrol*. 2015;10(5):875–887. doi:10.2215/CJN.10021014.
267. Angioi A, Cabiddu G, Conti M, et al. Metformin associated lactic acidosis: a case series of 28 patients treated with sustained low-efficiency dialysis (SLED) and long-term follow-up. *BMC Nephrol*. 2018;19(1):77. doi:10.1186/s12882-018-0875-8.
268. Ahmed S, Kaplan A. Therapeutic plasma exchange using membrane plasma separation. *Clin J Am Soc Nephrol*. 2020;15(9):1364–1370. doi:10.2215/CJN.12501019.
269. Padmanabhan A, Connelly-Smith L, Aqui N, et al. Guidelines on the use of therapeutic apheresis in clinical practice—Evidence-based approach from the writing committee of the American Society for Apheresis: The Eighth Special Issue. *J Clin Apher*. 2019;34(3):171–354. doi:10.1002/jca.21705.
270. George JN, Nester CM. Syndromes of thrombotic microangiopathy. *N Engl J Med*. 2014;371(7):654–666. doi:10.1056/NEJMra1312353.
271. Walsh M, Merkel PA, Peh CA, et al. PEXIVAS Investigators: Plasma exchange and glucocorticoids in severe ANCA-associated vasculitis. *N Engl J Med*. 2020;382(7):622–631. doi:10.1056/NEJMoa1803537.
272. Keith PD, Wells AH, Hodges J, Fast SH, Adams A, Scott LK. The therapeutic efficacy of adjunct therapeutic plasma exchange for septic shock with multiple organ failure: a single-center experience. *Crit Care*. 2020;24(1):518. doi:10.1186/s13054-020-03241-6.